

Third- and Higher-Order ODEs

Coefficient synthesis, identification, and physical realization

Hob Nilre & Bo C. Herlin

2026-09-28

Abstract

We develop a framework for synthesizing coefficients of third- and higher-order ordinary differential equations from declared reference quantities. Two dimensional constraints give the complete monomial family in a positive reference triple; constitutive laws, interconnection structure or identifiable observations determine particular combinations. The rotary impact driver is the principal worked application: exact hammer–anvil elimination yields coefficient laws in four independent dimensionless coordinates, with seven- and nine-term constant-weight representations on specified restricted parameter domains. Changing the joint connects these laws to coefficient prediction and contact–joint discrimination; release exposes a remaining contact store accessible to torque and motion measurements on a slow torsional analogue. An exact branch-addition recurrence constructs successive higher-order operators and their forcing terms; a separate rational-series recurrence generates higher expansion coefficients without adding physical states. Reference transport, support independence and observation uncertainty distinguish representation, recovery and prediction. Stability, initial-data compatibility and passive realization delimit the equations’ physical use. Four physical collision states can give a third-order initialized output and a second-order zero-state transfer. RF, battery, mechanical, finite-supply and distributed examples establish complementary conditions and limitations. Signed port works and independently evaluated stores audit declared realizations after synthesis: contact-store deletion and omitted coupling power produce exact model residuals, while a hidden RC discharge has a complete internal account. Coefficient-discrimination and physical measurements remain conditional on their calibration and preparation assumptions; no measured discrepancy is reported.

Latest PDF on GitHub:

<https://github.com/hobnilre/physics-ode-3rd-deg/blob/main/third-and-higher-order-odes.pdf>

1 Synthesizing coefficients for higher derivative orders

Given reference quantities and additional model information, how can coefficients of higher derivatives be generated, determined and justified? This question organizes the article. Dimensional constraints supply a family of admissible terms. A constitutive model and its connections can determine a coefficient law by exact elimination; sufficiently informative observations can instead identify a law within a declared candidate family. Each construction must state its parameter domain, normalization, forcing and remaining freedom.

An impact driver gives the coefficient problem a concrete setting. During one blow, the rotating hammer loads the anvil through a deforming contact, while the output attachment and tightened joint supply a second dynamic response. Starting from those component laws, we construct the coefficients of the observed hammer motion. Changing the joint or contact then gives a direct question for the theory: which coefficients change, and can the same law predict another blow?

The construction begins with the second-order reference triple and its complete monomial family in Section 2. Section 3 carries the impact driver from its component equations to a fourth-order observation law, with exact coefficient support on several parameter domains in Section 3.2. Section 4 constructs arbitrary successive orders by adding physical relaxation states and distinguishes this operation from extending a transfer series. Section 5 returns to the impact measurements to establish what independent parameter changes and observations can recover. Initial data, physical realizations and finite-domain limits then determine what the synthesized equations describe.

The contribution is the connected treatment of these constructions: dimensional freedom, exact parameter-dependent coefficient laws, constructive extension in order, and conditions for recovery and physical interpretation. Dimensional analysis, realization theory, passive synthesis and crossing theory supply established foundations; their derivations are retained where needed. No claim of historical priority is made for those foundations. The examples make the determined quantities and the remaining freedom explicit.

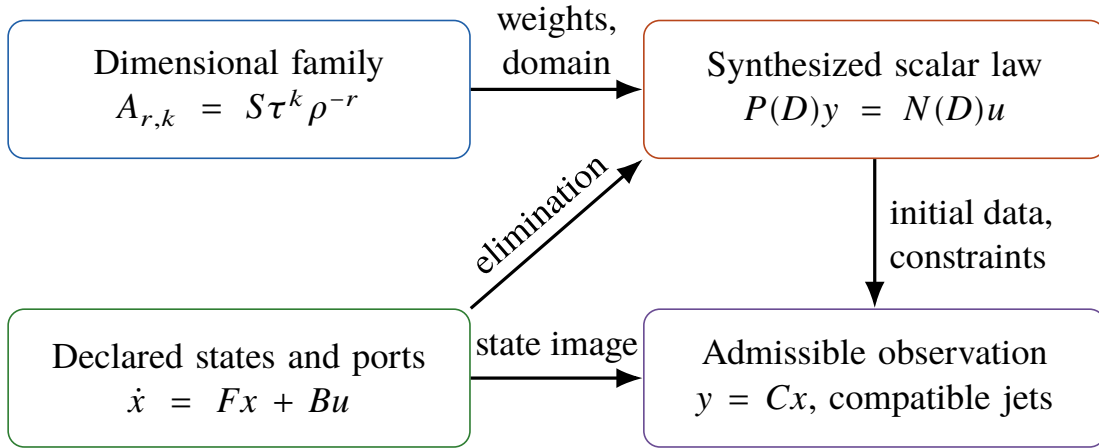
An ordinary differential equation has effective order n when its highest nonzero derivative coefficient multiplies $y^{(n)}$. Polynomial degree in y is a different property: a cubic contact force can occur in a second-order system, while a linear scalar equation can have fourth order. Throughout, $D = d/dt$, coefficients are constant on each declared interval unless stated otherwise, and the scalar observation y need not be a complete physical state.

Three questions must be answered separately. Are the coefficients dimensionally admissible? Does the selected scalar operator reproduce specified observables on a stated domain? Does a declared system of components, connections, ports and initial states realize that operator? Dimensional consistency answers the first question alone. A small discrepancy in one observable answers neither the third question nor the corresponding question about another observable.

For a finite linear realization $\dot{x} = Fx + Bu$, $y = Cx + D_0u$, elimination gives

$$\det(sI - F)Y(s) = C \operatorname{adj}(sI - F)B U(s) + D_0 \det(sI - F)U(s) + \text{initial-state terms}. \quad (1)$$

This polynomial equation can contain common factors. For a proper scalar transfer function, its reduced denominator degree equals the dimension of a minimal controllable and observable realization. Neither must equal the order of an unreduced annihilator or the number of physical storage coordinates in a particular construction. Initial states absent from the zero-state transfer can produce a free response at that same output, remain hidden there, or affect another port. Conversely, algebraic constraints can make nominal component coordinates dependent. These state and input–output distinctions belong to established realization theory [Kalman \(1963\)](#). They establish which equation and initialization a coefficient synthesis must retain; the examples are compared in Section 6. Figure 1 summarizes the construction.



Effective derivative order, observable order and physical state dimension are separate quantities.

Figure 1. Coefficient synthesis from a dimensional family or a declared realization. Weights and a parameter domain specify a candidate; elimination supplies both scalar and forcing operators. Arrows denote additional constructions or checks, not equivalences.

The article derives results for declared models. Ideal switches, linear contacts, lumped components and prescribed material laws are assumptions, not measurements. Exact integrations have zero numerical integration residual; a nonzero balance residual can nevertheless remain when an event deletes a store or the declared ports leave a coupling transfer unassigned. Missing physical mechanisms are kept as named, unevaluated model residuals. Neither work nor energy is used to choose coefficients, derivative order, release criteria, preparation or control strategy.

The primary measurement questions ask whether independent component changes distinguish coefficient contributions and whether one fixed law predicts independently specified configurations. The impact example makes the first question physical: which motion and torque measurements separate hammer–anvil contact from the joint against which the tool strikes? Section 5.1 gives a prediction-set separation criterion for these observations; the circuit and RF proposals supply complementary controls using calibrated voltage, current and phase. Parameter identification, candidate choice and final comparison remain separate. A small response discrepancy can coexist with an incorrectly recovered weak coefficient.

Physical audits test the declared realization after synthesis. The impact driver’s release question is developed beside its coefficient construction in Section 3.7: deleting a positive contact store without an event transfer produces a negative model residual. Its actual release law and destination remain unresolved, with the extended account in Section 15.3. Unequal effective transducer couplings produce the listed-port residual in Section 13.2, while the separately powered actuator in Section 13.2.2 supplies one explicit model account. The different finite bias supply retains independent coil, DC-link and thermal questions. A three-state RC loop in Section 8.4.2 instead has a complete resistor-work account despite zero loop current and changing internal stores. Imperfect observation still limits the hidden-energy bound of Section 8.4.1.

These supporting measurements have different attainable outcomes. Volt- and milliampere-scale branch observations can resolve the hidden RC change; finite transformer returns and a physical reference ramp expose additional measurable works at fixed capacitor endpoints. Sections 14.7 and 16.2 retain the compensation supplies, endpoint states and calibration bounds needed for broader accounts. Quiet terminals, zero named commands and a small combined residual leave distinct local questions. A heat channel, controller or reset identifies a physical transfer only when its power and state changes have been established independently. Every unresolved signed residual retains its interval, preparation, assumptions and uncertainty.

2 The second-order foundation and its dimensional extension

The reference triple fixes dimensions, time scale and a parameter chart. Source placement and topology fix the forcing interpretation. These foundations are kept explicit because a higher-order coefficient family must transport both when it is used across mechanical and electrical descriptions.

2.1 Variables, analogies and connections

Begin with

$$a\ddot{y} + b\dot{y} + cy = f, \quad v = \dot{y}, \quad x = av, \quad (2)$$

where $a, b, c > 0$ are reference quantities. The generalized effort is $f = a\dot{v} + bv + c \int v dt$, with the integration constant fixed by the initial y . The following correspondences include the state variable; agreement of coefficients alone is insufficient.

Table 1. Second-order mechanical and electrical templates. Linkage is $\lambda = N\phi$ for a winding of N turns; it is not generally the flux of one turn.

System	(a, b, c)	y	v	$x = av$	Effort f
Mass, damper, spring	(m, d, k)	displacement	velocity	momentum	force
Compliance, mobility, inverse mass	$(1/k, 1/d, 1/m)$	momentum	force	displacement	velocity
Series electrical circuit	$(L, R, 1/C)$	charge q	current i	linkage λ	voltage
Parallel electrical circuit	$(C, 1/R, 1/L)$	linkage λ	voltage v	charge q	current

Force balance at a mechanical connection corresponds to current balance under the appropriate mobility analogy, while common velocity corresponds to common voltage. Under an impedance analogy, the connection rules interchange. Arbitrary electrical graphs cannot be realized merely by replacing every inductor with an ordinary mass: a mass is referenced to an inertial frame, whereas a general two-terminal inertial element may require an inerter. Both kinematic constraints and port orientation must be transported.

The coefficient transformation

$$\mathcal{D}(a, b, c) = (1/c, 1/b, 1/a), \quad \mathcal{D}^2 = I \quad (3)$$

interchanges the two connection templates. It is an algebraic correspondence until topology and the effort–flow map are supplied. A useful experimental extension is a damped mechanical oscillator and a resistor–inductor–capacitor circuit with matched dimensionless dynamics, including preparation and reset [OP-LR50-01]. Displacement and current can be recorded with an encoder and a shunt at modest frequencies. Matching their normalized trajectories while independently auditing the driving force–velocity and voltage–current products would distinguish a complete port correspondence from a match of homogeneous equations. Lossless comparisons alone do not establish this damped correspondence: the coordinates below are singular at $b = 0$.

Specifically, with $\tau = a/b$, $\rho = b^2/(ac)$, $y = Yz$ and $\xi = t/\tau$, both driven templates reduce exactly to $z'' + z' + \rho^{-1}z = \varphi$, where $\varphi = f/(b^2Y/a)$ and primes denote ξ derivatives. Matching ρ , the normalized input, and the initial pair (z, z') predicts identical normalized responses by uniqueness. For mechanics observe velocity as $\dot{y} \tau/Y$; for the electrical charge coordinate observe current as $i \tau/Y$, and obtain charge from its known initial value and integrated current. A normalized trajectory difference exceeding the combined calibration and parameter-error enclosure rejects the proposed complete mapping. Agreement within it remains a bounded correspondence result. Preparation, sensor loading and each conjugate power product must still be audited independently.

2.2 Source placement and the forcing operator

A source completes the synthesis specification. An ideal battery maintains a voltage difference; its force–voltage counterpart maintains a force. Source location determines which coordinate is driven and which effort appears on the scalar equation's right-hand side. A real battery additionally requires its internal resistance and finite chemical state when those affect the chosen interval.

A voltage source V_b driving a series circuit gives $L\ddot{q} + R\dot{q} + q/C = V_b$. The corresponding mechanical equation is $m\ddot{x} + d\dot{x} + kx = F_b$, with force scale transported independently of the voltage unit. For uniform gravity on a mass referred to a fixed inertial support, downward displacement measured from the unloaded spring position gives $F_b = mg$. Constant effort still gives time-dependent power when its conjugate flow changes.

A battery across a parallel network instead prescribes its common voltage $V = V_b$. Its total delivered current is $i_b = C\dot{V}_b + V_b/R + i_L$, with $Li_L = V_b$. The mechanical series counterpart prescribes the common force through $\dot{p} = F_b$; its source-terminal velocity is $u_b = \ddot{p}/k + \dot{p}/d + p/m$. On a constant-force interval, $u_b = F_b/d + p/m$: spring deformation is fixed, damper velocity is fixed and the mass momentum changes. Substituting a force directly into a right-hand side with velocity units, or a voltage into one with current units, would change the equation being constructed.

Source placement within a branch gives another forcing law. For a voltage source V_g in series with the inductor of a parallel network, let $\lambda = Li_L$ be the inductor's own linkage and V the capacitor/resistor voltage. With the source polarity fixed by $\dot{\lambda} = V + V_g$, current balance gives

$$\begin{aligned} C\dot{V} + V/R + \lambda/L &= 0, & V &= \dot{\lambda} - V_g, \\ C\ddot{\lambda} + \dot{\lambda}/R + \lambda/L &= C\dot{V}_g + V_g/R. \end{aligned} \quad (4)$$

For constant V_g , the right-hand side is V_g/R . This reduction does not turn the physical voltage source into a current source or impose a voltage across the entire network. It is a simple example of an input operator generated by elimination.

Under the mechanical map $\lambda \leftrightarrow p$, $L \leftrightarrow m$, $C \leftrightarrow 1/k$ and $R \leftrightarrow d$, the branch source corresponds to a generalized force Q_g acting on the selected inertial coordinate. The common spring/damper force

is $f = \dot{p} - Q_g$, so their deformation and velocity are f/k and f/d . In a two-body relative coordinate, $m = \mu = m_1 m_2 / (m_1 + m_2)$ and $p = \mu(v_1 - v_2)$. Derive Q_g in that coordinate: uniform gravity on two freely falling bodies gives $Q_g = 0$, while $Q_g = mg$ applies to the fixed-reference case. Center-of-mass motion has separate storage and ports.

For a smooth interval $[t_0, t_1]$, enclose the passive elements, keep the source and heat reservoirs outside, and assume fixed supports with no other working ports. The signed source work is $\int F_b \dot{x} dt$ for the driven parallel mechanical system, $\int F_b u_b dt$ for its force-driven series counterpart, $\int V_b \dot{q} dt$ for the electrical series circuit, and $\int V_b i_b dt$ for the voltage-clamped parallel circuit. Each heat export is integrated separately from its actual damper velocity or resistor current. For the branch-source case in (4), the complete account is

$$\begin{aligned} W_g &= \int_{t_0}^{t_1} V_g \frac{\lambda}{L} dt, & W_R &= - \int_{t_0}^{t_1} \frac{V^2}{R} dt, \\ E_j &= \frac{\lambda(t_j)^2}{2L} + \frac{CV(t_j)^2}{2}, & E_1 - E_0 - W_g - W_R &= 0. \end{aligned} \quad (5)$$

Multiplication of the two physical state equations gives $\dot{E} = V_g \lambda/L - V^2/R$; separate integration proves the identity, with both endpoints evaluated using $V = \dot{\lambda} - V_g$. The mechanical counterpart uses source power $Q_g p/m$, heat power $-(\dot{p} - Q_g)^2/d$, and store $p^2/(2m) + (\dot{p} - Q_g)^2/(2k)$. Omitting the source offset from either endpoint changes the audit. Source attachment, incompatible initial states and ideal steps require their own finite or impulsive switching account. These results establish forcing and port conventions before any higher-order coefficient is interpreted.

2.3 All monomials of the required dimensions

Since $[a] = [c]T^2$ and $[b] = [c]T$, a coefficient of $D^k y$ must have dimensions $[c]T^k$. For a monomial $a^\alpha b^\beta c^\gamma$, the two constraints are $\alpha + \beta + \gamma = 1$ and $2\alpha + \beta = k$. With $r = \gamma$ their solution is

$$A_{r,k} = a^{k-1+r} b^{2-k-2r} c^r = S \tau^k \rho^{-r}, \quad S = \frac{b^2}{a}, \quad \tau = \frac{a}{b}, \quad \rho = \frac{b^2}{ac}. \quad (6)$$

This is the complete monomial family in the declared positive triple under the two stated dimensional constraints. Integer r, k give a Laurent family. Real r are also dimensionally admissible for positive references; dimensional analysis does not prove that physical exponents are integers. A continuous superposition over r would be a Mellin-type representation requiring its own measure, convergence conditions and identification assumptions.

For third order, $A_{0,3} = a^2/b$, $A_{-1,3} = ab/c$ and $A_{-2,3} = b^3/c^2$ are equally valid. An equation

$$P(D)y = f, \quad P(s) = \sum_{k=0}^n B_k s^k, \quad B_k = \sum_{r \in \mathcal{R}_k} w_{r,k} A_{r,k} \quad (7)$$

requires the dimensionless weights $w_{r,k}$ to be specified or identified. No dimensional argument fixes them. Negative k denote initialized integrals, treated in Section 7; they are not additional ordinary derivatives.

2.4 What the synthesis determines

For a constant-coefficient family, let $\rho, \eta_1, \dots, \eta_d$ be a complete set of independent dimensionless coordinates. Dimensional covariance permits

$$B_k = S\tau^k F_k(\rho, \eta_1, \dots, \eta_d). \quad (8)$$

Dividing by $S\tau^k$ removes the coefficient dimensions; invariance under a coherent change of units leaves dependence only on the independent dimensionless coordinates. The functions F_k are not fixed by this argument. Expressing one as a finite sum of $\rho^{-r} \prod_j \eta_j^{p_j}$ is an additional representation claim. Finite elimination may prove it; an arbitrary dimensionless function need not satisfy it. Integer, half-integer and unrestricted real exponents define distinct hypotheses. Evidence requiring a different class must be reported as a failure of the original class on the stated domain, rather than silently changing its definition.

Table 2. Information required to turn the dimensional family into a specified synthesis.

Information supplied	Consequence for the synthesis
Scalar observation, physical input and derivative convention	Defines the coefficients' dimensions and the equation being constructed
Positive reference triple and independent component ratios	Defines dimensional scales and parameter coordinates
Parameter domain and quantities held fixed	Defines whether the weights are constants across configurations
Exponent set, support and convergence conditions	Defines the candidate coefficient functions
Constitutive laws and connections, or informative observations	Determines coefficients or bounds their remaining freedom
Common equation normalization and forcing operator	Makes coefficient values and the retained physical drive comparable

A generated candidate family, an exact coefficient law, an identified law with uncertainty and a physically realized system are successive claims with different evidence. Synthesis may return a family with explicitly characterized freedom. Uniqueness requires the independence and observation conditions in Section 5.1. Allowing an arbitrary weight function of the varying coordinates would let that function absorb the unknown F_k and would remove the proposed constant-weight representation's predictive restriction.

For a driven realization the output can be $P(D)y = N(D)u$ with compatible initialized terms. In (7), $f = N(D)u$ is then the effective forcing, not automatically the physical input u . A nonzero normalization factor that is constant in time on the declared interval multiplies both P and N ; it also transports the initialized relation. If the factor varies in time, derivatives require the product rule and the constant-coefficient construction no longer applies unchanged. Monic normalization is valid only where the leading coefficient is nonzero. Combine signed contributions before assigning effective order, and use the original equations or a new reference chart at a zero reference value.

One important physical limit is already determined by the port assignment. For the direct electrical law $P(D)q = v$, $i = \dot{q}$, a polynomial degree of at least three gives an impedance $P(s)/s$ that cannot be positive real on the entire right half-plane. The proof and its assumptions are in Section 8. Rational input

operators, other scalar observations and finite-domain representations remain distinct possibilities. Thus a synthesis must say whether it specifies a formal operator, an initialized eliminated equation, a physical constitutive law or a finite representation with a stated discrepancy. Section 11 establishes the domain and preparation requirements for finite coefficient representations of distributed, fractional and delayed dynamics.

2.5 Conventional extensions and their limits

A conventional economical sequence is

$$r_k = \left\lfloor \frac{2-k}{2} \right\rfloor, \quad A_{r_{2j}, 2j} = \frac{a^j}{c^{j-1}}, \quad A_{r_{2j+1}, 2j+1} = \frac{a^j b}{c^j}. \quad (9)$$

The identities hold for all integer j , including negative values; increasing k by two multiplies the selected coefficient by a/c . In the (L, R, C) basis this choice minimizes the sum of absolute component exponents, with a tie resolved by placing R in the numerator. It is a convention dependent on the coordinate basis, as the following exact alternatives show.

Table 3. Basis dependence of monomial simplicity. Here $Z_0 = \sqrt{L/C}$, $\omega_0 = 1/\sqrt{LC}$, $\zeta = R/(2Z_0)$ and $\rho = 4\zeta^2$; integer restrictions introduce ties in the last basis.

Basis used to measure exponent size	Minimizing r under the stated convention
(L, R, C)	$\lfloor (2-k)/2 \rfloor$
(S, τ, ρ)	0
(L, τ, ρ)	0
(Z_0, ω_0, ζ) , real exponents allowed	$(2-k)/2$

The full electrical coefficient array is reproduced in Appendix A. It carries no automatic stability or passivity interpretation. For example the cubic selected by (9) factors exactly:

$$\frac{ab}{c}s^3 + as^2 + bs + c = (bs + c) \left(\frac{a}{c}s^2 + 1 \right). \quad (10)$$

It has an undamped imaginary pair. For $a = b = c = 1$, the fourth-order continuation is $1 + s + s^2 + s^3 + s^4 = (s^5 - 1)/(s - 1)$ and includes roots with positive real part. Positive coefficients are therefore not a stability certificate.

2.6 Transformations, support and convergence

The complete lattice relation and its electrical dual are

$$A_{r+\Delta r, k+\Delta k} = \rho^{-\Delta r} \tau^{\Delta k} A_{r, k}, \quad (11)$$

$$A_{r, k}(\mathcal{D}(a, b, c)) = \frac{ac}{b^4} A_{-r-k, k}(a, b, c) = \frac{1}{b^2} A_{1-r-k, k}(a, b, c). \quad (12)$$

Here $\tau' = \rho\tau$ and $\rho' = 1/\rho$. The two prefactors correspond to different normalizations; their support labels must not be mixed. Applying either complete transformation twice returns the original coefficient. A finite candidate set is generally not closed under $r \mapsto -r - k$. A comparison using unchanged labels after replacing the references compares different model classes.

A change of units represented by $(a, b, c) \mapsto (\mu \lambda^2 a, \mu \lambda b, \mu c)$ gives $A_{r,k} \mapsto \mu \lambda^k A_{r,k}$. A change of physical references is different. If a second independent dimensionless coordinate η is included, coefficients written as $S \tau^k \sum w_{r,s,k} \rho^{-r} \eta^s$ retain their values only if

$$\tilde{w}_{r,s,k} = w_{r,s,k} \frac{S}{\tilde{S}} \left(\frac{\tau}{\tilde{\tau}} \right)^k \left(\frac{\tilde{\rho}}{\rho} \right)^r \left(\frac{\eta}{\tilde{\eta}} \right)^s. \quad (13)$$

For an interchange that sends η to $1/\eta$, its exponent also changes sign. Reference singularities at zero damping, zero stiffness or zero inertia require another chart or the original equations; a divergent coordinate is not by itself a physical instability.

At fixed references $A_{r,k} = S(\rho^{-r})(\tau^k)$ is an outer product. Every finite such array has rank one. Different exponents at a fixed k change one coefficient; they do not add poles. For a modal signal $y = Y e^{st}$, an individual contribution is $S \rho^{-r} (\tau s)^k y$. Thus relative importance depends on the actual derivative content as well as on coefficient magnitude.

Infinite superpositions require convergence. Absolute convergence follows from $\sum_r |w_{r,k}| \rho^{-r} < \infty$; conditional sums need a declared ordering. Equal positive weights extending from r_0 downward give $\rho^{-r_0}/(1-\rho)$ for $0 < \rho < 1$, and upward give $\rho^{-r_0}/(1-1/\rho)$ for $\rho > 1$. The bilateral equal-weight sum diverges for every positive ρ , including $\rho = 1$. Cancellation cannot repair the positive case.

Useful diagnostics distinguish cancellation of coefficients from cancellation of evaluated terms:

$$\gamma_k = \frac{|B_k|}{\sum_r |w_{r,k} A_{r,k}|}, \quad \gamma_T = \frac{|\sum_k B_k y^{(k)}|}{\sum_k |B_k y^{(k)}|}, \quad \kappa(P, s) = \frac{\sum_k |B_k s^k|}{|P(s)|}. \quad (14)$$

A zero denominator makes the respective quotient undefined; a pole of $1/P$ makes the last condition measure unbounded. A normalized equation residual can instead use $\sum_k |B_k y^{(k)}| + |f|$ when nonzero. These quantities describe arithmetic sensitivity or cancellation, not physical loss or stored energy.

2.7 Independent dimensionless coordinates

Counting dimensional constraints identifies available coordinates, not which coordinates every coefficient must use.

Table 4. Dimensional counts for the principal applications.

Declared parameters	n	Rank	Dimensionless coordinates
(a, b, c)	3	2	ρ
Two inertias, two stiffnesses, two dampings	6	2	$J_h/J_a, k_c/k_j, d_j/d_c, d_j^2/(J_a k_j)$
Motional L_m, R_m, C_m and shunt C_0	4	2	$R_m^2 C_m / L_m, C_0 / C_m$
Ideal line Z_0, t_d	2	2	None until an excitation scale, such as ωt_d , is supplied

A restriction fixing contact and inertia values leaves a lower-dimensional collision manifold. Success there cannot establish a collapse of the full model onto ρ , or even onto (ρ, η) . Conversely, a count of four does not prove that all four enter every coefficient. Exact elimination settles that question.

3 The impact driver: from hammer–anvil contact to higher-order coefficients

The principal example is a rotary impact driver whose hammer engages the anvil to deliver a torsional blow to a fastener. Two synthesis tasks arise. **Constructing the observed-motion equation** starts from four physical coordinates, eliminates the anvil and represents the complete forced scalar law across declared parameter domains. **Constructing richer contact and joint laws** asks what additional internal dynamics or observations determine further coefficients. The first task is solved exactly for the component laws below; Section 3.6 formulates the second. The seven- and nine-atom identities establish representations of the specified model, not a new constitutive law for contact.

3.1 Contact, joint and observation boundary

Choose a driver bit coupled rigidly to the anvil on the modeled interval, driving an already seated fastener against a fixed workpiece. Include the bit and every co-rotating rigid output part in J_a . Place the effective twist and losses of the fastener, its interfaces and the workpiece attachment in the joint response (k_j, d_j) . In an impact-wrench configuration, a socket replaces the bit; its rigidly co-rotating inertia belongs in J_a , while any resolved socket twist requires its own compliance or additional states. Each configuration must declare that allocation once. The springs below represent effective deformation, not necessarily installed coil springs.

An incoming hammer state leads to engagement, compression, rebound and separation; the motor and clearance then prepare the next blow. The displayed constant-coefficient equations apply only to the smooth active-contact interval between engagement and release. Figure 2 connects the rotary mechanism to that interval's lumped model.

Let θ_h, θ_a be hammer and anvil angles, with inertias $J_h, J_a > 0$. A contact deformation is $\delta = \theta_h - \theta_a - g$, where g is the current clearance. During one active linear contact, choose its angular origin so that $g = 0$. Write

$$\begin{aligned} J_h \ddot{\theta}_h &= u - \tau_c, & J_a \ddot{\theta}_a &= \tau_c - \tau_j, \\ \tau_c &= k_c \delta + d_c \dot{\delta}, & \tau_j &= k_j \theta_a + d_j \dot{\theta}_a. \end{aligned} \tag{15}$$

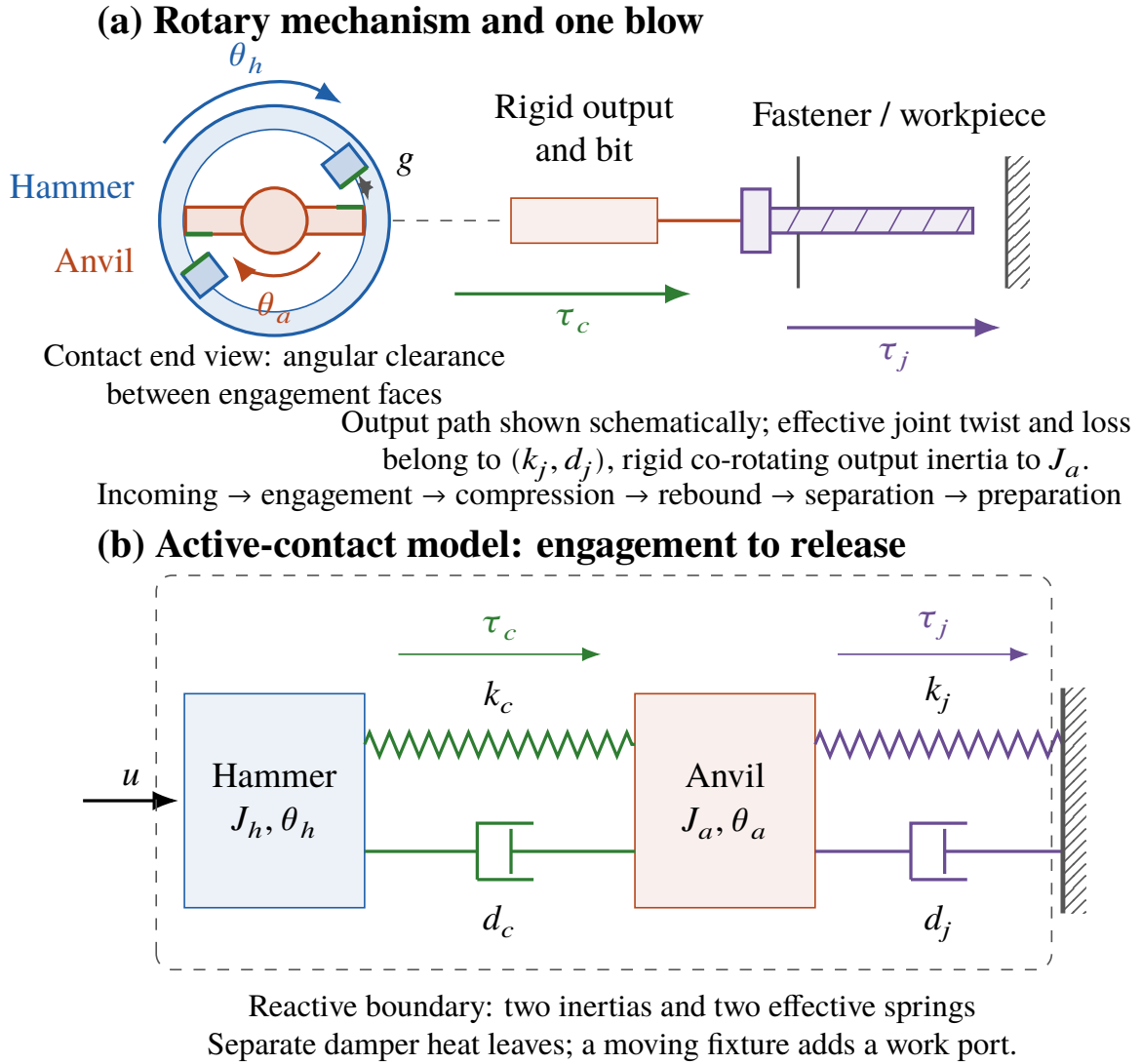


Figure 2. Schematic rotary impact-driver mechanism and its active-contact abstraction. Matching colors identify hammer, anvil and rigid output, deforming contact, and effective joint response. The angular face clearance closes at engagement; the spring-damper laws apply on the indicated contact interval. Separate damper channels export heat from the reactive boundary.

Figure 2 shows the active mechanical boundary. All four contact and joint constants are positive here. The drive torque u may be negligible during a short collision, but that is a declared interval assumption. A tightened joint can still deform elastically; absence of gross nut rotation does not imply $\theta_a = 0$. Contact geometry, fixture compliance and the torque measurement location are part of the model. The hammer-mechanism study of [Wettstein, Graubeger and Matthiesen \(2021\)](#) uses an impact-wrench setting. Its hammer–anvil and socket–joint distinction supplies the mechanism-level correspondence used here for the declared driver-bit boundary. This correspondence motivates a two-body spring–damper description; calibration must establish its parameters and response domain for the selected tool.

Eliminating θ_a gives the exact forced equation

$$\begin{aligned} P(D)\theta_h &= N(D)u, & N(s) &= J_a s^2 + (d_c + d_j)s + k_c + k_j, \\ P(s) &= J_h J_a s^4 + [J_h(d_c + d_j) + J_a d_c]s^3 \\ &\quad + [J_h(k_c + k_j) + J_a k_c + d_c d_j]s^2 + (d_c k_j + d_j k_c)s + k_c k_j. \end{aligned} \quad (16)$$

The determinant of the two-by-two dynamic stiffness matrix proves this expression. The highest coefficient, $J_h J_a$, records both resolved inertias. The coefficient of D^2 combines inertia–stiffness terms with $d_c d_j$, the product of the separate contact and joint dampings. Thus an apparently small coefficient contribution can encode an interaction between two physical regions. Both numerator and denominator matter: the physical hammer mobility is $Y_h(s) = sN(s)/P(s)$. Replacing it by $s/P(s)$ changes the driven system. With positive constants and a grounded joint spring, the state store $E = J_h \dot{\theta}_h^2/2 + J_a \dot{\theta}_a^2/2 + k_c \delta^2/2 + k_j \theta_a^2/2$ is positive definite. Its derivative for $u = 0$ is $-d_c \dot{\delta}^2 - d_j \dot{\theta}_a^2$. A trajectory on which this derivative stays zero has both velocities zero, and the equations then require both deflections zero. The finite linear system is therefore asymptotically stable. This proof concerns the physical realization, rather than a positivity rule for arbitrary polynomial coefficients.

3.2 Normalization and exact coefficient support

For a concrete dimensional illustration, set

$$J_h = \frac{1}{6250}, \quad J_a = \frac{1}{10000} \text{ kg m}^2, \quad k_c = k_j = 1500 \text{ N m}, \quad d_c = \frac{1}{100}, \quad d_j = \frac{1}{50} \text{ N m s}. \quad (17)$$

Angles are dimensionless radians. Dividing P by $J_h J_a$ gives exactly $s^4 + (725/2)s^3 + 39387500s^2 + 2812500000s + 14062500000000$ in the corresponding SI time units. The reduced contact inertia is $J_h J_a / (J_h + J_a) = 1/16250 \text{ kg m}^2$. An initially stationary anvil and hammer speed 600 s^{-1} have hammer kinetic energy $144/5 \text{ J}$. These are assumed values, not observations. Changing joint stiffness to 1000 or 2200 N m requires changing every dependent coefficient; changing initial speed to 400 or 800 s^{-1} does not change the linear operator.

The contact reference also gives an exact coefficient comparison. Put $J_* = J_h J_a / (J_h + J_a)$, use the economical coefficients $(k_c, d_c, J_*, J_* d_c / k_c, J_*^2 / k_c)$, and multiply P by $\gamma = J_*^2 / (k_c J_h J_a)$. The complete driven relation becomes $\gamma P(D)\theta_h = \gamma N(D)u$; the source torque and its differentiated contributions remain specified. Dividing the five resulting coefficients by those reference coefficients gives, for (17),

$$(w_0, w_1, w_2, w_3, w_4) = \left(\frac{40}{169}, \frac{120}{169}, \frac{3151}{1950}, \frac{29}{13}, 1 \right). \quad (18)$$

Here $w_4 = 1$ is an equation-normalization choice, not an independently recovered physical parameter. These weights describe one parameter point; transferring them to a different joint requires the parameter dependence derived below.

To construct a law across configurations, choose the joint references $(a, b, c) = (J_a, d_j, k_j)$ and let $h = J_h/J_a$, $\beta = k_c/k_j$, $\eta = d_j/d_c$, $\rho = d_j^2/(J_a k_j)$. Dividing both sides of (16) by k_c puts its left-hand coefficients in the dimensions of (2); its forcing remains $N(D)u/k_c$. An exact representation is

$$\begin{aligned} B_0 &= A_{1,0}, \\ B_1 &= A_{0,1} + \beta^{-1} \eta^{-1} A_{0,1}, \\ B_2 &= [1 + h + h/\beta] A_{0,2} + \beta^{-1} \eta^{-1} A_{-1,2}, \\ B_3 &= \beta^{-1} [h + (h+1) \eta^{-1}] A_{-1,3}, \\ B_4 &= h \beta^{-1} A_{-1,4}. \end{aligned} \tag{19}$$

This chart treats the four independent groups as independent. On a different experimental path, contact stiffness, contact damping and both inertias may be fixed while k_j, d_j vary. Then β is no longer independent: $k_j = d_j^2/(J_a \rho)$ and $d_j = d_c \eta$. Use the fixed normalization $\gamma = J_*^2/(k_c J_h J_a)$ from (18). The coefficient of s^k in γP is $\sum_{r,s} w_{r,s,k} A_{r,k} \eta^s$, with:

Table 5. Nine exact atoms on the fixed-contact, fixed-inertia manifold, with dimensionless constant weights under the stated normalization.

k	(r, s)	Exact weight $w_{r,s,k}$
0	(1, 0)	γk_c
1	(0, 0)	γk_c
1	(1, 1)	$\gamma d_c^2/J_a$
2	(0, 0)	$\gamma k_c (J_h + J_a)/J_a$
2	(0, 1)	$\gamma d_c^2/J_a$
2	(1, 2)	$\gamma J_h d_c^2/J_a^2$
3	(0, 1)	$\gamma d_c^2 (J_h + J_a)/J_a^2$
3	(0, 2)	$\gamma J_h d_c^2/J_a^2$
4	(0, 2)	$\gamma J_h d_c^2/J_a^2$

To verify the table directly, use $A_{1,0} = k_j$, $A_{0,1} = d_j$, $A_{1,1} = J_a k_j/d_j$, $A_{0,2} = J_a$, $A_{1,2} = J_a^2 k_j/d_j^2$, $A_{0,3} = J_a^2/d_j$, and $A_{0,4} = J_a^3/d_j^2$. Each term of P then has exactly the listed dependence. If damping is fixed as well and only k_j varies, the required supports are $\{1\}$, $\{0, 1\}$, $\{0, 1\}$, $\{0\}$, $\{0\}$, respectively: seven atoms suffice. Extending that one-coordinate conclusion to independently variable damping fails algebraically.

Writing the fixed joint damping as d_0 , the seven weights in that support order are

$$\begin{aligned} w_{1,0} &= \gamma k_c, & (w_{0,1}, w_{1,1}) &= \gamma \left(k_c, \frac{d_c d_0}{J_a} \right), \\ (w_{0,2}, w_{1,2}) &= \gamma \left(\frac{k_c (J_h + J_a) + d_c d_0}{J_a}, \frac{J_h d_0^2}{J_a^2} \right), \\ (w_{0,3}, w_{0,4}) &= \frac{\gamma}{J_a^2} ([J_h (d_c + d_0) + J_a d_c] d_0, J_h d_0^2). \end{aligned} \tag{20}$$

Here the two subscripts denote (r, k) . Distinct powers of positive k_j are linearly independent on an open interval, so a constant-weight support lacking one of these nonzero powers cannot represent that entire interval. The economical single coefficient at each order fails this requirement. Allowing extra atoms only at low orders still misses the constant high-order coefficients; shifting the high-order supports to include $r = 0$ repairs that particular defect. Two adjacent atoms at every order or three atoms around each economical choice contain redundant powers on this one-coordinate path. A wider sparse window can contain the seven-atom solution, but a finite search window is not a proof of global optimality.

The four independent groups also specify the normalized state dynamics exactly. With $\xi = t/\tau$, $\tau = J_a/d_j$, and $z = (\theta_h/Y, \tau \dot{\theta}_h/Y, \theta_a/Y, \tau \dot{\theta}_a/Y)^T$, the unforced equation is

$$\frac{dz}{d\xi} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\beta/(h\rho) & -1/(h\eta) & \beta/(h\rho) & 1/(h\eta) \\ 0 & 0 & 0 & 1 \\ \beta/\rho & 1/\eta & -(\beta+1)/\rho & -(1+1/\eta) \end{pmatrix} z. \quad (21)$$

Conversely any positive $J_a, \tau, h, \rho, \beta, \eta$ reconstruct $d_j = J_a/\tau$, $J_h = hJ_a$, $k_j = J_a/(\tau^2\rho)$, $k_c = \beta k_j$, and $d_c = d_j/\eta$. Thus the dimensionless groups leave dimensional scale freedoms. Changing the stiffness reference from joint to contact sends (β, ρ) to $(1/\beta, \rho/\beta)$ with the other physical ratios unchanged; physically interchanging two unequal springs is a different operation.

The contribution distinguishing the nine-atom construction from an eight-atom candidate is $d_c d_j$ in the coefficient of s^2 . In the table it has $(k, r, s) = (2, 0, 1)$. A candidate that omits it has a nonzero coefficient discrepancy on the positive parameter domain; it does not show that the physical term is absent. A small response error can coexist with an incorrectly recovered small coefficient, particularly when the response is insensitive to that coefficient.

Table 6. Exact coefficient representations under distinct parameter restrictions. Their source operators and compatible initial data are retained.

Parameter domain	Fixed description	Exact claim
All six positive component parameters	Four independent ratios and two dimensional scales; $P/k_c, N/k_c$	The coefficient functions in (19) reproduce the complete driven model
Fixed inertias and contact constants; variable k_j, d_j	Joint references, (ρ, η) , fixed γ	The nine atoms reproduce γP throughout this two-coordinate domain
The preceding domain with $d_j = d_0$ fixed	One varying coordinate and the same equation scale	The seven weights in (20) reproduce γP throughout the stiffness interval

These are symbolic identities over the stated domains, stronger than agreement at the illustrative component values. They establish representation and exact transfer within each domain without refitting. They do not establish recovery from noisy hammer measurements or transfer to a different topology. Changing the reference chart transports the weights by (13); changing the physical parameters outside the restriction requires the larger coefficient law. The following comparisons give those domains an impact-driver interpretation.

Table 7. From a change in the impact mechanism to a prediction under one coefficient law. The scaling claim concerns a common smooth interval; engagement and release conditions remain separate.

Physical change during a declared comparison	Coefficient law and prediction
Scale the complete incoming state by λ ; keep all components fixed and $u = 0$ on the common active interval	P and N stay fixed. Angles, rates and torque scale by λ ; physical powers scale by λ^2 . Changing speed alone gives this result only when the other prepared coordinates scale compatibly.
Vary k_j with J_h, J_a, k_c, d_c, d_j fixed	The seven weights predict γP throughout the stiffness interval; use γN and the declared preparation for each response.
Vary k_j, d_j independently with inertias and contact fixed	The nine atoms predict γP with the same fixed weights; γN changes according to the same component laws.
Vary contact properties independently of the joint	Use the full chart for P/k_c and N/k_c . Richer constitutive dynamics require their own internal-state or observation information.

A preload change is not automatically a known stiffness change: its relation to effective joint stiffness, damping and possible slip must be identified. For each comparison, establish the component parameters, sensor response and incoming state before the final prediction. Changing fitted weights independently at every joint setting would forfeit the claimed transferable law. The next subsection determines which initialized scalar order the resulting physical operator actually requires.

3.3 Generic order and exact exceptions

Four independent coordinates are present, but a single observed angle does not always have order four. A hidden motion with $\theta_h = 0$ must satisfy $(d_c D + k_c)\theta_a = 0$ and $(J_a D^2 + d_j D + k_j)\theta_a = 0$. Consequently the exact cancellation surface is

$$\Delta = k_c^2 - \frac{k_c d_c d_j}{J_a} + \frac{d_c^2 k_j}{J_a} = 0. \quad (22)$$

Away from this surface the hammer observation has four independent jets and the forced transfer is generically of degree four. On it the mode $s = -k_c/d_c$ is invisible at that observation. For the illustrative normalized choice $J_h = J_a = k_c = d_c = k_j = 1, d_j = 2$,

$$N(s) = (s+1)(s+2), \quad P(s) = (s+1)(s^3 + 3s^2 + 2s + 1). \quad (23)$$

Every component remains positive. Thus a universal claim of fourth-order necessity for all positive components is false; generic necessity and exact exceptional surfaces are the correct statements. A visible trajectory can also lack a mode because its amplitude happens to vanish, which does not reduce the system's generic order.

There is a further distinction within the cancellation surface. A double zero of N at $s = -k_c/d_c$ requires $N' = 0$ there as well. Solving these two conditions gives

$$k_j = \frac{J_a k_c^2}{d_c^2} - k_c, \quad d_j = \frac{2J_a k_c}{d_c} - d_c, \quad J_a k_c > d_c^2,$$

$$\begin{aligned}
N(s) &= \frac{J_a}{d_c^2} (d_c s + k_c)^2, \\
P(s) &= \frac{(d_c s + k_c)^2}{d_c^2} (J_a J_h s^2 + J_a d_c s + J_a k_c - d_c^2).
\end{aligned} \tag{24}$$

The inequality makes both joint coefficients positive. The reduced zero-state angle transfer has degree two on this subset; it has degree three generically elsewhere on the cancellation surface. This additional cancellation does not remove every initialized natural contribution.

For example, in consistent normalized units choose $J_h = 1$, $J_a = 2$, $k_c = d_c = k_j = 1$ and $d_j = 3$. Then

$$\frac{\Theta_h(s)}{U(s)} = \frac{2}{2s^2 + 2s + 1}, \quad (D+1)(2D^2 + 2D + 1)\theta_h = 0 \quad (u=0). \tag{25}$$

In state order $(\theta_h, \dot{\theta}_h, \theta_a, \dot{\theta}_a)$ the realization matrix is

$$F = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1/2 & 1/2 & -1 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad C = (1 \ 0 \ 0 \ 0). \tag{26}$$

The controllability and observability matrices each have rank three. Direct multiplication gives $C(F + I)(2F^2 + 2F + I) = 0$, whereas $C(2F^2 + 2F + I) = (-1, 0, 2, 2)$. The preparation $(1, -1, 0, 1)$ has the exact free motion $\theta_h = e^{-t}$, $\theta_a = te^{-t}$, verified by substitution into both body equations. Its hammer motion is absent from the zero-state quadratic transfer. Thus four physical states, a third-order initialized scalar output and a second-order driven transfer coexist with positive components. The anvil and contact stores remain in the physical work account.

A fixed anvil gives $J_h \ddot{\theta}_h + d_c \dot{\theta}_h + k_c \theta_h = u$, a second-order equation, with a generally nonzero fixture reaction. A massless hammer with $u = 0$ instead imposes $\tau_c = 0$. For $d_c > 0$, this leaves $\delta(t) = \delta(0)e^{-k_c t/d_c}$ alongside the second-order anvil motion. The hammer angle can therefore remain third order. It becomes identical to the second-order anvil motion only with compatible $\delta(0) = 0$, or under a further contact constraint. Setting an inertia to zero after assuming four arbitrary initial jets is invalid.

A compliant boundary with its own inertia adds another displacement and velocity, hence up to six states. A rational joint impedance likewise adds its internal states. Fixing the anvil suppresses the motion needed to identify those joint dynamics. The limiting model must be derived from the constrained equations, with preparation and observation specified; a small positive inertia does not justify a uniform lower-order law near resonance.

3.4 Competing contact descriptions

An instantaneous rigid-impact map predicts a velocity jump and restitution but no resolved contact waveform. For a one-dimensional free pair with initial velocities u_1, u_2 , restitution e gives

$$v_1^+ = \frac{(m_1 - em_2)u_1 + (1+e)m_2u_2}{m_1 + m_2}, \quad v_2^+ = \frac{(1+e)m_1u_1 + (m_2 - em_1)u_2}{m_1 + m_2}. \tag{27}$$

These follow from signed momentum and the specified relative-velocity law, not an energy premise. For $u_2 = 0$, $u_1 > 0$, the striker stops at $m_1 = em_2$ and retains forward motion above that ratio. A fixture or accumulating penetration boundary invalidates the free-pair assumption. An oblique frictionless contact

applies the impulse along its declared normal and leaves the tangential relative component unchanged; offset rotation requires (159). For a finite compliant regularization of a constant-mass body, integrating $F \cdot v$ gives the impulse work $J \cdot (v^- + v^+)/2$. Multiplying a delta impulse by an unspecified discontinuous velocity is not an independent work definition. A linear compliant contact predicts a duration, a waveform and reversible spring storage with viscous dissipation. A nonlinear contact such as

$$\tau_c = k_c \delta + d_c \dot{\delta} + \beta k_c \frac{\delta^3}{\delta_*^2}, \quad U_c = \frac{k_c \delta^2}{2} + \frac{\beta k_c \delta^4}{4 \delta_*^2} \quad (28)$$

retains four first-order mechanical states but has no global constant-coefficient fourth-order scalar law. Its tangent stiffness is $k_c [1 + 3\beta \delta^2 / \delta_*^2]$; illustrative values $\beta = 1/5$, $\delta_* = 1/25$ define one model only. A Hertz-type fractional power law has a local tangent only on its active deformation domain and needs a separate release rule.

A candidate scalar extension may use $B_k(\delta, p)$ depending on deformation and preload p , or an explicit nonlinear remainder $g(\delta, \dot{\delta}, p)$. It must be derived or identified with its own regularity and initialization conditions. Eliminating a nonlinear internal coordinate can introduce products of derivatives, local inversion conditions and singularities; assigning variable weights to a linear polynomial does not prove equivalence to the four-state contact model.

A higher-order linear scalar model can represent eliminated linear modes or a finite-band rational contact response. It cannot, merely by adding derivatives, replace engagement gates, backlash, partial-edge geometry, dry friction, adhesion, plastic deformation, evolving preload or an amplitude-dependent force law. Resolved inertias must not be counted again in a contact correction. If an exact contact transfer is expanded as a derivative series, convergence of that series and admissible initial history must be proved; retaining a finite internal-state realization is often the clearer construction.

These distinctions determine the comparisons that are meaningful:

Table 8. Collision descriptions and their distinct scopes.

Description	What can be predicted within its assumptions	What remains unspecified
Rigid restitution	Outgoing velocities and impulse	Contact duration, peak torque, within-contact power
Linear compliant two-body model	Motion, torque and signed power during active contact	Release physics and unmodeled boundary modes
Nonlinear compliant model	Amplitude-dependent waveform under a declared force law	Constitutive transfer outside the declared law
Higher-order scalar linear model	A chosen observation and compatible derivatives	Component interpretation, other ports and preparation unless realized

For a linear system with proportionally scaled initial state and input, motions and torques scale by the amplitude factor, while physical powers scale by its square. Failure of this scaling can distinguish nonlinear contact, altered boundary conditions or instrumentation effects; it does not identify the cause uniquely. Compare contact duration, peak torque, outgoing speeds, compression and rebound phases, instantaneous signed power and its peak and root-mean-square values. A global work match can conceal a wrong waveform and is not an order-selection criterion.

3.5 Initial states, repeated engagement and measurement

The four jets of $q = \theta_h$ determine the hidden anvil coordinates only when the following matrix is nonsingular:

$$\begin{pmatrix} k_c & d_c \\ -d_c k_j/J_a & k_c - d_c d_j/J_a \end{pmatrix} \begin{pmatrix} \theta_a \\ \dot{\theta}_a \end{pmatrix} = \begin{pmatrix} J_h \ddot{q} + k_c q + d_c \dot{q} \\ J_h q^{(3)} + k_c \dot{q} + d_c (1 + J_h/J_a) \ddot{q} \end{pmatrix}. \quad (29)$$

Here $u = 0$ and the angle origin is the active-contact origin. The determinant is (22). Close to that surface a formally invertible reconstruction can be practically useless. Noisy jerk is especially dangerous: differentiating the angle three times amplifies high-frequency errors, and stable coefficients do not bound the response to an arbitrarily corrupted initial jet.

During free flight set the contact torque to zero, evolve both bodies and the joint, and include the motor if present. Re-engagement occurs at a root of the gap condition with the appropriate approaching velocity. States remain continuous at an uncompressed engagement without an imposed impulse; higher jets generally jump because the vector field changes. A motor acting between contacts does not justify a motor-free model of an entire impact sequence. More than a small number of engagements, frictional changes and evolving joint properties require their own constitutive model.

Which measurements separate contact dynamics from the frequency-dependent joint against which the impact driver strikes [OP-LR08-01]? This is the principal application-specific identification question. A practical discriminating setup is a low-energy torsional impact between two instrumented disks and an interchangeable compliant support. Two encoders provide synchronized hammer and anvil angles, with rates reconstructed within their calibrated response band; a calibrated transducer observes contact torque at the declared interface. A support-torque channel adds a direct observation of joint loading. Choose the slow contact so its discrimination frequencies lie inside the common calibrated amplitude and phase band, and include rate-reconstruction error in the uncertainty. Hold inertia and contact geometry fixed while changing support stiffness, then change contact compliance independently. Agreement of hammer motion with disagreement in anvil motion rejects a claimed internal reconstruction. Trigger offset, probe loading, anti-alias response, colored uncertainty and fixture variation must enter the comparison. For a selected commercial driver, record the bit or socket allocation, sensor location, contact-duration range and usable calibrated bandwidth anew; the slow rig does not establish these. A fully synchronized industrial impact measurement is a stronger, separately unfulfilled requirement.

The exact hidden-motion alternative on (22) makes the required internal sensor concrete. Two preparations differing by $\Delta\theta_a(t) = Ae^{-k_c t/d_c}$, with $A \neq 0$ and the corresponding anvil velocity, have identical hammer angle and contact torque: $k_c \Delta\theta_a + d_c \Delta\dot{\theta}_a = 0$. An anvil encoder separates them at time t if $|A|e^{-k_c t/d_c}$ exceeds the sum of their angle-error bounds. A contact-torque channel alone cannot detect this particular motion. Further derivatives of the identical hammer-angle trace cannot recover it either. Away from the cancellation surface, compare the complete predictions $Ce^{Ft}x_0$ and the driven convolution from each calibrated contact/boundary model using (48); allowing fixture parameters to vary can cause those prediction sets to overlap even when a nominal observability matrix has full rank.

3.6 Separate contact and joint constitutive families

The second synthesis task supplies dynamics within the contact or joint. Its coefficients require extra information: a declared internal-state model or observations that identify its parameters independently of the resolved body motion. A scalar equation for hammer angle obtained by eliminating the anvil differs from a higher-derivative constitutive proposal for the contact itself. To formulate the latter, introduce positive unresolved-contact reference scales (J_*^c, d_*^c, k_*^c) and independent joint scales (J_*^j, d_*^j, k_*^j) . The

superscripts identify different physical regions, not derivatives. For $\ell \in \{c, j\}$ define

$$A_{r,k}^\ell = (J_*^\ell)^{k-1+r} (d_*^\ell)^{2-k-2r} (k_*^\ell)^r, \quad B_k^\ell = \sum_r w_{r,k}^\ell A_{r,k}^\ell. \quad (30)$$

Each coefficient has torque times time to power k per angular deformation. The economical contact sequence has the six coefficients $k_*^c, d_*^c, J_*^c, J_*^c d_*^c / k_*^c, (J_*^c)^2 / k_*^c, (J_*^c)^2 d_*^c / (k_*^c)^2$ at orders zero through five. It remains a convention, not six new components. In particular, J_*^c must represent an unresolved local inertia or an explicitly modeled internal contact mode; assigning either resolved body inertia to it again counts that inertia twice.

On each smooth active interval, the candidate laws are

$$\tau_c = \chi_c \sum_{k=0}^{N_c} B_k^c \delta^{(k)}, \quad \tau_j = \sum_{k=0}^{N_j} B_k^j \theta_a^{(k)}, \quad \chi_c \in \{0, 1\}. \quad (31)$$

The contact gate χ_c is fixed within that interval and supplied by an engagement/release law at its boundaries. Derivatives of a switching gate are not silently included in the smooth constitutive equation. Orders zero and one recover separate spring–damper descriptions. Varying their two sets of coefficients together would conceal whether a missing mode belongs to contact or support. The two-body balances still have to be solved together with these constitutive laws; their combined scalar order is not simply either constitutive truncation index.

A constructed linear internal contact model instead has, for example, $\dot{z}_c = F_c z_c + G_c \delta$ and $\tau_c = H_c z_c + K_c \delta$. Its zero-state dynamic stiffness is $\mathcal{H}_c(s) = K_c + H_c(sI - F_c)^{-1} G_c = P_c(s)/Q_c(s)$. For initialized elimination take $Q_c(s) = \det(sI - F_c)$ and $P_c(s) = K_c Q_c(s) + H_c \text{adj}(sI - F_c) G_c$ before cancelling common factors. Eliminating the states on a smooth interval gives

$$Q_c(D) \tau_c = P_c(D) \delta, \quad Q_j(D) \tau_j = P_j(D) \theta_a, \quad (32)$$

with initial derivatives restricted by the respective state-to-jet maps. The joint uses the analogous unreduced construction. A reduced pair is sufficient only when every removed natural contribution is invisible at this output or excluded by the preparation. Zero-state reduction alone cannot remove a separately prepared natural response. A Kelvin–Voigt direct term or an explicit inertial term can be retained alongside the internal-state model when its port realization is declared.

If $\mathcal{H}_c$ is analytic at the expansion point, its infinite Taylor series is an exact analytic identity only inside the disk to the nearest uncanceled pole. Identifying that series with a time-domain derivative operator further requires convergence on the chosen signals and transport of the initial state or history. For a finite polynomial T_N , the exact transfer discrepancy is $(P_c - Q_c T_N)/Q_c$. Thus a finite derivative family is a conditional candidate on a declared response domain, not an exact replacement of arbitrary rational memory. Negative derivative indices additionally require the initialized histories of Section 7. A repeatable slow tail may motivate their investigation, but cannot determine their integration constants.

The physical contact power remains $\tau_c \dot{\delta}$. Its formal decomposition into $B_k^c \delta^{(k)} \dot{\delta}$ does not make each summand a separately accessible port or a physical store. An instantaneous contribution fraction can be defined as the absolute value of one summand divided by the sum of absolute values, only when that denominator is nonzero; it is a diagnostic of the chosen representation and normalization. Model comparison can use independently measured torque, motion, phase and instantaneous physical power, with coefficient estimation and final conditions kept separate. It cannot use the integrated formal term

works to choose weights or derivative order.

The practical comparison asks which repeatable features of the impact response are explained by additional linear dynamics and which require a changing contact or joint law. Tuned linear pulse shape, phase and unresolved linear modes may then receive a rational constitutive explanation; resolved rebound may predict restitution rather than impose it. None of these constructions removes release logic, partial engagement, backlash, preload-dependent microslip or an amplitude-dependent deformation law. Stable response and port passivity require the whole declared interconnection. A direct physical contact law of polynomial degree above two faces the same global passivity obstruction as the electrical law proved in Section 8, with torque and angular rate replacing voltage and current.

3.7 What remains when the blow releases?

The blow ends when contact releases. What remains stored, what returns to the mechanism, and what leaves through the support [OP-LR03-01]? At zero contact torque in the linear spring–damper model,

$$\delta = -\frac{d_c}{k_c} \dot{\delta}, \quad U_c(t_e^-) = \frac{d_c^2 \dot{\delta}(t_e^-)^2}{2k_c}. \quad (33)$$

This store is positive unless relative speed vanishes. Consider the explicit deletion model: the two body states and joint store remain continuous at t_e , the contact store is removed, and no finite or impulsive release transfer is supplied. Bounded regular port powers have vanishing integrals on a shrinking event interval. Independent event-side store evaluation therefore gives

$$\Delta E_e = -U_c(t_e^-), \quad \sum_{\ell} W_{\ell,e}^{\text{declared}} = 0, \quad r_{E,e} = -\frac{d_c^2 \dot{\delta}(t_e^-)^2}{2k_c}. \quad (34)$$

The removed store has positive magnitude; the residual is negative with power positive into the boundary. The calculation checks the torque-zero condition, event-side continuity, the spring store and the regular port integrals. Its numerical integration residual is zero. The physical destination and a constitutive law for the release remain unresolved.

The first accessible comparison uses the slow torsional rig of Section 3.5. Synchronized encoders give δ and its rate within the calibrated band; the torque transducer locates the zero-torque event. Independently calibrated contact stiffness and measured pre-release deformation determine $k_c \delta^2/2$. Resolve a positive lower bound on this store after propagating stiffness, angle and event-timing uncertainty, then follow retained deformation and torque–motion work through release. A support-transfer measurement additionally needs support torque and motion: a fixed support can exert torque while doing zero work.

This comparison connects the synthesized active-contact law to the event at which its description changes. It audits an independently specified realization and trajectory; the energy outcome does not choose the coefficient weights or order. Section 15 derives the separate signed port works, and Section 15.3 develops the destination alternatives, repeated-event residual and two-sided observation window. Thermal, acoustic and fracture shares require their own measured channels.

4 Constructive synthesis at arbitrary order

There are three different continuations of a coefficient construction: extend its dimensional basis, add an independently specified physical state, or retain more coefficients of one rational transfer series. Only the second necessarily adds a physical coordinate, and even then the order visible at a selected output can decrease through cancellation. The following construction determines complete coefficient laws at

successive orders before their physical work is audited. It gives an explicit answer beyond the impact driver's fourth-order example: each added relaxation branch supplies the extra physical information that its next coefficients require. The rational contact and joint models of Section 3.6 pose the analogous state-addition question. Applying the electrical construction mechanically requires the actual connection topology and effort–flow map; renaming an electrical parameter does not derive a contact law.

4.1 Adding physical relaxation states

Let positive current enter a battery's positive terminal. An energized coil of inductance $L > 0$ transfers current through an ideal diode into a constant open-circuit voltage $V_{oc} \geq 0$, total series resistance $R_\Sigma \geq 0$ and N polarization branches with $R_j, C_j > 0$:

$$Li + R_\Sigma i + V_{oc} + \sum_{j=1}^N v_j = 0, \quad C_j \dot{v}_j = i - v_j/R_j, \quad \tau_j = R_j C_j. \quad (35)$$

The equation applies on a conducting interval with $i > 0$, up to its first downward zero if one exists. Continuing it through negative current removes the diode constraint and describes a different system. An initial negative current is incompatible with this diode orientation: a reversed diode or a finite commutation circuit is a different declared topology. At zero current, the battery terminal voltage determines whether blocking or renewed conduction is admissible.

Eliminating the voltages gives, with $Q(s) = \prod_j (1 + \tau_j s)$,

$$\mathcal{P}(s) = (Ls + R_\Sigma)Q(s) + \sum_j R_j \frac{Q(s)}{1 + \tau_j s}, \quad \mathcal{P}(D)i = -Q(D)V_{oc}. \quad (36)$$

For distinct active time constants the generic current order is $N + 1$; three branches give fourth order. A current equation is shifted by one derivative index relative to the charge convention in (62). The affine forcing and arbitrary initial branch voltages must be retained in a state-to-jet comparison.

Write Q_N and $\mathcal{P}_N$ for the two polynomials with N branches. A separately imposed series voltage u changes the zero on the right of the first state equation to u . The complete eliminated driven law is then

$$\mathcal{P}_N(D)i = Q_N(D)(u - V_{oc}), \quad \frac{I(s)}{U(s)} = \frac{Q_N(s)}{\mathcal{P}_N(s)} \quad \text{for zero-state perturbations.} \quad (37)$$

The constant operating offset and prepared branch states remain separate from that perturbation transfer. Specifying only $\mathcal{P}_N$ would omit the input operator. The current cannot be continued past a diode event without selecting the appropriate post-event law.

Hold L, R_Σ and the existing branches fixed, and add a branch with independently declared $R_{N+1} > 0$ and $\tau_{N+1} > 0$. Splitting the new branch from the sum in (36) gives

$$\begin{aligned} Q_{N+1}(s) &= (1 + \tau_{N+1}s)Q_N(s), \\ \mathcal{P}_{N+1}(s) &= (1 + \tau_{N+1}s)\mathcal{P}_N(s) + R_{N+1}Q_N(s), \\ Q_0(s) &= 1, \quad \mathcal{P}_0(s) = Ls + R_\Sigma. \end{aligned} \quad (38)$$

Every old summand acquires the factor $1 + \tau_{N+1}s$; the new summand is $R_{N+1}Q_N$. This proves the construction for any finite N . Define $Q_N = \sum_k q_k^{(N)} s^k$ and $\mathcal{P}_N = \sum_k p_k^{(N)} s^k$, with zero coefficients

outside their polynomial ranges. Coefficient comparison yields

$$\begin{aligned} q_k^{(N+1)} &= q_k^{(N)} + \tau_{N+1} q_{k-1}^{(N)}, \\ p_k^{(N+1)} &= p_k^{(N)} + \tau_{N+1} p_{k-1}^{(N)} + R_{N+1} q_k^{(N)}. \end{aligned} \quad (39)$$

Thus adding a relaxation state changes lower-order coefficients as well as supplying a new highest derivative. Induction gives $q_N^{(N)} = \prod_j \tau_j$ and $p_{N+1}^{(N)} = L \prod_j \tau_j$. For positive finite parameters the unreduced degree is exactly $N + 1$; initialized and reduced transfer orders still require the cancellation analysis in Section 9.

For one branch, the construction is already explicit:

$$\mathcal{P}_1(s) = (R_\Sigma + R_1) + (L + R_\Sigma \tau_1)s + L\tau_1 s^2, \quad Q_1(s) = 1 + \tau_1 s. \quad (40)$$

Adding a second branch gives a cubic current operator whose four coefficients are

$$\begin{aligned} p_0^{(2)} &= R_\Sigma + R_1 + R_2, \\ p_1^{(2)} &= L + R_\Sigma(\tau_1 + \tau_2) + R_1\tau_2 + R_2\tau_1, \\ p_2^{(2)} &= L(\tau_1 + \tau_2) + R_\Sigma\tau_1\tau_2, \quad p_3^{(2)} = L\tau_1\tau_2. \end{aligned} \quad (41)$$

These coefficient identities hold across the positive parameter family. They are exact representations supplied by component laws; they are not recoveries from measured current alone. The resistance and time constant of each added branch are information that the second-order reference triple does not determine.

4.2 Expressing the recurrence in the coefficient family

For $N \geq 1$ and $R_\Sigma > 0$, choose $(a, b, c) = (L, R_\Sigma, 1/C_1)$ and write $\tau_0 = L/R_\Sigma$ to distinguish the reference time from branch times. Let

$$\rho = \frac{R_\Sigma^2 C_1}{L}, \quad \mu_j = \frac{R_j}{R_\Sigma}, \quad \chi_j = \frac{C_j}{C_1}, \quad \chi_1 = 1, \quad \alpha_j = \frac{\tau_j}{\tau_0} = \mu_j \chi_j \rho. \quad (42)$$

The $2N$ independent component ratios can be taken as ρ , all μ_j , and $\chi_2, \dots, \chi_N$. Together with L and τ_0 they reconstruct $R_\Sigma = L/\tau_0$, $C_1 = \rho\tau_0^2/L$, $R_j = \mu_j R_\Sigma$ and $C_j = \chi_j C_1$. This explicitly identifies the dimensional freedoms and prevents one ratio from being mistaken for a complete topology description. The current coefficient $p_k^{(N)}$ has resistance times time to power k as its unit. Its family therefore uses the shifted atoms

$$\mathcal{A}_{r,k}^{(i)} = A_{r,k+1} = R_\Sigma \tau_0^k \rho^{-r}, \quad \hat{p}_k^{(N)} = \frac{p_k^{(N)}}{R_\Sigma \tau_0^k}, \quad \hat{q}_k^{(N)} = \frac{q_k^{(N)}}{\tau_0^k}. \quad (43)$$

The shift is a derivative convention, not an extra physical state. Applying it to (39) gives

$$\begin{aligned} \hat{q}_k^{(N+1)} &= \hat{q}_k^{(N)} + \alpha_{N+1} \hat{q}_{k-1}^{(N)}, \\ \hat{p}_k^{(N+1)} &= \hat{p}_k^{(N)} + \alpha_{N+1} \hat{p}_{k-1}^{(N)} + \mu_{N+1} \hat{q}_k^{(N)}. \end{aligned} \quad (44)$$

The nonzero normalized base coefficients are $\hat{q}_0^{(0)} = \hat{p}_0^{(0)} = \hat{p}_1^{(0)} = 1$. The $N \geq 1$ chart can be held fixed while this base is used for the algebraic induction. Every resulting coefficient is a finite polynomial in ρ, μ_j, χ_j . A term ρ^ℓ multiplies $\mathcal{A}_{-\ell,k}^{(i)}$; its remaining monomial in the independent ratios is the additional

support. For example the cubic coefficient becomes

$$p_3^{(2)} = \mu_1 \mu_2 \chi_2 \mathcal{A}_{-2,3}^{(i)}, \quad p_0^{(2)} = (1 + \mu_1 + \mu_2) \mathcal{A}_{0,0}^{(i)}. \quad (45)$$

Expanding these finite polynomials specifies constant numerical weights on the full independent-coordinate domain. Holding some ratios fixed consolidates their factors into different constant weights. The physical recurrence remains valid at $R_\Sigma = 0$, but this divided reference chart does not; the original coefficients or another positive reference must then be used. Deleting a branch or imposing a singular capacitance also requires its physical state and preparation to be reconsidered.

4.3 Higher expansion coefficients of a fixed rational system

Now let $H(s) = U(s)/V(s)$ with real polynomials $U = \sum_n u_n s^n$, $V = \sum_{j=0}^m v_j s^j$ and $v_0 \neq 0$. On its disk of analyticity about zero, $H(s) = \sum_{n \geq 0} d_n s^n$. Comparing coefficients in $VH = U$ gives

$$d_n = \frac{u_n - \sum_{j=1}^{\min(m,n)} v_j d_{n-j}}{v_0}, \quad u_n = 0 \text{ above the degree of } U. \quad (46)$$

This determines successively higher expansion coefficients from one fixed rational model. For the two-state inductor of Section 8, $U = R + sL$ and $V = 1 + RC_p s + LC_p s^2$ give $d_0 = R$, $d_1 = L - R^2 C_p$ and $d_n = -RC_p d_{n-1} - LC_p d_{n-2}$ for $n \geq 2$. The full coefficients and exact finite remainder are retained in (66) and (67). No additional state is inferred from the expansion index. The nearest uncanceled pole bounds the Taylor disk; applying an infinite derivative series to a signal also requires convergence and its initialized interpretation.

Finally, $A_{r,k+1} = \tau A_{r,k}$ extends a dimensional basis element. It supplies neither the new branch data in (38) nor the rational denominator in (46). These three recurrences answer different synthesis questions. Their exact domains and state interpretations must accompany any proposed higher-order continuation. Section 11 extends the comparison to finite ladders and rational candidates for infinite-dimensional laws, retaining the discrepancy and field or history data that increasing coefficient order alone does not determine.

5 Identifying coefficients without mistaking coordinates for physics

5.1 What representation and identification establish

For fixed derivative order, normalize the coefficient as $\bar{B}_k = B_k / (S\tau^k)$. Let $\mathcal{U}$ be a declared positive parameter domain, and let a finite support $\mathcal{R}_k$ contain exponent tuples $(r, p_1, \dots, p_d)$. With

$$\phi_{r,\mathbf{p}} = \rho^{-r} \prod_{j=1}^d \eta_j^{p_j}, \quad \bar{B}_k \in \text{span}\{\phi_{r,\mathbf{p}} : (r, \mathbf{p}) \in \mathcal{R}_k\} \quad \text{on } \mathcal{U}, \quad (47)$$

the span condition is exactly the claim that constant weights reproduce that coefficient function throughout the domain. Finite pointwise agreement alone does not establish this functional identity. Allowing arbitrary parameter-dependent weights changes the claim.

Lemma 1. *Distinct finite Laurent monomials in independent positive coordinates are linearly independent on any nonempty open coordinate domain. Consequently a fixed finite Laurent dictionary has unique constant weights whenever it represents a coefficient function on that domain.*

Proof. Multiply a proposed vanishing linear combination by a monomial large enough to remove all negative exponents. The result is a polynomial that vanishes on an open set. Restricting successively to open intervals in each coordinate proves that all of its coefficients vanish. Distinct original exponent tuples give distinct resulting monomials, so every original weight is zero. The difference of two representations proves uniqueness. $\square$

Restricting the parameter domain can destroy this independence. For example, 1 and $\rho^{-1}\eta$ coincide on $\eta = \rho$. At fixed ρ , changing r merely rescales an otherwise identical observation column. A finite set of configurations must therefore establish its own evaluation rank. Even full rank may have inadequate sensitivity at the declared uncertainty, and uniqueness within one dictionary does not identify that dictionary among competing supports.

Table 9. Different conclusions supported by coefficient synthesis. A successful comparison at one level does not establish the next.

Result being claimed	Evidence required
Representation of a known operator	A coefficient identity on the stated domain, including normalization and forcing
Recovery of support, exponents or weights	An identifiable observation map with the remaining ambiguity and uncertainty stated
Prediction on independent conditions	The same fixed law and nuisance assumptions, with an exact identity or a justified error enclosure
Physical realization	Components, connections, admissible states and ports that give the asserted input–output law

The impact-driver identities prove the first conclusion on their full and restricted domains, and predict coefficients elsewhere on the same domains without adjusting their weights. The branch recurrence proves an analogous statement as its declared component list grows. Recovering either law from measured motion or current is a separate question about the observation operator, forcing and preparation. Unknown initial states and sensor parameters must enter that question rather than being fitted away after a final comparison. Appendix I collects the exact checks on these constructions and their realization conditions; those checks establish the stated model identities, while observation uncertainty requires the separate comparison below.

For the proposed measurements, let $\mathcal{G}_A, \mathcal{G}_B$ be the prediction sets of two alternatives after allowing every declared nuisance parameter and preparation. Let W be a fixed calibration-based observation scaling, and bound the remaining observation error in the corresponding norm by ϵ_A, ϵ_B . A sufficient separation condition is

$$d_W(\mathcal{G}_A, \mathcal{G}_B) = \inf_{g_A \in \mathcal{G}_A, g_B \in \mathcal{G}_B} \|W(g_A - g_B)\| > \epsilon_A + \epsilon_B. \quad (48)$$

The triangle inequality proves that the two error neighborhoods are then disjoint. An observation rejects an alternative only when its distance from that alternative's prediction set exceeds the applicable error bound. If the sets overlap, the comparison is unresolved. Covariance alone does not supply a deterministic bound; probabilistic regions require a stated distribution and confidence level. The local proposals give exact predictions to which this criterion can be applied.

For the impact driver, take g to contain the synchronized hammer angle, anvil angle and contact torque over the common calibrated band in Section 3.5. Include incoming preparation, fixture parameters and sensor response in each prediction set. Varying joint stiffness follows the seven-atom path; varying damping independently adds the nine-atom discrimination. The law and uncertainty established before comparison must then predict an independently specified joint configuration without new weights. On the cancellation surface, the exact hidden motion leaves hammer angle and contact torque identical but separates anvil angle. This is an observation choice that functional independence of coefficient atoms alone cannot supply. Circuit and RF measurements below provide complementary controlled parameter changes and phase observations.

5.2 Rank, exponent recovery and uncertainty

For $b(x) = \sum_{j=1}^m a_j x^{p_j}$ with distinct real exponents, consider exact evaluations on a geometric grid $x_\ell = x_0 q^\ell$, $q > 0$, $q \neq 1$. Then

$$b(x_\ell) = \sum_{j=1}^m \alpha_j z_j^\ell, \quad \alpha_j = a_j x_0^{p_j}, \quad z_j = q^{p_j}, \quad H_{uv} = b(x_{u+v}) = \sum_j \alpha_j z_j^u z_j^v. \quad (49)$$

A sufficiently large Hankel matrix has rank m when all $\alpha_j \neq 0$ and the z_j are distinct. Its annihilating polynomial recovers the z_j , hence $p_j = \log z_j / \log q$. This proves identifiability in exact arithmetic under those conditions, not reliable recovery from arbitrary noisy observations. Coincident exponents combine; exact signed cancellation removes an atom before the rank is counted.

The logarithmic slope $xb'(x)/b(x) = \sum_j p_j a_j x^{p_j} / \sum_j a_j x^{p_j}$ need not be an integer on a finite interval, even when every exponent is integer. Only a proved dominance limit justifies an endpoint exponent. A noninteger estimate must not be rounded into agreement. Integer, half-integer and unrestricted-real candidates provide different model classes; their comparisons must share the same reference and domain.

Holding ρ fixed makes all ρ^{-r} columns proportional. Varying η cannot recover missing ρ dependence. This exact null is distinct from poor but nonzero sensitivity. For a vector of observables $g(\vartheta)$, scaled parameters ϑ and a declared observation covariance Σ , the local information is in

$$J_w = \Sigma^{-1/2} \frac{\partial g}{\partial \vartheta}, \quad \|J_w \Delta \vartheta\| \geq \sigma_{\min}(J_w) \|\Delta \vartheta\| \quad (50)$$

on the identifiable subspace. Structural rank, a chosen practical threshold, and prediction on untouched conditions are three separate requirements. Singular values at exact rank deficiency are exactly zero; a condition number there is infinite, not a finite empirical characteristic. Values below a calculation's precision or below physical uncertainty cannot establish additional states.

The unresolved question is whether independent coordinates and realistic correlated uncertainty make the weak coefficient contributions identifiable [OP-LR31-01]. Interchangeable resistors and capacitors on an ordinary low-frequency circuit provide inexpensive independent parameter changes, while simultaneous voltage and current channels provide an empirical covariance estimate. Holding ρ fixed supplies a deliberate null control; changing it independently of a second ratio can remove that particular proportionality. The chosen support and voltage/current observation map still require a full rank and uncertainty check. The first attainable result is separation of specified competing laws on independently chosen conditions. Recovering every weak contribution requires the stronger support-wide sensitivity condition; a complete physical account additionally needs the internal states and ports. Failure despite adequate sensitivity indicates an inadequate model or an incorrect uncertainty specification. Terminal measurements alone do

not identify internal stresses.

For an explicit exponent comparison, let two normalized coefficient laws agree at ρ_0 , with common nonzero value b_0 , and differ only in exponents r, r' . At ρ_1 their predicted difference is

$$\Delta \bar{B}_k = b_0 \left[(\rho_1 / \rho_0)^{-r} - (\rho_1 / \rho_0)^{-r'} \right], \quad \bar{B}_k = \frac{B_k}{S \tau^k}. \quad (51)$$

It is zero at the deliberate null $\rho_1 = \rho_0$. At other positive ratios it is nonzero for distinct exponents, and separates the two laws only if it exceeds the sum of the coefficient-error bounds inferred from the voltage/current observation model. For mixtures, allow all candidate weights and additional coordinates to vary within their calibrated sets before applying (48); a nominal nonzero slope alone is insufficient.

5.3 Selection, metrics and derivative measurements

Use separate conditions for coefficient estimation, candidate selection and final prediction. A final condition cannot re-enter the first two steps after its discrepancy is seen. A support search may begin with one-coordinate integer atoms, then introduce an independently motivated coordinate if the first family fails. One bounded class uses $r \in \{r_k - 2, \dots, r_k + 2\}$, at most three atoms per coefficient, and extra-coordinate exponents $s \in \{0, 1, 2\}$. Its illustrative conditioning ceiling 10^4 , contribution floor 10^{-9} , eight retained coefficient candidates per order, thirty complete finalists, near-best tolerance 10^{-4} and expansion trigger $1/100$ are definitions of that comparison, not physical thresholds or evidence that it succeeds. Finite exponent windows, maximum atom counts, contribution cutoffs and condition-number limits define a bounded search. They do not prove global minimality. Results can depend on whether the criterion is coefficient error, trajectory error, instantaneous signed power error, peak error or phase error, on absolute versus relative normalization, and on the time or frequency interval. Work and energy are excluded from this choice.

Physical power may be a waveform observable only when it is formed from independently declared effort and flow. Multiplying a scalar equation by a chosen derivative does not supply that observable. The coefficient of a small physical term can be unstable under noise, partition changes or contribution thresholds even when predicted waveforms are stable. A fixed-support confidence region and a region conditional on selecting that support answer different questions. Learning curves against independent conditions, repeated calibration and full covariance are needed before interpreting an omitted contribution as a physical absence.

A continuous-frequency derivative and a discrete difference have different symbols. With normalized step $h = T_s / \tau$, dimensionless frequency Ω and $z = e^{i\Omega h}$,

$$q_c = i\Omega, \quad q_f = (z - 1)/h, \quad q_b = \frac{2z - 1}{h(z + 1)}. \quad (52)$$

These correspond to the continuous derivative, forward difference and bilinear transformation. Replacing one by another changes the design matrix; the bilinear symbol is singular at $z = -1$. An apparent change of exponents can be an observation-operator effect. First differences, local polynomial estimates and regularized polynomial derivative estimates impose different filters and initial biases. Sensor bandwidth, timing and colored noise must be propagated through the chosen estimator. Measuring hidden positions and velocities can be more informative than estimating a high derivative from one channel.

6 Initial data, effective order and physical preparation

The coefficient laws determine an operator for a declared observation. Its admissible preparations and reduced transfer order require the additional state information summarized below.

Table 10. State dimension and transfer degree for specified observations. A prescribed constant input is not itself a zero-state transfer experiment.

Declared input and output	Physical states	Reduced zero-state transfer degree	Initialization or topology qualification
Hammer torque to hammer angle	Four	Four generically	Three generically on the cancellation surface; two on its double-cancellation subset
Series voltage perturbation to coil current during battery conduction	$N + 1$	$N + 1$ for distinct active polarization times	Equal times hide difference states; diode blocking changes the equations
Terminal current to voltage in the cancelling network of Section 10	Six	Zero	An incompatible preparation gives one observable natural decay
Primary clamp voltage to primary current with a secondary capacitor	Three	Three for nonzero mutual inductance	One when uncoupled; opening leaves the secondary LC system
Input to a nontrivial exact delay or a reflecting distributed line	History or field	No unrestricted finite rational representation	Finite candidates require a domain and a history or field map

6.1 Four levels of admissibility

Let $Y > 0$ be an independently declared observation scale and use $\xi = (t - t_0)/\tau$. The normalized jets and their dimensional basis are

$$v_j = \frac{\tau^j y^{(j)}(t_0)}{Y}, \quad G_{qj} = Y \rho^{-q} \tau^{-j}, \quad y^{(j)}(t_0) = \sum_q v_{qj} G_{qj}. \quad (53)$$

A numerical list with these dimensions is merely a dimensional jet. For a regular order- n scalar ODE, the first n derivatives may be assigned, but higher derivatives must satisfy the operator recurrence. A realization-consistent jet must additionally come from a physical state. A prepared jet must come from a feasible past path, including switch states and source limits. These four levels must remain separate.

If $p_k = B_k/(S\tau^k)$ and the normalized forcing is $\varphi = f/(SY)$, operator compatibility is

$$\sum_{k=0}^n p_k v_{k+m} = \varphi^{(m)}(0), \quad m \geq 0, \quad (54)$$

where forcing derivatives are with respect to ξ . At a zero of p_n , solve the lower-order equation and its constraints; do not divide through the zero. For example $p_3 = 1 - x$ defines an order-three equation for

$x \neq 1$ and an order-two equation at $x = 1$ if the next coefficient remains nonzero. Signed atoms must be combined before this decision.

If coefficient and initial supports are finite Laurent polynomials in $x = \rho^{-1}$, the recurrence numerator is another finite Laurent polynomial. Division by a monomial preserves finite Laurent support. Division by a general polynomial need not. The units of the Laurent polynomial ring are precisely nonzero monomials, which proves generic finite closure only in the former case; exact divisibility supplies the exceptions.

Table 11. Finite-support closure and its exceptions. Cancellation of a quotient does not restore the missing highest derivative at $x = 1$.

Leading factor	Recurrence numerator	Exact quotient or remainder
$2x^2$	$6x + 4x^3$	$3x^{-1} + 2x$
$1 + x$	$1 + x^2$	$x - 1 + 2/(1 + x)$
$1 + x$	$(1 + x)(1 + 2x)$	$1 + 2x$
$(1 - x)^2$	$(1 - x)^2 h(x)$	$h(x)$ away from $x = 1$

The mirror-family dual transformation is $q' = j - q$ when Y is transported as the same observation scale: $G'_{q,j} = G_{j-q,j}$. This differs from the coefficient transformation $r' = -r - j$. A paired product has exponent $r + q$, and complete dual transport reverses that sum. Products use the Minkowski sum of supports; predicted support sites and sites removed by signed cancellation should be recorded separately. Changing units, reference parameters and Y transports the weights by the ratios in (53), rather than resetting them.

For example, $(1 + x)(x^{-1} - 1) = x^{-1} - x$: the predicted support $\{-1, 0, 1\}$ loses its middle site by exact cancellation. By contrast, $1/(1 + x)$ is not a finite Laurent polynomial, because its pole at $x = -1$ cannot occur in such a polynomial. Finite-support recovery on finitely many positive parameter values cannot remove that distinction.

An anchor change obeys $y^{(j)}(t_0 + \Delta) = \sum_{\ell \geq 0} \Delta^\ell y^{(j+\ell)}(t_0)/\ell!$ when the Taylor series converges. It is a finite identity only for a polynomial trajectory or an explicitly vanishing remainder. Thus finite support in a parameter coordinate does not imply a finite time-shift formula.

6.2 The state image

For an affine realization $\dot{x} = Fx + b$, $y = Cx$, with constant b on the interval,

$$\begin{pmatrix} y \\ \dot{y} \\ \vdots \\ y^{(m)} \end{pmatrix}_{t_0} = a + \mathcal{O}_m x_0, \quad \mathcal{O}_m = \begin{pmatrix} C \\ CF \\ \vdots \\ CF^m \end{pmatrix}, \quad a_0 = 0, \quad a_j = CF^{j-1}b \quad (j \geq 1). \quad (55)$$

The compatible physical jets form an affine image. Rank determines its dimension; the nullspace identifies hidden state freedom. A scalar annihilator containing unobservable factors admits operator-compatible jets outside this image. The constant-forcing term must be retained: an absolute jet is generally not in the homogeneous image through the origin. Adding a constant coordinate to the state turns the affine formula into a homogeneous augmented one, but that coordinate is constrained to one.

Exact rank should be checked before assessing scaled practical conditioning. The collision image is generically four-dimensional and loses rank at (22). On the double-cancellation example (25), its rank remains three while the reduced zero-state transfer degree is two. A resonator with three reactive coordinates

can be exactly observable and still have a very poorly conditioned terminal map. Equal-time-constant polarization branches can have an exactly hidden difference coordinate even though each capacitor has a physically independent initial voltage. These examples distinguish algebraic order from state availability.

Manufactured solutions separate operator error from initial-map error. A stable operator with roots $-1, -2, -3$ admits sums of those exponentials; a control with roots $-2, -1, 1/4$ contains an unstable mode. A repeated root -1 admits te^{-t} , while t^2e^{-t} requires multiplicity at least three. For $Q_\delta(s) = (s+3)[(s+1)^2 - \delta^2]$, changing the sign of δ permutes roots without changing the operator. Root labels are therefore unsuitable for defining a continuous initial map through coalescence. A forced example $y = e^{7t/10}$ has exactly $f = P(7/10)e^{7t/10}$; a discrepancy from this trajectory is then either operator, forcing or initial-map error.

6.3 Raising order and crossing singular limits

A controlled extension

$$P_\lambda(s) = Q(s)(1 + \lambda s) \quad (56)$$

adds the root $-1/\lambda$ when $\lambda \neq 0$. Every old solution of $Q(D)y = 0$ remains a solution and defines an invariant subspace with the new amplitude zero. At $\lambda = 0$ the order drops; for $\lambda < 0$ the added mode is unstable. Four comparisons are distinct: retain an old trajectory; continue a specified modal subspace; transport a common physical augmented state where one exists; or prescribe a normalized ensemble of jets. Setting the added derivative to zero is one ensemble convention, not a consequence of the old equation. A companion-state augmentation used only for comparison is mathematical until components, ports and a connection/preparation path are supplied. Most- and least-observable old directions may be defined from the observability map; energy, work, passivity and terminal error do not choose those directions. For monic degree- n Q , let two degree- $(n+1)$ solutions have the same derivatives through $n-1$ at zero and differ in $y^{(n)}(0)$ by α . Their difference has Laplace transform

$$\Delta Y(s) = \frac{\lambda \alpha}{P_\lambda(s)}. \quad (57)$$

At a simple new root its amplitude is $\alpha/Q(-1/\lambda)$. This gives an exact sensitivity statement, without assuming a small λ . If that root coincides with a root of Q , use the confluent exponential basis. A finite-window error may become unbounded on an unstable continuation even when coefficient transport is exact; a finite arithmetic residual then cannot independently establish trajectory closure.

6.4 Finite preparation and event transport

A physically prepared state requires sources and a path. For a controllable linear realization, the endpoint condition is

$$x(T) = e^{FT}x(0) + \int_0^T e^{F(T-t)}Bu(t) dt. \quad (58)$$

The controllability span determines which targets are reachable. A Gram matrix can establish reachability algebraically without using energy as an objective. Component voltage, current, force and slew limits may make an algebraically reachable state inaccessible in a specified time.

For a series RLC branch, a polynomial charge path chosen to satisfy the required endpoint charge and current gives the exact preparation voltage $v = L\ddot{q} + R\dot{q} + q/C$. Hermite interpolation supplies such a path when no incompatible limits are imposed. Independent shunt states may require an additional source or switch. For an RL branch the smooth path $i(t) = I[1 - \cos(\pi t/T)]/2$ has $u = Li + Ri$ and, by separate

integration,

$$W_u = \int_0^T ui \, dt = \frac{LI^2}{2} + \frac{3RI^2T}{8}, \quad W_R = -\frac{3RI^2T}{8}, \quad \Delta E = \frac{LI^2}{2}. \quad (59)$$

The source and resistor powers have been integrated before the independent endpoint store is compared. Imposing I without this path assigns no preparation work. Finite relaxation generally leaves a nonzero state; an exact reset requires a declared finite control or an explicitly analyzed limiting event.

At a switch, carry the physical state through the stated reset map and recompute the jets from the new vector field. The pre-event fourth derivative is not a freely transportable coordinate of a post-event second-order system. If a state is removed, specify whether it remains in another subsystem or transfers work through a port. Simultaneous events need a joint finite model or an ordering rule. The rapid two-capacitor connection in Appendix C gives an exact example in which the final state is order-independent while the individual port-work partitions depend on the edge ordering. No universal instantaneous partition follows from the endpoint alone.

7 Initialized integrals and histories

The coefficient array also includes negative derivative indices. Their use extends the synthesis to initialized integral operators; the required constants and histories are additional information, not coefficients determined by dimensions.

Negative derivative indices require a base time t_p and initialized integrator states. Set $h'_1 = y$ and $h'_j = h_{j-1}$ for $j > 1$. Repeated integration gives

$$h_m(t) = \sum_{j=0}^{m-1} h_{m-j}(t_p) \frac{(t-t_p)^j}{j!} + \int_{t_p}^t \frac{(t-v)^{m-1}}{(m-1)!} y(v) \, dv. \quad (60)$$

These m initial values are additional history coordinates. A present derivative jet does not determine them. For a constant input and zero integrator initialization, $h_m = y(t-t_p)^m/m!$; there is no finite DC phasor equal to $(i\omega)^{-m}y$ at $\omega = 0$.

Suppose an equation containing D^{-m} has residual $R(t)$. Multiplication by D^m produces $D^m R = 0$, which permits an arbitrary polynomial residual of degree at most $m-1$. Equivalence to the original initialized equation requires

$$R^{(j)}(t_p) = 0, \quad 0 \leq j < m. \quad (61)$$

Thus a derivative-cleared equation can have extra solutions even when its coefficients are exact. Anchor changes must transport the integrator states through (60). Preparation extending farther into the past can amplify compatibility sensitivity without changing present derivatives.

A finite chain is not a complete delay history. To see this exactly, add to any history a perturbation equal to 1, -4, 6, -4, 1 on the five consecutive intervals $[-1 + j/8, -1 + (j+1)/8)$, $j = 0, \dots, 4$, and zero elsewhere. Its first four moments against $(-t)^j/j!$ vanish: integration produces a fourth finite difference of a polynomial of degree at most four. It also vanishes near the present, so every present derivative agrees for smooth base histories. Nevertheless its delayed value at lag 11/16 differs by 6. A second perturbation equal to 2 on $[-2, -3/2)$ changes no history integral based at -1, but changes the delayed value at lag 7/4 by 2. Both controls work with a constant or a sinusoidal base history. Neither any finite present jet nor these four moments uniquely represents arbitrary delay history.

8 RF response and the limits of polynomial impedance

Frequency-domain observations connect synthesized derivative coefficients to measurable impedance. The charge/current convention, forcing operator and rational realization determine that connection; the same formal polynomial can have a different physical meaning under a different port assignment.

8.1 Harmonic substitution and a precise onset criterion

For an equation driven by terminal voltage with $y = q$ and $i = \dot{q}$, its zero-state impedance is $Z(s) = P(s)/s$. With $(a, b, c) = (L, R, 1/C)$ and $x = \omega\tau$,

$$\frac{Z_2(i\omega)}{R} = 1 + ix + \frac{1}{i\rho x}, \quad \frac{\Delta Z_{r,k}}{R} = w_{r,k} \rho^{-r} (ix)^{k-1}. \quad (62)$$

The derivative index shifts by one because charge is integrated current.

Theorem 1. *Let P be a real polynomial of degree $n \geq 3$ with nonzero leading coefficient. If the physical terminal law is $P(D)q = v$ and $i = \dot{q}$, the zero-state impedance $Z(s) = P(s)/s$ is not positive real on the entire right half-plane.*

Proof. Choose an arbitrary inverse-time scale $s_0 > 0$ and write $z = s/s_0$, so $Z(s_0 z) = \sum_{k=0}^n c_k z^{k-1}$, where $c_k = B_k s_0^{k-1}$ all have impedance units. Put $m = n - 1 \geq 2$ and $M = \sum_{k=0}^{n-1} |c_k|$. For $z = r e^{i\theta}$ and $r \geq 1$, the lower terms have absolute sum at most $M r^{m-1}$. If $c_n < 0$, take $\theta = 0$ and $r > M/|c_n|$. If $c_n > 0$, take $\theta = 3\pi/(4m)$ and $r > \sqrt{2}M/c_n$. In the latter case $|\theta| < \pi/2$, yet the leading real part is $-c_n r^m / \sqrt{2}$, larger in magnitude than the bound on all remaining terms. In either case $\operatorname{Re} s > 0$ and $\operatorname{Re} Z(s) < 0$, contradicting positive-realness. $\square$

This obstruction is specific to an unrestricted physical effort–flow assignment. It does not rule out a passive third- or higher-order realization with $P(D)q = N(D)v$, whose impedance is $P(s)/[sN(s)]$, or a polynomial surrogate restricted to a stated band. The physical collision numerator in (16) illustrates why the eliminated input dynamics matter. An arbitrary formal scalar observation need not be a physical port at all.

Table 12. Harmonic phase and magnitude of a charge-equation contribution.

k	0	1	2	3	4	5	6
Factor (ix) $^{k-1}$	$-i/x$	1	ix	$-x^2$	$-ix^3$	x^4	ix^5

An added atom exceeds relative magnitude ε exactly when

$$\frac{|w_{r,k}| \rho^{-r} x^{k-1}}{\sqrt{1 + [x - 1/(\rho x)]^2}} \geq \varepsilon, \quad x > 0. \quad (63)$$

This avoids replacing the denominator by a high-frequency estimate. For the illustrative values $\rho = 1$, $r = 0$, $k = 4$, $w = 1/100$, the ratio at $x = 1$ is exactly $1/100$. Near a cancellation in the baseline reactance, a small atom can be important without implying a newly resolved physical mode.

One device at fixed ρ identifies only the combined coefficient at each k . Independent devices or controlled components are required to infer exponent dependence. The practical unresolved question is whether calibrated multi-device impedance measurements support a transferable coefficient law [OP-LR12-01]. A

low-power VNA measurement of several resonators with independently varied shunt capacitance supplies an accessible complex observable and a controllable second coordinate. Fixture calibration and compensation are essential, as described by [Keysight \(n.d.\)](#). The predicted alternatives are a common rational state model versus coefficients that change with fixture or amplitude. Such a measurement does not establish nonlinear, temperature or aging behavior, and no acquisition is asserted here.

For the quartz model derived below, changing only the known shunt capacitance by ΔC_0 adds $s\Delta C_0$ to admittance. If Z_a, Z_b are the two impedances, this gives the exact finite-change identity

$$Z_b - Z_a = -s\Delta C_0 Z_a Z_b, \quad |Z_b - Z_a| = \omega |\Delta C_0| |Z_a Z_b| \quad (s = i\omega). \quad (64)$$

This is a predicted complex relation across devices, not a small-capacitance approximation. At frequencies where it is defined, compare the measured pair with this relation using joint gain, phase, fixture and capacitance-error enclosures. Distinct alternative prediction sets must meet (48); a measured discrepancy within the enclosure cannot establish changed device physics. The fixture must either be held fixed and calibrated or enter the nuisance set explicitly.

8.2 Inductor self resonance

A series R, L branch in parallel with C_p has the exact impedance

$$Z_L(s) = \frac{R + sL}{1 + sRC_p + s^2LC_p}, \quad (1 + RC_pD + LC_pD^2)v = (R + LD)i. \quad (65)$$

There are two reactive states. A derivative expansion of the terminal impedance does not create additional physical states. If $Z_L(s) = \sum_{n \geq 0} d_n s^n$ within its convergence disk, coefficient comparison gives

$$\begin{aligned} d_0 &= R, & d_1 &= L - R^2 C_p, \\ d_n &= -RC_p d_{n-1} - LC_p d_{n-2} \quad (n \geq 2), \\ d_2 &= R^3 C_p^2 - 2LRC_p, & d_3 &= -L^2 C_p + 3LR^2 C_p^2 - R^4 C_p^3. \end{aligned} \quad (66)$$

For $T_N(s) = \sum_{n=0}^N d_n s^n$, the exact remainder is

$$Z_L(s) - T_N(s) = \frac{R + sL - (1 + sRC_p + s^2LC_p)T_N(s)}{1 + sRC_p + s^2LC_p}. \quad (67)$$

No truncation is assumed exact. The Taylor disk ends at the nearest uncanceled denominator pole; improving a polynomial locally cannot remove that boundary.

An illustrative inductor has $L = 100$ nH, $R = 3/5$ Ω , $C_p = 1/4$ pF. The lossless frequency $1/(2\pi\sqrt{LC_p})$ is a reference, not automatically the lossy impedance maximum. Winding capacitance and the fixture both affect measured self resonance; [Coilcraft \(n.d.\)](#) specifically explains the fixture dependence. A real-valued empirical loss law such as $R(\omega) = R_{dc}[1 + \alpha\sqrt{\omega/\omega_0}]$ specifies neither the associated reactive dispersion nor a causal transient model. A causal complex material response and its state or memory realization remain necessary.

8.3 Quartz, branch cancellation and observation geometry

The motional branch R_m, L_m, C_m in parallel with a holder capacitance C_0 is the standard single-mode equivalent circuit described by Bible (2002). With $P_m(s) = L_m C_m s^2 + R_m C_m s + 1$,

$$Z_q(s) = \frac{P_m(s)}{s[C_0 P_m(s) + C_m]}, \quad \eta = \frac{C_0}{C_m}. \quad (68)$$

It is a third-order terminal model with three reactive coordinates before exceptional cancellations or source constraints. Illustrative values $R_m = 30 \, \Omega$, $L_m = 11/500 \, \text{H}$, $C_m = 9/500 \, \text{pF}$ and $C_0 = 9/2 \, \text{pF}$ give $\eta = 250$. They define a near-8 MHz resonator through the exact formula, not a claim of exactly 8 MHz. The lossless reference frequencies obey

$$\omega_s = \frac{1}{\sqrt{L_m C_m}}, \quad \frac{\omega_p}{\omega_s} = \sqrt{1 + \frac{1}{\eta}}. \quad (69)$$

For nonzero R_m , zero phase, peak impedance and other resonance definitions need not coincide with these references. Varying C_0/C_m and motional damping independently cannot generally be collapsed onto one ρ .

For branch-current peak phasors I_j , distinguish $A = \sum_j |I_j|$, $C = |\sum_j I_j|$, and $\chi = C/A$ when $A > 0$. Geometry gives

$$\max(0, 2 \max_j |I_j| - A) \leq C \leq A, \quad \left\langle \sum_j |i_j(t)| \right\rangle = \frac{2}{\pi} A, \quad (70)$$

for real sinusoidal currents $i_j(t) = \text{Re}(I_j e^{i\omega t})$. The instantaneous maximum is $\max_{\sigma_j = \pm 1} |\sum_j \sigma_j I_j|$, generally different from A . These relations explain large cancelling branch currents without treating a sum of magnitudes as conserved charge, energy or a universal quality factor. Topology and physical branch definitions matter. A modal-coordinate norm cannot be substituted for a component rating.

Two further geometric diagnostics also require explicit conventions. The largest pairwise phase separation is $\max_{j,k} |\text{Arg}(I_j I_k^*)|$, omitting zero phasors whose phase is undefined. If $S_0 = 0$ and $S_j = \sum_{\ell=1}^j I_\ell$, the signed area of the ordered polygon closed from S_n to the origin is $\frac{1}{2} \text{Im} \sum_{j=1}^n S_{j-1}^* S_j$. Its sign changes on reversing the ordering; unlike the resultant and polygon bounds, its value can depend on that ordering.

For Z and Y denoting series impedance and shunt admittance, respectively, the two-port blocks

$$M_Z = \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix}, \quad M_Y = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \quad M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \quad (71)$$

have unit determinant and so does their product. With output current directed toward a load Z_0 and equal reference impedances,

$$S_{21} = \frac{2}{A + B/Z_0 + CZ_0 + D}, \quad \frac{V_2}{V_1} = \frac{1}{A + B/Z_0}, \quad \frac{V_2}{I_1} = \frac{1}{C + D/Z_0}. \quad (72)$$

These are different functions. Separate unconstrained fits can violate reciprocity or passivity, and accuracy in one does not bound the other two. Joint matrix fits respecting topology and passivity, calibrated two-port observations and uncertainty across disjoint frequency intervals remain open requirements.

A series RLC path uses M_Z with $Z = R + sL + 1/(sC)$. A shunt- $C/2$ -series-RL-shunt- $C/2$ section uses the product $M_Y M_Z M_Y$ with $Y = sC/2$ and $Z = R + sL$; two identical sections use the square of

that product. Their entries may be finite polynomials or Laurent polynomials while their loaded transfer functions are rational. Thus an exact finite matrix representation and a failed polynomial driving-point representation are compatible outcomes.

8.4 A controllably hidden internal mode

Place a series RC branch behind an ideal transformer with secondary-to-primary voltage ratio $\nu > 0$, and add a terminal conductance $G_0 > 0$. Orient primary branch current into the transformer and secondary current j toward the RC load, so that $v_s = \nu v_p$ and $i_{p,\text{branch}} = \nu j$. For capacitor voltage x , the initialized realization is

$$j = \frac{\nu v_p - x}{R}, \quad C\dot{x} = j, \quad i_p = G_0 v_p + \nu j, \quad R, C > 0. \quad (73)$$

Eliminating x with zero initial state gives the admittance

$$Y(s) = G_0 + \frac{\nu^2 s C}{1 + s R C}, \quad h = \frac{\nu^2}{R}, \quad \delta_h = \frac{h}{G_0 + h}. \quad (74)$$

For $x(0) = V_0$, the terminal current additionally contains $-\nu V_0 e^{-t/(RC)}/R$. The poles and initialized response therefore follow from the same orientation. The primary-to-secondary ratio used in Section 10 is $n = 1/\nu$; reflecting a fixed secondary admittance with that convention divides it by n^2 .

As $\nu \rightarrow 0$, both the driven branch contribution and its terminal initial-state signal vanish. The endpoint of (73) is a decoupled terminal with a closed internal RC discharge loop, not an ordinary finite-turns transformer. An open series RC branch would not have the same discharge. Enclose the capacitor and exclude the two heat reservoirs: the inward powers are $v_p i_p$, $-G_0 v_p^2$, and $-R j^2$. Integrating them separately gives their sum $\int_0^T x j dt = C[x(T)^2 - x(0)^2]/2$, agreeing with the independently evaluated store $C x^2/2$. In the decoupled endpoint with $v_p = 0$, the sole nonzero transfer and endpoint store are

$$W_R(0, T) = - \int_0^T \frac{V_0^2 e^{-2t/(RC)}}{R} dt = - \frac{C V_0^2}{2} (1 - e^{-2T/(RC)}), \quad E(T) = \frac{C V_0^2}{2} e^{-2T/(RC)}. \quad (75)$$

Exact terminal cancellation therefore does not imply absence of an internal prepared mode. The changing store in (75) is accounted for by the independently integrated resistor work, even though the terminal work is zero. Its relevance to a higher-order realization is that eliminated internal coordinates can carry the same hidden change.

8.4.1 Measuring a changing store behind a quiet terminal

How tightly can terminal uncertainty bound hidden prepared energy when coupling and preparation are only partly known [OP-LR23-01]? With $v_p = 0$, known finite $\nu > 0$ and the state law (73),

$$|i_p(0)| = \frac{\nu |V_0|}{R}, \quad E(0) = \frac{C V_0^2}{2}, \quad |i_p(0)| \leq \epsilon_i \implies E(0) \leq \frac{C R^2 \epsilon_i^2}{2 \nu^2}. \quad (76)$$

Here ϵ_i bounds the true initial current after calibration and timing uncertainty, not merely the displayed reading. The bound follows by eliminating $|V_0|$. If $C \leq C_{\max}$, $R \leq R_{\max}$ and $\nu \geq \nu_{\min} > 0$, the corresponding bound is $C_{\max} R_{\max}^2 \epsilon_i^2 / (2 \nu_{\min}^2)$. Without a positive coupling lower bound or an independent preparation bound, there is no uniform finite ceiling: choosing $|V_0| = R \epsilon_i / \nu$ preserves the terminal limit while the store grows as ν^{-2} . Those arbitrarily large states belong to the ideal parameter family; finite

voltage and preparation limits must be supplied for an actual device.

A low-voltage RC branch with an independently calibrated finite coupling permits ordinary capacitor-voltage, resistor-voltage/current and terminal-current measurements. Prepare a specified V_0 , establish the operation-side state after switching, and observe a finite interval $[0, T]$ with $v_p = 0$. For the boundary enclosing the capacitor and ideal coupling, source work is zero and the resistor power is $-v_R j = -Rj^2$. Integrating it gives (75); evaluate $CV_0^2/2$ and $Cx(T)^2/2$ independently from the endpoint voltages. The exact residual is zero for this declared discharge. A measured residual exceeding its independently bounded uncertainty remains open, while a resolved capacitor change with an unresolved terminal signal establishes the narrower hidden-state result. Probe loading supplies an additional port if it is appreciable in the declared model.

The equal-and-opposite polarization preparation in Section 9 offers a low-frequency emulator of terminal cancellation with separately accessible stores. Multiple nearly cancelling modes, nonideal transformers, preparation limits, sensor loading and finite observation bandwidth still need bounds. A resolved single-branch account does not close those broader questions.

8.4.2 A three-state loop with a directly measurable hidden store

A closed loop containing an inductor, a series resistance $R_\Sigma \geq 0$ and two series-connected parallel RC sections gives a direct control for that observation question. With identical $R, C > 0$, no applied voltage and consistently oriented section voltages,

$$Li = -R_\Sigma i - v_1 - v_2, \quad C\dot{v}_j = i - v_j/R, \quad j = 1, 2. \quad (77)$$

The compatible preparation and exact response are

$$(i(0), v_1(0), v_2(0)) = (0, V_0, -V_0), \quad i = 0, \quad v_1 = V_0 e^{-t/(RC)}, \quad v_2 = -V_0 e^{-t/(RC)}. \quad (78)$$

There are three independent physical states. The driven pair $(i, v_1 + v_2)$ has two states; $v_1 - v_2$ relaxes independently and is absent from the loop-current transfer. A branch-voltage observation generally retains all three initialized modes. Unequal time constants couple the difference back to the terminal observation. Thus a quiet terminal does not remove the third state.

Enclose the inductor and both capacitors, excluding heat reservoirs, on the smooth operating interval $[0, T]$. The separate inward resistor powers are $-R_\Sigma i^2$ and $-v_j j_{R,j}$, where $j_{R,j} = v_j/R$. Direct integration and independent endpoint evaluation give

$$\begin{aligned} W_\Sigma &= 0, & W_{R,j} &= -\int_0^T \frac{v_j^2}{R} dt = -\frac{CV_0^2}{2}(1 - e^{-2T/(RC)}), \\ E_0 &= CV_0^2, & E_1 &= CV_0^2 e^{-2T/(RC)}, \\ r_E &= E_1 - E_0 - W_\Sigma - W_{R,1} - W_{R,2} = 0. \end{aligned} \quad (79)$$

The two nonzero works and decreasing stores are invisible to a terminal-only account. This exact model resolves their destination; it does not report an unexplained gain. Preparation is a separate interval. Closing the loop at zero inductor current with continuous capacitor voltages has no impulse under these laws; actual switch and probe transfers need their own account.

For the illustrative component choices $R = 1 \text{ k}\Omega$, $C = 1 \text{ mF}$, $V_0 = 1 \text{ V}$, and $T = 1 \text{ s}$, the initial store is $1/1000 \text{ J}$ and each resistor work is $-(1 - e^{-2})/2000 \text{ J}$. Ordinary voltage channels and shunt-current

measurements resolve the internal volt and milliampere scales over seconds. The open experimental alternatives are a resolved internal change with unresolved terminal signal, an independently closed resistor account, or a signed residual beyond the declared resolution. Only the last leaves the energy account unresolved.

An assumed calibration budget makes that distinction quantitative. Take capacitance estimates $\hat{C} = 1/1000$ F with bounds $\epsilon_C = 1/100000$ F, voltage error $\epsilon_v = 1/1000$ V, resistor-current error $\epsilon_j = 1/10^6$ A, and observed bounds $|\hat{v}_j| \leq 11/10$ V, $|\hat{j}_{R,j}| \leq 11/10000$ A throughout the one-second interval. The channel and endpoint formulas in Section 16.2 give

$$\epsilon_{\text{cap}+R} = \frac{1652401}{50000000000} \text{ J} < \frac{1}{25000} \text{ J}, \quad \epsilon_r \leq \epsilon_{\text{cap}+R} + \epsilon_{\text{rest}}. \quad (80)$$

This is the sum of four capacitor endpoint bounds and two resistor-work bounds, not an achieved apparatus specification. The independently determined ϵ_{rest} must cover inductor endpoints, series-resistance work, timing, acquisition integration, probe loading and any event account. Terminal quietness does not set it to zero. Compare the predicted capacitor decrease $(1 - e^{-2})/1000$ J with its endpoint error, and compare the measured full residual with its separate total error. Deliberately unequal relaxation times provide the contrasting observation derived in (89).

8.4.3 Preparing and comparing RF transients

Steady phasor agreement cannot determine how a polynomial surrogate should be initialized. For an inductor with shunt capacitance, use physical states (i_L, v) ; for quartz use (i_m, v_{C_m}, v) . Charged shunt capacitor, energized inductor and combined preparations are distinct initial states, even at a common chosen scale. Placing either device behind a finite source resistance changes the released homogeneous operator. The ideal source clamp that imposes an endpoint does not specify its own preparation work or impulse.

Three explicit initial maps illustrate the ambiguity. The physical derivative map uses $y^{(j)}(0) = C(F/\omega_{\text{ref}})^j x_0$ in normalized time. A convenience map retains $y(0), y'(0)$ and sets all higher derivatives to zero. A third retains those two values and chooses a minimum Euclidean-norm higher-derivative vector to cancel only the anchor equation. The last condition is one linear constraint; it does not impose the differentiated hierarchy (54), uniform parameter compatibility, or physical preparation. The minimum depends on coordinate scaling.

Let y_{phys} be the physical response, y_{der} the surrogate response from the physical derivative map, and y_{map} its response from another map. Then

$$y_{\text{map}} - y_{\text{phys}} = (y_{\text{der}} - y_{\text{phys}}) + (y_{\text{map}} - y_{\text{der}}). \quad (81)$$

The first difference isolates the surrogate operator effect under that reference map; the second isolates the initial-map effect. Vector errors add exactly, but their norms need not add or combine as independent variances. Retain the norm triangle gap and both absolute and reference-normalized errors on an explicit finite prefix; a vanishing reference norm makes the relative error undefined. Alternative initialization can reduce total error by cancellation without repairing an unstable operator.

An unstable polynomial impedance fit may match a finite frequency interval while producing an unstable free transient. Its stability and active frequency domain must be established separately; it is unsuitable for an unrestricted physical transient. Conversely, a time-domain collision fit need not transfer to low, modal

and high frequency intervals that played no part in its selection. A physical state map must be checked through (55), and a preparation path through (58), before a transient has a physical interpretation.

Finite RF preparation can prescribe paths to each of the charged-capacitor, energized-inductor and combined states, including a nonzero prehistory and a finite clamp interval. The boundary encloses every named inductor and capacitor; external powers comprise the preparation source, gate/switch channel, passive losses, load clamp, clamp losses and any environmental-reference channel. Each is integrated separately before comparison with the independently evaluated operation-side stores and jets. A continuous finite clamp has no ideal impulse by construction. It neither assigns the impulse of a different zero-time clamp nor turns zero higher derivatives into the prepared physical map. Arbitrary source limits, switching nonlinearity, stochastic prehistories, temperature, aging and realization of the polynomial surrogate remain unresolved.

On a truly periodic interval of length $T = 2\pi/\omega$, a port with peak phasors V, I has

$$W = \int_0^T v i \, dt = \frac{\pi}{\omega} \operatorname{Re}(VI^*). \quad (82)$$

Each source and resistor contribution is integrated separately. Component states independently return to their initial values; this follows from the periodic solution, not from assuming constant instantaneous store. A finite settling interval generally has nonzero endpoint change and must retain its homogeneous transient. No component energy is assigned to bare polynomial terms.

8.5 A passive reference for finite representations

A useful exact normalized reference is

$$Z_*(q) = \frac{9}{50} + \sum_{j=1}^3 \frac{a_j q}{q^2 + 2\zeta_j \omega_j q + \omega_j^2},$$

$$(\omega_j) = \left(\frac{7}{10}, 2, 6\right), \quad (\zeta_j) = \left(\frac{2}{25}, \frac{3}{20}, \frac{3}{10}\right), \quad (a_j) = \left(\frac{9}{10}, \frac{3}{5}, \frac{7}{20}\right). \quad (83)$$

Each summand is a parallel RLC impedance with $C_j = 1/a_j$, $L_j = a_j/\omega_j^2$ and $R_j = a_j/(2\zeta_j\omega_j)$. The series connection is therefore passive. Its nearest pole has modulus 7/10, fixing the Taylor convergence radius at the origin.

Different finite representations answer different mathematical questions. Taylor coefficients match derivatives at an anchor. A real-coefficient polynomial fit minimizes a chosen finite-band norm. A discrete min-max construction certifies only its declared evaluation set unless an interval bound is supplied. A Padé rational function matches moments but has no automatic passive component interpretation. Fixed-pole residue adjustment differs from moving-pole rational identification. One-sided Krylov projection matches specified moments; stable balanced truncation does not in general preserve positive-realness. A positive Foster construction enforces a realizable cone but may exclude the best unconstrained fit. Increasing order need not reduce an unrelated error criterion monotonically. For every method, stability, positive-realness, fit domain, residual bound and initial-state map are separate obligations. The monotonic reactance property of Foster (1924) applies to lossless reactance functions, not arbitrary lossy impedances.

9 Eliminated polarization states in a switched battery circuit

The coefficient construction in Section 4 supplies the conducting-state equations (35), the complete forced operator (37) and the coefficient recurrence (39). This application determines what those equations retain

under preparation, coincident relaxation times and diode switching. The positive current convention and the conducting-interval restrictions stated there remain in force.

For coincident $\tau_1 = \tau_2 = \tau$, the sum $v_1 + v_2$ obeys $\tau(v_1 + v_2)' = (R_1 + R_2)i - (v_1 + v_2)$. The difference state associated with $(i, v_1, v_2) = (0, 1, -1)$ is terminally hidden. Its internal resistor losses and capacitor stores need not vanish. Nearly coincident branches are exactly distinct but can be poorly identifiable. A relaxed vanishing-residue limit differs from a fixed-state limit: taking $R_j \rightarrow 0$ at fixed τ_j sends $C_j = \tau_j/R_j$ to infinity. If $v_j(0)$ is fixed and nonzero, $v_j(0)e^{-t/\tau_j}$ remains in the terminal voltage and its initial store diverges. Dropping that branch is justified only with an appropriate state and preparation limit.

For $N = 0$, $I_0 > 0$, $V_{oc} > 0$ and $R_\Sigma = R > 0$, the solution and cutoff are exact:

$$i(t) = \left(I_0 + \frac{V_{oc}}{R}\right) e^{-Rt/L} - \frac{V_{oc}}{R}, \quad t_c = \frac{L}{R} \log \left(1 + \frac{RI_0}{V_{oc}}\right). \quad (84)$$

Integrating the OCV power first yields

$$W_{oc} = \int_0^{t_c} V_{oc} i dt = V_{oc} \left(\frac{LI_0}{R} - \frac{V_{oc} t_c}{R}\right). \quad (85)$$

At $R = 0$ with $V_{oc} > 0$, the separately derived solution is $i = I_0 - V_{oc}t/L$, $t_c = LI_0/V_{oc}$, and the same integral is $LI_0^2/2$. The illustrative $L = 1$ mH, $I_0 = 100$ A, $V_{oc} = 12$ V gives $t_c = 1/120$ s and $W_{oc} = 5$ J in the ideal model. No claim about chemical efficiency follows from this ideal voltage-source work.

The zero-OCV boundary is separate. For $N = 0$, $V_{oc} = 0$, $R > 0$ and $I_0 > 0$, $i(t) = I_0 e^{-Rt/L}$ never reaches zero at finite time. Its OCV work is zero and its signed resistance work on $[0, T]$ is $-LI_0^2(1 - e^{-2RT/L})/2$, directly from $\int_0^T -Ri^2 dt$; the independent coil endpoint store is $LI_0^2 e^{-2RT/L}/2$. If also $R = 0$, current is constant, every external power is zero and the two endpoint stores agree. These cases have no duration selected by a finite first-zero condition. At $I_0 = 0$ the unpolarized circuit can remain blocked from the outset, which is not a downward crossing.

For the multi-branch realization, write $z = (i, v_1, \dots, v_N)^T$, so $z(t) = e^{Ft}z(0) + \int_0^t e^{F(t-u)}b du$ exactly. The first-zero event is an implicit root of its current component. This closed form defines all interval integrals without reporting a numerical trajectory. An illustrative three-branch family uses $(R_j, \tau_j) = (3/200, 1/5000), (3/100, 1/1000), (3/50, 1/200)$ in ohms and seconds, coil resistance $3/100 \Omega$ and battery ohmic resistance $1/50 \Omega$. Changing the number, spacing or preparation of branches changes the cutoff and internal partition; there is no universal fourth-order battery law.

The coil-only boundary has inward powers $-v_b i$ and $-R_{coil} i^2$, with $v_b = V_{oc} + R_{b0} i + \sum v_j$. For the combined coil and polarization-store boundary, use instead

$$P_{oc} = -V_{oc} i, \quad P_{coil} = -R_{coil} i^2, \quad P_{b0} = -R_{b0} i^2, \quad P_j = -v_j^2/R_j, \quad E = Li^2/2 + \sum_j C_j v_j^2/2. \quad (86)$$

Integrate every channel on a finite conducting interval $[0, t_c]$, then compare the independent endpoint stores. Including both the battery terminal work and its internal decomposition in this combined sum would count the same transfer twice. A permanently opened connection after $i(t_c) = 0$ leaves each polarization branch to relax independently, with continuous capacitor voltages and no ideal opening impulse.

If the connection instead retains only the diode, the candidate blocked solution must satisfy

$$i = 0, \quad v_j(t) = v_j(t_c)e^{-(t-t_c)/\tau_j}, \quad V_b^{\text{off}}(t) = V_{\text{oc}} + \sum_j v_j(t_c)e^{-(t-t_c)/\tau_j} \geq 0. \quad (87)$$

Nonnegative branch voltages at cutoff are sufficient for continued blocking. Arbitrarily prepared signed voltages need not satisfy the guard forever. At a crossing of V_b^{off} into negative values, the conducting equations restart with $i = 0$, since $Li = -V_b^{\text{off}} > 0$. A tangency that leaves the guard nonnegative does not trigger that transition.

For an exact counterexample, let $V_{\text{oc}} = V_b > 0$, $(v_1(t_c), v_2(t_c)) = (8V_b, -8V_b)$ and $(\tau_1, \tau_2) = (T_b, 2T_b)$, with $T_b > 0$. Writing $\vartheta = (t - t_c)/T_b$, the proposed off-state voltage is $V_b(1 + 8e^{-\vartheta} - 8e^{-\vartheta/2})$. It starts positive, first vanishes at

$$t_r - t_c = -2T_b \log\left(\frac{2 + \sqrt{2}}{4}\right), \quad (88)$$

and its inadmissible continuation would equal $-V_b$ at $\vartheta = 2 \log 2$. The cutoff state is locally attainable from positive current: the conducting derivative at the event is $-V_b/L < 0$. Thus the first downward current zero does not prove permanent isolated relaxation.

On an opened interval, or on a blocked interval ending before the guard fails, each resistor integral is exactly $-C_j v_j(t_c)^2 [1 - e^{-2(T-t_c)/\tau_j}]/2$. The independently evaluated remaining store is $C_j v_j(t_c)^2 e^{-2(T-t_c)/\tau_j}/2$. It is not set to zero at finite time. A later conducting interval requires its own source, coil and polarization ledger; the off-state formula cannot be extended through it.

A pulse selected to have duration T in the simple model with $V_{\text{oc}} > 0$ needs $I_0 = (V_{\text{oc}}/R)(e^{RT/L} - 1)$, or $I_0 = V_{\text{oc}}T/L$ when $R = 0$. This uses the current-zero condition, followed by current, voltage and instantaneous-power limits. It does not use energy to choose the pulse. In a polarization model duration selection remains a root condition on the exact state solution. Geometry-derived inductance, coil resistance and temperature cannot be varied independently unless the construction permits it.

A scheduled interruption before the current zero leaves $Li^2/2$ in the coil. A finite clamp capacitor, dissipative clamp or recovery converter must receive the corresponding signed work. Finite preparation sources, branch charging resistors, switch losses and reset conductances similarly require their own powers. Ideal imposed initial states establish an observation-interval model only. Reverse recovery, temperature, aging, continuous relaxation spectra and material nonlinearity are additional mechanisms, not evidence for a universal order inferred from a finite branch family.

Recovering order from noisy data with arbitrary branch preparations remains a practical open question [OP-LR01-04]. A low-voltage battery emulator made from a DC source and independently accessible RC branches allows ordinary shunt-current and capacitor-voltage measurements without uncertain electrochemistry. Prepare equal and opposite voltages on equal-time-constant branches: the terminal transient is unchanged while branch probes reveal relaxation. Split their time constants controllably and compare the uncertainty-scaled sensitivity with (50). This distinguishes exact hidden freedom from inadequate resolution. It does not validate an actual battery's temperature- and age-dependent spectrum.

For a quantitative version, disconnect the external current path during the observation, so that the diode guard is irrelevant. Prepare $(v_1(0), v_2(0)) = (V_0, -V_0)$ with $V_0 \neq 0$. The terminal departure from the known OCV and its largest absolute value occur according to

$$\Delta v(t) = V_0(e^{-t/\tau_1} - e^{-t/\tau_2}), \quad t_* = \frac{\tau_1 \tau_2}{\tau_2 - \tau_1} \log\left(\frac{\tau_2}{\tau_1}\right), \quad 0 < \tau_1 < \tau_2. \quad (89)$$

Differentiating the two exponentials gives the unique positive stationary time; their difference is zero at both interval endpoints 0 and infinity, so it is the maximum in magnitude. Equal time constants instead give $\Delta v = 0$ throughout. If each alternative has a total voltage error bound ϵ_v , the predictions separate when $|\Delta v(t_*)| > 2\epsilon_v$. The bound must include initial-voltage mismatch, OCV drift, timing, component tolerances and probe loading, with their allowed correlations. Otherwise the outcome remains unresolved under (48). Internal branch probes establish the prepared hidden relaxation even when the terminal comparison cannot resolve it.

10 Passive realizations and internal nonuniqueness

A rational terminal coefficient law can admit several physical constructions. This section gives explicit realizations and exact passivity conditions, then identifies the internal properties that terminal coefficients leave undetermined. These are conditions on a declared physical interpretation of a synthesized law.

10.1 A rational family with an exact certificate

Consider

$$Z_N(s) = R_\infty + \sum_{j=1}^N \frac{R_j}{1 + s\tau_j}, \quad \tau_j > 0, \quad (90)$$

with distinct active poles. The real part on $s = i\sqrt{x}$ has positive denominator and numerator

$$q_N(x) = R_\infty \prod_j (1 + x\tau_j^2) + \sum_j R_j \prod_{\ell \neq j} (1 + x\tau_\ell^2), \quad x \geq 0. \quad (91)$$

Since the poles lie strictly in the left half-plane, nonnegative q_N on this half-line, together with its behavior at infinity, is equivalent to positive-realness of this family. A polynomial certificate checks $x = 0$, every nonnegative stationary point and the leading behavior; isolated frequency checks cannot replace it.

Positive residues give the constructive cone $R_\infty \geq 0, R_j > 0$. They are sufficient, not necessary. For example $Z = 1 - \alpha/(1 + s)$ is positive real exactly for $\alpha \leq 1$ with $\alpha \geq 0$; for $0 < \alpha \leq 1$, $Z = (1 - \alpha) + \alpha s/(1 + s)$ explicitly realizes a resistor in series with a parallel RL section. A negative residue is therefore not itself an active component claim.

A more detailed signed family takes $R_\infty = 7/20 \, \Omega$ and

$$\begin{aligned} (\tau_j) &= (2/25, 9/50, 21/50, 19/20, 21/10, 24/5) \, \text{s}, \\ (R_j) &= (7/10, 11/20, 43/100, 17/50, 27/100, 11/50) \, \Omega, \end{aligned} \quad (92)$$

using the first N terms for $2 \leq N \leq 6$. Replace the first residue by $-\alpha R_1$. Its exact boundary is

$$\alpha_*^{(N)} = \inf_{x \geq 0} \frac{1 + x\tau_1^2}{R_1} \left[R_\infty + \sum_{j=2}^N \frac{R_j}{1 + x\tau_j^2} \right]. \quad (93)$$

The infimum is characterized by the endpoint and stationary polynomial equations obtained by differentiating the rational function. Values $(99/100)\alpha_*^{(N)}$, $\alpha_*^{(N)}$ and $(101/100)\alpha_*^{(N)}$ lie inside, on and outside this particular positive-real boundary, respectively. This exact characterization replaces unsupported decimal boundary claims. It does not map the full correlated signed-residue region. General passive synthesis of those ports and the minimum number and arrangement of physical stores remain separate constructive questions; absence of a supplied construction is not a refutation of a valid port certificate.

10.2 Three constructions and what each preserves

For positive residues, connect R_∞ in series with parallel R_j , C_j sections, $C_j = \tau_j/R_j$. The state equations and store are

$$\dot{v}_j = i/C_j - v_j/\tau_j, \quad v = R_\infty i + \sum_j v_j, \quad E_F = \frac{1}{2} \sum_j C_j v_j^2. \quad (94)$$

This is a Foster RC realization. Euclidean extraction instead alternates a high-frequency series resistance and a shunt capacitance to form a Cauer ladder when the extracted elements are positive. For example the exact identity

$$1 + \frac{1}{1+s} + \frac{1}{1+2s} = 1 + \frac{1}{\frac{2}{3}s + \frac{1}{\frac{9}{5}s + \frac{1}{\frac{25}{3}s + 5}}} \quad (95)$$

exhibits two different passive two-capacitor circuits with identical terminal impedance. The terminating conductance is 5, hence its resistance is $1/5$, in the normalized units.

A third construction places each Foster section behind an ideal transformer:

$$v_p = n_j v_s, \quad i_s = n_j i_p, \quad R_{s,j} = R_j/n_j^2, \quad C_{s,j} = n_j^2 C_j. \quad (96)$$

The reflected impedance and instantaneous power are unchanged, and $C_{s,j} v_s^2/2 = C_j v_p^2/2$. But secondary capacitor charge and current scale by n_j , and voltage by $1/n_j$. Thus an ideal ratio tending to infinity gives unbounded internal charge or current at a fixed terminal response; a ratio tending to zero gives unbounded secondary voltage. A finite value such as $n_j = 8$ illustrates the exact factor without a numerical transient. Winding resistance, bandwidth, saturation, insulation and tolerance constraints are needed for finite engineering bounds.

The terminal rational function fixes its zero-state voltage for a declared current and therefore its signed terminal work on each fixed interval. It fixes the McMillan degree N when all poles are distinct and residues nonzero. The explicit positive-residue construction attains N independent capacitor states, so this cone's minimal state count is exactly N . It does not fix arbitrary internal magnitude sums, resistor-loss partitions or nonminimal hidden stores. Independently prepared decoupled stores can be arbitrarily large and are excluded from zero-state comparisons.

For all three passive constructions, enclose the reactive stores, exclude the source and heat reservoirs, and use source power vi and each resistor power $-v_R i_R$. Ideal transformer powers cancel internally. On preparation, operation, relaxation and reset intervals, integrate these products separately and compare $\sum C v^2/2 + \sum L i^2/2$ at both ends. A smooth current such as $\sin^2(\pi t/4)$ on $0 \leq t \leq 4$ and zero outside supplies a compact exact input; since it vanishes at enable and release, finite states are continuous and there is no imposed impulse. Finite relaxation still leaves an endpoint store. Equal terminal work does not by itself force equal internal energy at intermediate times across every passive realization.

Universal storage bounds require more than comparisons between a few constructions. For a realization $\dot{x} = Fx + Bu$, $y = Cx + D_0 u$, a quadratic candidate $E = x^T H x/2$ is dissipative if

$$\begin{pmatrix} -F^T H - HF & C^T - HB \\ C - B^T H & D_0 + D_0^T \end{pmatrix} \geq 0, \quad H \geq 0. \quad (97)$$

Indeed the associated quadratic form is twice $u^T y - \dot{E}$. The relation between quadratic supply, storage inequalities and available or required storage is established in [Willems \(1972\)](#); extremal interpretations

require the corresponding reachability assumptions. Determining their bounds over all minimal physical syntheses, and bounding individual resistor shares, remains unresolved here. The terminal function alone supplies no unique physical H .

10.3 A finite network grammar and exact cancellation

Take one-store atoms $Z_C = R/(1 + sRC)$ and $Z_L = sLR/(R + sL)$, and compose them recursively in series and parallel. Quotient associative and commutative composition, keeping the two atom types distinct. If $U(z)$ counts series-rooted expressions, symmetry gives the same count for parallel roots. With $B(z) = 2z + U(z) = \sum b_n z^n$,

$$U(z) = \prod_{n \geq 1} (1 - z^n)^{-b_n} - 1 - B(z), \quad T(z) = 2z + 2U(z). \quad (98)$$

The product counts unordered multisets of eligible children; subtraction removes zero and one child. Recursive coefficient comparison gives the finite counts below exactly.

Table 13. Exact counts for this restricted grammar, totaling 2016. They are not a count of all passive circuits.

Number of atoms	2	3	4	5	6
Canonical expressions	6	20	80	340	1570

Each rational impedance must be reduced by polynomial common factors before quoting zero-state transfer degree. In normalized units with unit component values, $Z_C = 1/(1 + s)$ and $Z_L = s/(1 + s)$, so $Z_C + Z_L = 1$. A series chain containing two of each has impedance 2; in parallel with a chain containing one of each it has impedance $2/3$. Six reactive component coordinates thus coexist with a degree-zero zero-state transfer. Equal-value cancellation is not a generic unequal-component theorem.

The initialized terminal law needs a separate derivation. An RC atom driven by current j has $\dot{v}_C = j - v_C$; an RL atom has $\dot{i}_L = j - i_L$ and voltage $v_L = j - i_L$. A series pair therefore has $v = j + w$, with $w = v_C - i_L$ and $\dot{w} = -w$. Let a_0 be the sum of initial capacitor voltages minus the sum of initial inductor currents in the four-atom chain, and b_0 the corresponding difference in the two-atom chain. These sums are in the declared normalized units. With branch currents j_A, j_B ,

$$v = 2j_A + a_0 e^{-t}, \quad v = j_B + b_0 e^{-t}, \quad i = j_A + j_B, \quad 3v = 2i + (a_0 + 2b_0) e^{-t}. \quad (99)$$

Consequently $(D + 1)(3v - 2i) = 0$ describes the initialized terminal behavior, whereas $3v = 2i$ holds precisely when $a_0 + 2b_0 = 0$. Zero initial state is sufficient but not necessary. The illustrative preparation $a_0 = 1, b_0 = 0$ produces $v = e^{-t}/3$ at zero terminal current. This observable natural decay is uncontrollable from the terminal current, while further internal directions can be hidden. A purely resistive zero-state fit therefore establishes neither zero free-response order nor absent physical stores.

For arbitrary parameters, a modified nodal model is

$$C_n \dot{e} + Ge + B_L i_L = bu, \quad L_L \dot{i}_L = B_L^T e. \quad (100)$$

If U_0 spans the nullspace of C_n , first impose $U_0^T (Ge + B_L i_L - bu) = 0$. Only after solving the algebraic constraints should one form an ordinary state equation and examine controllability, observability and terminal minimality. Unit inductances can suppress L_L in notation, but not in dimensions.

There are $2N$ component parameters in this grammar, dimensional rank two and therefore $2N - 2$ independent component ratios; adding a load adds one. One ρ leaves $2N - 3$ component coordinates unrepresented. Graph cycle rank, bridges, spanning trees and automorphisms describe topology, but do not alone determine terminal derivative support. Port placement must be quotiented by the actual graph automorphisms, including drive, load, observation and polarity jointly. A zero transfer at one frequency or at one placement is not absence of a state. Withholding entire series-rooted or parallel-rooted classes asks a stronger transfer question than changing values within one graph. Extension to all placements, unequal components, grounded common-mode ports, bridges, multigraphs, transformers, gyrators and larger graphs remains open.

Singular component values require separate topologies. For the normalized two-atom series circuit, $Z(1) = R_C/(1 + R_C C) + LR_L/(R_L + L)$. With $R_C = R_L = L = 1$, $C \rightarrow 0$ gives $3/2$, whereas $C = \varepsilon$, $R_C = \varepsilon$ gives $1/2$ in the limit. With the capacitor atom fixed at $1/2$, $L \rightarrow \infty$ at $R_L = 1$ gives $3/2$, while $L = R_L = 1/\varepsilon$ diverges as $1/(2\varepsilon) + 1/2$. These exact paths prove that a multivariate boundary needs a declared coscaling. Incompatible initial capacitor voltages or inductor currents require a finite parasitic path before any impulse work can be assigned.

11 Distributed, fractional and delayed limits

For field, fractional and delayed dynamics, a finite coefficient construction is a representation on a declared domain. The finite-state and rational-series constructions of Section 4 motivate two distinct questions here: which finite realization supplies the coefficients, and how its law differs from the infinite-dimensional target. The following exact laws and discrepancies delimit that domain and state the additional field or history preparation that a finite ODE cannot infer.

11.1 Transmission lines and finite passive ladders

For a uniform line of length ℓ , per-length constants $L', C' > 0$, $R', G' \geq 0$ and current directed toward the load, the telegrapher equations are

$$\partial_x v = -(R' + sL')i, \quad \partial_x i = -(G' + sC')v, \quad \gamma = \sqrt{(R' + sL')(G' + sC')}, \quad Z_c = \sqrt{\frac{R' + sL'}{G' + sC'}}. \quad (101)$$

The load-to-input matrix is $\begin{pmatrix} \cosh \gamma \ell & Z_c \sinh \gamma \ell \\ Z_c^{-1} \sinh \gamma \ell & \cosh \gamma \ell \end{pmatrix}$, with branches chosen by physical continuation from $\text{Re } s > 0$. Equivalently it is the exponential of $\ell \begin{pmatrix} 0 & R' + sL' \\ G' + sC' & 0 \end{pmatrix}$. The load reflection coefficient is $\Gamma_L = (Z_L - Z_c)/(Z_L + Z_c)$. Forward and reflected waves define local reflection magnitude and phase. For a lossless line the standing-wave ratio is $(1 + |\Gamma|)/(1 - |\Gamma|)$; in a lossy line a local ratio does not generally equal the global envelope ratio along an arbitrary length. Group delay is $-\text{d arg } H(i\omega)/\text{d}\omega$ where the phase derivative exists; transfer zeros require separate treatment.

An ideal shorted lossless line has

$$Z_{\text{short}}(s) = Z_0 \tanh(st_d), \quad \frac{Z_{\text{short}}(s)}{Z_0} = st_d - \frac{(st_d)^3}{3} + \frac{2(st_d)^5}{15} - \frac{17(st_d)^7}{315} + \dots, \quad (102)$$

where the infinite series converges for $|st_d| < \pi/2$. Its coefficients follow exactly by dividing the entire-function series for $\sinh$ and $\cosh$; no finite displayed polynomial is asserted equal to the line. For illustrative $Z_0 = 50 \, \Omega$, $t_d = 5 \, \text{ns}$, the first imaginary-axis pole has frequency $1/(4t_d) = 50 \, \text{MHz}$. References $L = Z_0 t_d$, $R = Z_0$, $C = t_d/Z_0$ give $\rho = 1$; this R is an impedance scale, not a dissipative resistor. A finite

polynomial has no distributed pole sequence, and a finite rational function has only finitely many poles. Neither is the exact line on an unrestricted frequency domain.

A physical finite model subdivides the line into N series branches of $L' \Delta x, R' \Delta x$ and $N + 1$ shunt nodes. With $\Delta x = \ell/N$, use half weights at both endpoints for C' and G' and full weights inside. Generically it has $2N + 1$ reactive state coordinates, subject to termination constraints, and exact finite transfer

$$H_N(s) = c^T (sI - F_N)^{-1} b. \quad (103)$$

The corresponding stores and physical spatial diagnostics are

$$E_N = \frac{1}{2} \sum_j L' \Delta x i_j^2 + \frac{1}{2} \sum_j C_j v_j^2, \quad \Lambda_{\text{abs}} = \sum_j L' \Delta x |i_j|, \quad Q_{\text{abs}} = \sum_j C_j |v_j|. \quad (104)$$

Raw sums without spatial weights change just because the subdivision changes. Linkage and charge sums have different units and are not added.

For the continuum boundary, $E = \int_0^\ell (L' i^2 + C' v^2) dx/2$. Multiplying the time-domain telegrapher equations by the conjugate fields and integrating by parts gives

$$\dot{E} = v(0)i(0) - v(\ell)i(\ell) - \int_0^\ell (R' i^2 + G' v^2) dx. \quad (105)$$

For a Thevenin source and resistive load these are the separately integrated powers $V_s i_s, -R_s i_s^2, -G_L v(\ell)^2$ and the two distributed loss integrals. The finite ladder gives the same identity by telescoping internal port products. Endpoint electric and magnetic fields or node states are independently evaluated. A post-pulse reflection tail need not have vanished at a chosen finite time. Deleting it by an ideal reset requires outward event work equal to its actual store and does not describe a constructed hardware reset.

A degree-six Taylor matrix or a $[3/3]$ Padé matrix is a finite candidate with its own exact residual against the matrix exponential. Neither the order nor a small low-frequency discrepancy certifies a near-open high-frequency corner. An alternating zero-weighted-mean capacitance change preserves total capacitance but alters the medium; it cannot be called mere refinement of a fixed uniform line. A continuum comparison needs either a fixed material profile with refinement or a proved homogenization limit. Ideal open and short terminations are separate boundary conditions. Long reflection tails, dispersive loss, discontinuous edges, exterior fields, radiation and measured terminations remain unresolved physical limits. A separately normalized line model does not automatically audit the field and termination realization of a dimensional RF device.

11.2 Fractional response

For $0 < \alpha < 1$, take the principal branch $H_\alpha(s) = (1 + s\tau)^{-\alpha}$. The exact integral

$$H_\alpha(s) = \frac{\sin(\pi\alpha)}{\pi} \int_1^\infty \frac{(t-1)^{-\alpha}}{t+s\tau} dt \quad (106)$$

is obtained by substituting $u = t - 1$ in the beta integral. It represents a continuous positive relaxation spectrum. A finite positive quadrature defines a positive-residue rational candidate and thus a finite RC realization after the port scale is supplied; its exact error is the difference between the integral and that quadrature. Positivity alone does not bound this error or ensure monotonic improvement with order.

At $1 + s\tau = -r, r > 0$, the two boundary values differ by $-2i \sin(\pi\alpha) r^{-\alpha}$. A finite rational function has no such branch discontinuity and cannot equal the fractional function globally. The binomial Taylor series is

exact only as an infinite convergent series for $|s\tau| < 1$; a finite polynomial is generally nonproper and has no automatically assigned physical stores. Finite-band bounds, nonprincipal paths, material identification and the continuous-spectrum initial history remain open. The stores of a chosen finite RC approximation belong to that approximation, not to the bare fractional formula.

11.3 Exact delay and feedback

A delay has transfer e^{-sT} , requiring the input history on an interval of length T . In a feedback relation $y(t) = u(t) + gy(t - T)$, the characteristic roots for $g \neq 0$ satisfy

$$s_n = \frac{\log |g| + i(\arg g + 2\pi n)}{T}, \quad n \in \mathbb{Z}. \quad (107)$$

The homogeneous dynamics decay for $|g| < 1$, are neutral at $|g| = 1$, and grow for $|g| > 1$. By repeated substitution, on each successive length- T interval the response is a finite sum of delayed inputs plus a known factor g^m multiplying the original history. This method-of-steps identity is exact and keeps the preparation explicit.

A diagonal Padé delay candidate can be written $Q_n(-sT)/Q_n(sT)$, where

$$Q_n(z) = \sum_{k=0}^n \frac{(2n-k)! n!}{(2n)! k! (n-k)!} z^k. \quad (108)$$

Real coefficients make numerator and denominator conjugate at imaginary s , proving unit modulus there wherever the quotient is defined. This candidate is rational and all-pass on the imaginary axis; this scattering property does not make it a positive-real impedance. No finite proper rational function approximates the unit delay with arbitrarily small absolute error on the entire imaginary axis: its limit at infinite frequency is a constant, whereas the delay traverses the unit circle endlessly. The worst limiting error is at least one, and is two for a unit-modulus limiting constant. An improper rational function is unbounded there. Thus every finite approximation requires a finite active band and a history-to-state map. Section 7 proves that an arbitrary history has no unique map to a finite moment chain. A prepared transmission-line field can realize a physical delay with distributed storage, but the bare delay or Padé expression has no uniquely specified store or radiation boundary.

12 Driven, nonlinear and time-varying realizations

Driven and changing systems test which parts of a coefficient law remain fixed when excitation, material state or control is varied. The input operator, time dependence and additional physical states must be specified before a constant-coefficient construction is transferred.

12.1 Common and differential electrical modes

Consider two reciprocal coupled windings with $\mathbf{L} = L \begin{pmatrix} 1 & \kappa \\ \kappa & 1 \end{pmatrix}$, $|\kappa| < 1$, winding resistance $R > 0$, and differential load resistance $R_\ell \geq 0$:

$$\mathbf{L}\dot{\mathbf{i}} + \begin{bmatrix} RI + R_\ell & -1 \\ -1 & RI + R_\ell \end{bmatrix} \mathbf{i} = \begin{pmatrix} V_0 + V_d \sin \omega t \\ V_0 - V_d \sin \omega t \end{pmatrix}. \quad (109)$$

With $i_\pm = (i_1 \pm i_2)/2$ the modes are exactly

$$L(1 + \kappa)\dot{i}_+ + Ri_+ = V_0, \quad L(1 - \kappa)\dot{i}_- + (R + 2R_\ell)i_- = V_d \sin \omega t. \quad (110)$$

Each solution is its exact particular solution plus a decaying exponential. At $\kappa = \pm 1$ one equation becomes an algebraic constraint; incompatible initial currents cannot be transported through that boundary without another physical path. An open differential load, DC and infinite-frequency limits also require their exact constraints, rather than a large finite replacement.

The physical store is $E = i^T Li/2$. For the two-winding boundary, the separate inward powers are $v_1 i_1$, $v_2 i_2$, $-Ri_1^2$, $-Ri_2^2$ and $-R_\ell(i_1 - i_2)^2$. Integrate them on preparation, driven operation and relaxation intervals and compare endpoint currents. Exact phasors describe operation from an exactly prepared periodic state, or the limiting periodic solution of a stable system; they do not remove a finite settling tail.

For settled common current $I_+ = V_0/R \geq 0$ and differential amplitude A_- , opposite current signs occur when $|i_-| > I_+$. Their fraction of a period is

$$d_{\text{opp}} = \begin{cases} 0, & A_- \leq I_+, \\ 1 - \frac{2}{\pi} \arcsin(I_+/A_-), & A_- > I_+. \end{cases} \quad (111)$$

The zero-common-current case has unit fraction apart from isolated zeros, provided $A_- > 0$; if both amplitudes vanish, there is no opposite-sign interval. Coupled linkage is Li , so an absolute linkage sum is not an absolute current sum. Neither is the sum of branch apparent powers $\sum V_{j,\text{rms}} I_{j,\text{rms}}$, which has units of volt-amperes. With the peak phasors used here, complex power is $VI^*/2$ and reactive power is $\Im(VI^*)/2$; RMS phasors instead give $\Im(V_{\text{rms}} I_{\text{rms}}^*)$. Neither reactive quantity is signed transferred work, and the peak convention agrees with (82).

A fitted admittance $G_\infty + \sum a_j/(1 + s\tau_j)$ with $a_j > 0$ is realized by a conductance and series RL branches with $R_j = 1/a_j$, $L_j = \tau_j/a_j$. Negating a branch's terminal response while keeping that same positive-store internal core leaves a transfer to identify. If the core absorbs uj but its terminal current is $s_b j$, $s_b = -1$, the unmatched inward power is $(1 - s_b)uj = 2uj$. Its signed integral on the declared interval is the required additional work. A controller is one candidate account, whose supply, state and losses must be constructed and evaluated independently. Stable poles alone do not certify the resulting terminal passivity. A useful finite response band does not certify global transfer or identify a fitted model's physical states.

12.2 Nonnormal coordinates and finite waveform families

A stable system $\dot{z} = -\text{diag}(p_1, p_2)z + bu$, $p_j > 0$, $b = (1, 7/10)^T$, can be written as $x = Vz$ with nearly parallel eigenvectors. Its coordinate norm may exhibit a transient increase. The positive physical quadratic metric transported from z is

$$H = V^{-T}V^{-1}, \quad E = x^T H x/2 = z^T z/2, \quad \dot{E} = -p_1 z_1^2 - p_2 z_2^2 + ub^T z. \quad (112)$$

The loss channels have effort $p_j z_j$ and flow z_j in the normalized realization, and the source port has effort u , flow $b^T z$. A coordinate burst does not identify an increase of physical store. As eigenvectors coalesce, the coordinate transformation may become singular; one must distinguish a change of basis from an actual Jordan-block system.

The exact forced response is Duhamel's formula (58). It applies to a commensurate multitone, a Gaussian packet, a carrier under a separately declared modulation envelope, a band-limited Gaussian random input, or any specified integrable waveform. These are distinct classes. A finite-window Fourier estimate retains the homogeneous term and the window convolution; zero-padding a missing past is an additional initial-history assumption. Increasing a model's order or a window length is not itself a proof of phase, peak or tail accuracy.

A linear system with jointly Gaussian forcing and Gaussian initial state has a Gaussian ensemble, but adjacent observations on one finite path are correlated. A fitted normality statistic based on independent observations does not have its nominal interpretation. For a stationary differentiable zero-mean Gaussian output, the exact total zero-crossing rate of [Rice \(1945\)](#) is

$$\nu_0 = \frac{1}{\pi} \sqrt{\frac{\mathbb{E}\dot{y}^2}{\mathbb{E}y^2}}. \quad (113)$$

It follows by integrating $|\dot{y}|$ against the joint Gaussian density at $y = 0$, using stationarity to obtain zero covariance of y and $\dot{y}$. Nondifferentiable forcing/output, nonstationarity, fitted means and finite-mode tails require different assumptions. Unresolved finite-path departures, correlation, non-Gaussian drives and uncertainty in the physical metric cannot be replaced by asserted ensemble conclusions.

12.2.1 Signed work and unresolved drift under random drive

A changing mean store under a finite-band random drive raises a separate question from Gaussianity or zero-crossing rate: what signed transfer accounts for the change, and does a residual remain after that transfer is independently evaluated? On each finite operation interval $[t_0, t_1]$, retain the physical source effort e , its conjugate flow f , all loss ports and the prepared initial state. Zero mean effort does not set mean power to zero, since

$$\mathbb{E}[ef] = \mathbb{E}[e] \mathbb{E}[f] + \text{Cov}(e, f). \quad (114)$$

This follows by expanding $(e - \mathbb{E}e)(f - \mathbb{E}f)$. With finite second moments and $\int_{t_0}^{t_1} \mathbb{E}|e_\ell f_\ell| dt < \infty$, expectation can pass through each separate integral. For every declared port define $W_\ell = \int_{t_0}^{t_1} e_\ell f_\ell dt$ before forming

$$\mathbb{E}[r_E] = \mathbb{E}[E(t_1)] - \mathbb{E}[E(t_0)] - \sum_\ell \mathbb{E}[W_\ell]. \quad (115)$$

The stores are evaluated from the states and the identified physical metric, not from the expected powers. For the normalized realization (112), the explicit products are $u(b^T z)$ and $-(p_j z_j) z_j$, with $E = z^T z / 2$. Duhamel's formula supplies the state path and retains any correlation between preparation and drive. A random input to a merely fitted terminal law supplies no internal store until a realization is specified.

The unresolved operation question is whether a persistent positive or negative residual survives the joint uncertainty of source work, losses and endpoint states. A source-off relaxation interval is audited separately with its actual remaining ports; its initial state is the operation endpoint. Agreement of a relaxation identity or an operation endpoint check does not bound uncertainty in each operation port integral. Shared paths, finite-window leakage, unresolved high-frequency modes and correlated gain or timing errors require their own bounds. A constant bias in one power channel can accumulate with interval duration; extending the interval without resolving that bias does not establish a physical drift.

For a finite damped realization and a specified forcing covariance, exact state and work integrals give conditional predictions. Their zero numerical integration residual does not resolve an independently unbounded finite-precision or physical observation. Section 16.1 gives the signed outcome criterion. Non-Gaussian drives, longer tails, nonlinear or time-varying plants and a physical implementation of the metric remain open beyond these conditions.

12.3 Material memory and parameter modulation

A concrete nonlinear inductive model uses flux ϕ , internal material coordinate m and potential

$$U(\phi, m) = \frac{\phi^2}{2} + \frac{\alpha\phi^4}{4} + \frac{h}{2}(\phi - m)^2, \quad i = \phi + \alpha\phi^3 + h(\phi - m), \quad \dot{\phi} = v - Ri, \quad \dot{m} = \frac{h}{\zeta}(\phi - m), \quad (116)$$

with $\alpha \geq 0, h, \zeta, R > 0$ in a declared normalized system. Direct differentiation yields $\dot{U} = vi - Ri^2 - h^2(\phi - m)^2/\zeta$. The material loss is the product of generalized effort $h(\phi - m)$ and its conjugate flow $\dot{m}$. Integrate that loss and electrical powers separately. A loop integral $\oint i d\phi$ describes a closed physical cycle only when the internal coordinate also returns to its initial state. This rate-dependent lag law is not a universal hysteresis model. Thermodynamically admissible measured material laws, saturation, longer settling and wider excitation remain open.

For a mechanical oscillator with prescribed positive $m(t), k(t)$, $\dot{q} = p/m$ and $\dot{p} = F - cp/m - kq$, the actual store satisfies

$$E = \frac{p^2}{2m(t)} + \frac{k(t)q^2}{2}, \quad \dot{E} = F\frac{p}{m} - c\left(\frac{p}{m}\right)^2 - \frac{p^2\dot{m}}{2m^2} + \frac{q^2\dot{k}}{2}. \quad (117)$$

An electrical counterpart uses linkage λ and charge q with $i = \lambda/L(t)$, $v_C = q/C(t)$, $\dot{\lambda} = u - Ri - v_C$ and $\dot{q} = i$. Its independently defined store obeys

$$E = \frac{\lambda^2}{2L(t)} + \frac{q^2}{2C(t)}, \quad \dot{E} = ui - Ri^2 - \frac{i^2}{2}\dot{L} - \frac{v_C^2}{2}\dot{C}. \quad (118)$$

The coefficients of $\dot{L}$ and $\dot{C}$ are the efforts conjugate to the imposed parameter rates. They are physical modulation ports only when the parameter-changing mechanism is included in the declared realization. For the mechanical model, the last two products are modulation-port powers, with efforts $-p^2/(2m^2)$ and $q^2/2$ conjugate to flows $\dot{m}$ and $\dot{k}$. Stability of a periodic modulation is determined by the exact monodromy matrix and its Floquet multipliers, not by frozen instantaneous poles. Positive coefficients at every instant do not exclude parametric instability. Modulation phase, depth, frequency, tangent boundaries and separate zero or negative parameter regimes require explicit continuation or proof.

12.4 Feedback and finite supplies

Closing feedback changes the pole equation to $1 + K(s)H(s) = 0$, including every controller, observer, sensor and actuator state. A first-order lag $1/(1 + sT_d)$ is not an exact delay. Replacing an oscillator by a frozen first-order fit can change closed-loop stability even when an open-loop band fit is good. Proportional control, lead-lag compensation, observers, polarity changes and saturation have different state structures. Stable/passive reduction must be assessed in the intended interconnection, using trajectory or transfer criteria rather than work or energy to select it.

An explicit supply can include a DC-link capacitor C_b with $C_b V_b \dot{V}_b = P_{\text{source}} - P_{\text{converter}}$, a bias winding, converter loss and a thermal state. The corresponding capacitor current is P/V_b only while $V_b \neq 0$; a zero bus voltage is a boundary requiring another model. Return power may charge the bus or bias winding even after the external source is disabled. Sensor supply, magnetic reaction and thermal ports cannot be suppressed while claiming a complete physical controller. Finite source ratings, nonlinear observer uncertainty, multivariable or adaptive control, long delays and hardware validation remain unresolved. An unstable trajectory can still have a correct signed work balance on a finite prefix; it has no stationary

phasor interpretation.

13 Mechanical absorption and electromechanical third order

Mechanical elimination supplies further fourth- and third-order coefficient laws. Their determinants expose which masses, couplings and source terms determine the coefficients; the subsequent work accounts assess those declared realizations.

13.1 A two-mass absorber

Let x_1 be the primary displacement, x_2 the absorber displacement and y the driven base displacement. Define $g(s) = k_0 + c_0s$ and $b(s) = k_2 + c_2s$. The exact equations are

$$M\ddot{x}_1 + g(D)(x_1 - y) + b(D)(x_1 - x_2) = F, \quad m\ddot{x}_2 + b(D)(x_2 - x_1) = 0. \quad (119)$$

For positive masses and stiffnesses, with sufficient damping to damp every mode, this is an asymptotically stable four-state system. Define $a_{11} = Ms^2 + g + b$, $a_{22} = ms^2 + b$ and $\mathcal{D} = a_{11}a_{22} - b^2$. The support reaction $R = g(D)(x_1 - y)$ satisfies

$$\mathcal{D}(s)R(s) = g(s)a_{22}(s)F(s) + g(s)[g(s)a_{22}(s) - \mathcal{D}(s)]Y(s). \quad (120)$$

Initial-state terms are supplied by the physical four-state realization. The internal coupling force is $b(D)(x_1 - x_2)$, and total momentum obeys $(M\dot{x}_1 + m\dot{x}_2)' = F - R$. A reaction fit alone does not reconstruct either internal quantity.

The boundary enclosing both masses and both springs has

$$\begin{aligned} E &= \frac{1}{2}M\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 + \frac{1}{2}k_0(x_1 - y)^2 + \frac{1}{2}k_2(x_1 - x_2)^2, \\ P_F &= F\dot{x}_1, \quad P_y = -[k_0(x_1 - y) + c_0(\dot{x}_1 - \dot{y})]\dot{y}, \\ P_0 &= -c_0(\dot{x}_1 - \dot{y})^2, \quad P_2 = -c_2(\dot{x}_1 - \dot{x}_2)^2. \end{aligned} \quad (121)$$

The base power includes the full force on the moving support. Integrate all four powers over the stated interval, then evaluate E at both endpoints. Omitting the base port while driving it produces a spurious residual.

With $y = 0$ and $c_2 = 0$, the frequency $\omega_a^2 = k_2/m$ makes $a_{22}(i\omega_a) = 0$. Provided the coupling is nonzero, x_1 vanishes while x_2 remains finite and balances the applied force. This exact antiresonance explains why a quiet terminal can hide internal stress. With both base and force excitation, their relative phase can also cancel reaction without eliminating either motion. The unresolved practical question is how much internal coupling stress a reaction measurement can identify [OP-LR58-01]. A small adjustable two-mass shaker with a support load cell, two accelerometers and a strain gauge on the coupling spring provides ordinary accessible measurements. At the predicted antiresonance, a vanishing primary response with nonzero coupling strain rejects terminal-only stress inference. Independent mass and damping changes distinguish parameter ambiguity from sensor noise. Calibrated material stress conversion, probe loading and wider nonlinear regimes remain unestablished.

The quantitative particular solution for force peak phasor $\hat{F} \neq 0$ is $\hat{x}_1 = 0$, $\hat{x}_2 = -\hat{F}/k_2$ and $\hat{R} = 0$. For a uniform axial coupling element of reference length ℓ_2 , its extension strain is $\hat{\varepsilon}_2 = \hat{F}/(k_2\ell_2)$. A zero-strain alternative is distinguishable if this magnitude exceeds the combined calibrated strain uncertainty, including the finite settling correction derived below. With nonzero c_2 , use the exact determinant

expressions instead of claiming an exact node. Low support reaction alone cannot decide between those internal predictions.

At $m = 0$, a compatible massless absorber satisfies $b(D)(x_2 - x_1) = 0$; its internal relaxation must be considered before identifying a locked displacement. At zero coupling, the absorber's free modes disconnect and become hidden from the primary. A rigidly locked absorber adds its mass to M . These are distinct boundary models.

The small-mass resonant limit is nonuniform. Set $k_2 = m\omega_a^2$, $c_2 = 2\zeta m\omega_a$ and evaluate at $s = i\omega_a$. Eliminating x_2 contributes the exact dynamic stiffness

$$b - \frac{b^2}{a_{22}} = -m\omega_a^2 + \frac{im\omega_a^2}{2\zeta}. \quad (122)$$

Thus $m \rightarrow 0$ at fixed $\zeta > 0$ removes the contribution, whereas $\zeta = m/m_*$ gives a finite imaginary contribution and faster vanishing ζ makes it unbounded. The dimensioned reference m_* keeps the scaling meaningful. This resolves one analytic path dependence without claiming all frequency, forcing or damping limits commute. Other coscalings, nonlinear/contact absorbers, base acceleration constraints and nonideal switching remain open.

A finite measurement after turn-on includes $Ce^{Ft}(x_0 - x_{\text{per}}(0))$. Its Fourier integral over $[0, T]$ is exactly $C(F - i\omega I)^{-1}[e^{(F - i\omega I)T} - I](x_0 - x_{\text{per}}(0))$ when the inverse exists, with continuous extension at a singular argument. This term supplies an analytic settling correction or bound. No finite waiting time is exactly steady unless the homogeneous amplitude is zero. Disjoint frequency intervals and independently varied fit structures are needed to establish transfer beyond one antiresonance neighborhood.

13.2 A third-order transducer and unassigned coupling work

An electrical winding and a mechanical oscillator form a natural third-order system:

$$L\dot{i} = v_s - Ri - g_e v, \quad \dot{x} = v, \quad m\dot{v} = g_m i - dv - kx + F_s, \quad (123)$$

where $d = c + c_\ell$ may include internal and load damping. With $Z_e = R + sL$, $Z_m = d + sm + k/s$,

$$\begin{pmatrix} v_s \\ F_s \end{pmatrix} = \begin{pmatrix} Z_e & g_e \\ -g_m & Z_m \end{pmatrix} \begin{pmatrix} i \\ v \end{pmatrix}, \quad \mathcal{D} = Z_e Z_m + g_e g_m, \quad \frac{v}{v_s} = \frac{g_m}{\mathcal{D}}, \quad \frac{i}{F_s} = -\frac{g_e}{\mathcal{D}}. \quad (124)$$

The minus sign follows the chosen mechanical source orientation. Reciprocity has $g_e = g_m$ in consistent effort–flow units; the antisymmetric off-diagonal signs do not mean nonreciprocity. Its scalar characteristic polynomial is

$$s\mathcal{D} = Lms^3 + (Rm + Ld)s^2 + (Rd + Lk + g_e g_m)s + Rk. \quad (125)$$

The cubic Routh condition applies to these complete coefficients. Positive reciprocal coupling with positive damping gives a passive stable realization; altering one coupling independently requires a new physical explanation.

For $E = Li^2/2 + mv^2/2 + kx^2/2$,

$$\dot{E} = v_s i + F_s v - Ri^2 - dv^2 + (g_m - g_e)iv. \quad (126)$$

The last term is an unassigned coupling power for the sources and losses listed in (123). Enclose the winding, mass and spring and exclude the source and loss reservoirs. On a smooth interval $[t_0, t_1]$ without

resets, integrate the four inward powers separately:

$$\begin{aligned}
 W_V &= \int_{t_0}^{t_1} v_s i \, dt, & W_F &= \int_{t_0}^{t_1} F_s v \, dt, \\
 W_R &= - \int_{t_0}^{t_1} (Ri) i \, dt, & W_d &= - \int_{t_0}^{t_1} (dv) v \, dt, \\
 r_E^{\text{listed}} &= E(t_1) - E(t_0) - W_V - W_F - W_R - W_d = \int_{t_0}^{t_1} (g_m - g_e) i v \, dt.
 \end{aligned} \tag{127}$$

When $d = c + c_\ell$, the last loss is split into the internal damper and load integrals before combination. A controller, bias reaction or revised constitutive law may account for the residual. Its signed work must be established independently. Assigning the rightmost integral to a named port does not establish the transfer. One-way coupling $g_m = 0$, reversed $g_m = -g_e$ and attenuated $g_m = g_e/2$ remain distinct controls, with stability and port passivity assessed for their own parameters.

13.2.1 Zero named commands with unequal effective coupling

An illustrative finite-prefix solution makes the positive residual explicit. Choose

$$\begin{aligned}
 L &= 1 \text{ H}, & m &= 1 \text{ kg}, & k &= 1 \text{ N/m}, \\
 R &= 1 \text{ } \Omega, & d &= 1 \text{ N s/m}, & g_e &= -2 \text{ V s/m}, & g_m &= 3 \text{ N/A},
 \end{aligned} \tag{128}$$

and set $v_s = F_s = 0$. With $\vartheta = t/(1 \text{ s})$, the prepared state and its evolution are

$$i = e^\vartheta \text{ A}, \quad x = e^\vartheta \text{ m}, \quad v = e^\vartheta \text{ m/s}. \tag{129}$$

Substitution gives $L\dot{i} = -Ri - g_e v$ and $m\dot{v} = g_m i - dv - kx$ exactly, together with $\dot{x} = v$. In the dimensionless Laplace variable $z = s(1 \text{ s})$, the characteristic polynomial is

$$z^3 + 2z^2 - 4z + 1 = (z - 1)(z^2 + 3z - 1). \tag{130}$$

The solution is an unstable finite prefix, with a specified initial state; it is not a periodic operating state. Preparation lies before $t = 0$ and has no assigned work here. Its constitutive initial store is nevertheless included. No switching occurs on $[0, T]$, $T = (\log 2) \text{ s}$.

Direct signed integration gives $W_V = W_F = 0$ and

$$\begin{aligned}
 W_R &= W_d = - \int_0^{\log 2} e^{2\vartheta} d\vartheta \text{ J} = -\frac{3}{2} \text{ J}, \\
 \int_0^T (g_m - g_e) i v \, dt &= 5 \int_0^{\log 2} e^{2\vartheta} d\vartheta \text{ J} = \frac{15}{2} \text{ J}.
 \end{aligned} \tag{131}$$

Independent state evaluation yields the following account.

Table 14. Increasing declared storage with both listed drives zero. The coupling transfer remains unassigned for this model's listed ports.

Quantity on $[0, (\log 2) \text{ s}]$	Exact value
Initial winding, kinetic and spring store combined	$3/2 \text{ J}$
Final combined store	6 J

Quantity on $[0, (\log 2) \text{ s}]$	Exact value
Store increase	$9/2 \text{ J}$
Electrical-source and mechanical-source works, each	0
Resistor work into the boundary	$-3/2 \text{ J}$
Damper work into the boundary	$-3/2 \text{ J}$
Residual for the listed ports	$+15/2 \text{ J}$

The trajectory, individual integrals, polynomial and endpoint stores have exact checks; numerical integration residual is zero. The missing physical coupling account remains open. A proposed implementation must identify how the unequal couplings arise and determine its additional work, stores and losses. This example fixes the magnitude and sign of that question without reporting a physical observation.

For a candidate realization, synchronized winding voltage/current and mechanical force/velocity provide the listed-port works; bias and DC-link channels provide separate candidate transfers. Independently calibrated L, m, k and state measurements determine both endpoint stores. Retain (127) beside the expanded-boundary result, and apply the uncertainty criterion in Section 16.1. Neither defining the controller work by subtraction nor choosing its parameters to cancel the residual identifies a physical account.

13.2.2 A separately instrumented coupling control

A reciprocal transducer of coupling g , together with a series voltage actuator commanded by velocity, realizes an explicit candidate port. Let $u_c = Kv$ and

$$Li = v_s + u_c - Ri - gv, \quad \dot{x} = v, \quad m\dot{v} = gi - dv - kx + F_s. \quad (132)$$

The reduced coefficients are $g_m = g$ and $g_e = g - K$. Taking $g = 3 \text{ N A}^{-1}$ and $K = 5 \text{ V s m}^{-1}$ reproduces the unequal coefficients above. The extra physical power is $u_c i$, positive into the plant and negative out of the actuator. Setting $v_s = 0$ leaves that port active.

On the exact exponential trajectory and interval already specified, direct integration gives $W_c = \int u_c i \, dt = 15/2 \text{ J}$. The reduced listed-port residual remains $+15/2 \text{ J}$; the enlarged plant account has zero residual under this declared actuator law. The output port was specified before the integral, not defined from the remainder. This is a realizable control hypothesis, not a measurement of the original apparatus. Measure actuator output voltage and current independently of its supply/DC-link channels, losses and endpoint stores. Equal output and supply work cannot be presumed.

Linearity permits an ordinary low-amplitude control: multiplying the original state by $\lambda = 1/100$ gives displacement 1 to 2 cm, current 10 to 20 mA and commanded voltage 1/20 to 1/10 V on the same interval. Its plant store increases by $9/20000 \text{ J}$ and the candidate output work is $3/4000 \text{ J}$. These exact predictions assume the same coefficients and feasible initial preparation; actual signal-to-error ratios require calibration. Force, velocity, voltage and shunt-current channels can distinguish output transfer, supply depletion and unexplained residual without assigning any of them by subtraction. Finite actuator and sensor dynamics add states and may change the effective order and closed-loop stability.

13.2.3 Passivity and a finite supply remain separate questions

For real constants, the Hermitian dissipative part of the impedance matrix is $\begin{pmatrix} R & (g_e - g_m)/2 \\ (g_e - g_m)/2 & d \end{pmatrix}$. After consistent port scaling it is positive semidefinite exactly when

$$R \geq 0, \quad d \geq 0, \quad (g_e - g_m)^2 \leq 4Rd. \quad (133)$$

Thus unequal coupling is not automatically a nonpassive terminal map; damping can satisfy this inequality. A model implemented with an active controller and a passive external map are compatible descriptions at different boundaries. Electrical linkage and mechanical momentum have different units and must not be added as a scalar stress measure.

A finite bias coil adds its own store, copper loss and source path. To make one conditional account explicit, let $I_* > 0$ be a reference bias current and prescribe $g_e = \bar{g}_e i_b / I_*$, $g_m = \bar{g}_m i_b / I_*$. Define the reaction effort $A = (\bar{g}_m - \bar{g}_e) i v / I_*$, which has voltage units, and the plant transfer $P_a = i_b A$. For a nonnegative converter loss $\ell(P) = \epsilon_\ell P^2 / (P_* + |P|)$, with $\epsilon_\ell \geq 0$ and $P_* > 0$, a candidate bias-winding equation is

$$\begin{aligned} L_b \dot{i}_b &= u_b - R_b i_b - A - v_\ell, & v_\ell &= \frac{\epsilon_\ell i_b A^2}{P_* + |i_b A|}, \\ E_b &= \frac{1}{2} L_b i_b^2, & \dot{E}_b &= u_b i_b - R_b i_b^2 - P_a - \ell(P_a). \end{aligned} \quad (134)$$

Here u_b is the converter voltage applied to the bias coil, and $v_\ell i_b = \ell(P_a) \geq 0$. The formula is continuous at $i_b = 0$: $v_\ell = 0$ there, while the reaction effort A may remain nonzero. The illustrative choice $\epsilon_\ell = 1/50$, $P_* = 1$ W gives the stated loss law without asserting measured converter behavior. This supplies a declared dynamical hypothesis, not an identified physical supply.

Enclose the bias winding and exclude the plant, DC link and heat reservoirs. On a smooth source-off interval $[t_0, t_1]$ with $u_b = 0$, its separately integrated inward works are

$$\begin{aligned} W_{\text{reaction}} &= - \int_{t_0}^{t_1} A i_b \, dt, & W_{\text{copper}} &= - \int_{t_0}^{t_1} (R_b i_b) i_b \, dt, \\ W_{\text{converter}} &= - \int_{t_0}^{t_1} v_\ell i_b \, dt, & r_b &= \frac{L_b}{2} [i_b(t_1)^2 - i_b(t_0)^2] - W_{\text{reaction}} - W_{\text{copper}} - W_{\text{converter}}. \end{aligned} \quad (135)$$

The endpoint coil store increases in this candidate exactly when

$$- \int_{t_0}^{t_1} P_a \, dt > \int_{t_0}^{t_1} R_b i_b^2 \, dt + \int_{t_0}^{t_1} \ell(P_a) \, dt. \quad (136)$$

It is the signed reaction, not the source-off label, that determines this prediction. Disabling an external charger does not imply $u_b = 0$; retain $\int u_b i_b \, dt$ if the converter continues to drive. The constant-coupling exponential example is a different model from this bias-dependent system and is not silently reused as its trajectory.

Appendix E supplies a finite DC link, sensor and thermal account with seven continuous states. Each state has its own endpoint evaluation and each transfer its own integral. The open question is whether a physical bias supply realizes these couplings and the source-off increase with consistent coil, DC-link, thermal and plant accounts [OP-LR59-02]. Measure coil current at both endpoints with independently calibrated L_b , winding voltage/current throughout the interval, and the separately accessible converter-output and DC-link pairs. Reaction effort requires an independently identified transduction law or a direct conjugate

measurement; the coil remainder alone does not identify it. The measurable reciprocal control above does not settle this different nonlinear seven-state model. Interval balance, DC-link work and thermal endpoint consistency remain separate unresolved checks beyond the exact conditional identities. A small combined-path residual can conceal opposite local residuals and does not settle them. Thermal laws, saturation, hysteresis, switching slew, feedback, multi-axis coupling and calibrated cross-domain sensors remain additional physical requirements. Finite-prefix work identities do not give an unstable control a steady-state interpretation.

14 Larger networks, observation geometry and switching

Network interconnection extends coefficient synthesis beyond one component pair. Incidence, observation and constraints determine which states survive scalar elimination. Switching and finite reference implementations can change both the coefficient law and its admissible preparation.

14.1 A common three-state conservative template

Two primary states s_1, s_2 coupled by a mediator z obey

$$\dot{s}_1 = -z, \quad \dot{s}_2 = z, \quad h_z \dot{z} = h_1 s_1 - h_2 s_2, \quad E = \frac{1}{2}(h_1 s_1^2 + h_2 s_2^2 + h_z z^2). \quad (137)$$

Here h_j are inverse masses for momenta, spring stiffnesses for extensions, inverse capacitances for charges, or inverse inductances for linkages. The mediator coefficient is respectively inverse spring stiffness, mass, inductance or capacitance. Multiplication of each state equation by its constitutive effort shows that the three separately integrated internal powers cancel; endpoint stores independently agree on the isolated operating interval. Preparation and reset are additional intervals.

The signed sum $c = s_1 + s_2$ is constant. With $d = s_2 - s_1$,

$$|s_1| + |s_2| = \max(|c|, |d|), \quad h_z \ddot{d} + (h_1 + h_2)d = (h_1 - h_2)c. \quad (138)$$

A scalar primary state generally satisfies a third-order homogeneous annihilator containing the zero common mode; specifying c instead yields a second-order affine equation. This is an elementary example of how elimination and initialization change the apparent order.

For two masses 1 and 19 kg, initial velocities 10 and 0 m s⁻¹, and contact stiffness 19/20 N m⁻¹, the exact compliant interval is $0 \leq t \leq \pi$ s:

$$p_1 = \frac{1}{2} + \frac{19}{2} \cos t, \quad p_2 = \frac{19}{2} (1 - \cos t), \quad \delta = 10 \sin t. \quad (139)$$

The contact force is $(19/2) \sin t$ in newtons. The signed momentum stays 10, while the endpoint magnitude sum is 28, a ratio 14/5. Its first sign crossing is $t = \arccos(-1/19)$. Direct integration of $-k \delta \dot{x}_1$ and $k \delta \dot{x}_2$ gives works $-19/2$ and $19/2$ J; the spring starts and ends unstrained, and both endpoint total stores are 50 J. In the center-of-mass frame the magnitude sum is 19 at both endpoints. The change of observer modifies kinetic stores and individual works, not the physical contact law.

For an uncoupled inductive pair with ratio $r = L_2/L_1$, the corresponding zero-mediator endpoint ratio is $(3r - 1)/(r + 1)$ for $r > 1$, and one for $r \leq 1$. The ceiling three belongs to that particular pair and preparation. It is not a general multi-state limit. An exact three-body counterexample consists of two separated compliant elastic encounters: masses 1, 19, 361, initially velocities (10, 0, 0), first let masses 1, 19 exchange, then 19, 361. At the end the velocities are $(-9, -9/10, 1/10)$, with momenta

$(-9, -171/10, 361/10)$. The magnitude ratio is $311/50 > 3$. Each encounter is realized by the sinusoidal relative solution with its own positive stiffness and reduced mass, and the gaps are chosen so they do not overlap. This establishes a wider counterexample, while maxima for a specified continuously connected chain still require its own exact trajectory and extremum analysis.

14.2 Coupled pickups and hidden common preparation

For opposed primary windings a differential pickup can sense $\lambda_3 = \kappa_d(\lambda_2 - \lambda_1)$. In the canonical excursion from $(10, 0)$ to $(-9, 19)$, its open-circuit volt-second integral is $38\kappa_d$. Its work is zero when its current is zero. Loading the pickup changes the full inductance matrix and back-reacts on the primary trajectory. If $K_d^2 = \kappa_d^2(L_1 + L_2)/L_3$, positive semidefiniteness requires $0 \leq K_d \leq 1$; the endpoint is a descriptor boundary, not an ordinary invertible inductance matrix.

A receiver winding adds a state; a receiver capacitor adds another. A figure-eight pickup followed by one rectifier and two separately rectified pickups feeding one bus have different state and connection graphs. For the declared smooth bridge law $v_b = V_b \tanh(i/i_\varepsilon)$, receiving bus power is $V_b i \tanh(i/i_\varepsilon) \geq 0$. Copper and bus powers are integrated separately. This positivity does not decide which architecture has higher peak bus power at a specified voltage or which preserves a specified flux excursion. A receiver's natural frequency, load phase and damping change the extrema and their ordering; a peak at the end of an observation interval is not an interior extremum.

For the canonical pair, let the initial currents be $(10+u, u)$ with $L_1 = 1, L_2 = 19$. The common linkage is $10+20u$ while the differential terminal evolution can remain unchanged in an ideal differential coupling. Unloaded endpoint linkages are $(-9+u, 19+19u)$. Individual winding works become $W_1 = -19/2 - 19u$, $W_2 = 19/2 + 19u$, by integrating their own powers along the shifted solution. The initial magnetic store is $50 + 10u + 10u^2$, evaluated from the prepared state. For $u = -1/2$ the common sum is zero and the magnitude sum is unchanged, even though a differential receiver can see the same trajectory. This changes physical preparation; it is not merely a relabeling of the observer. Preparing and resetting the common current requires actual sources and finite paths.

Normalize the differential extremum by $\mathcal{R} = (d_{\max} + 10)/38$ for the initial canonical state $c = 10, d_0 = -10$. Entry into positive counterflow is exactly $\mathcal{R} > 10/19$, not $\mathcal{R} > 1/2$. With another initial state this threshold must be recomputed. Load-phase and state interactions, disappearing extrema, tangent roots, folds, extreme inductance ratios and exact open/short or perfect-coupling boundaries remain unresolved beyond this algebraic criterion. Work does not select any of those operating conditions.

14.3 Exact coupled-winding crossing boundaries

For the connected lossless pair, let $\mathbf{L} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix}$, $M = \sigma \kappa \sqrt{L_1 L_2}$, $\sigma = \pm 1$, $0 \leq \kappa < 1$, and initial current $(I, 0)$ with $I > 0$. The connection vector is $(1, -1)^T$, so $c = \lambda_1 + \lambda_2 = I(L_1 + M)$. At the zero-mediator equilibrium,

$$\lambda_{\text{eq}} = \frac{I(L_1 + M)}{L_1 + L_2 + 2M} \begin{pmatrix} L_1 + M \\ L_2 + M \end{pmatrix}, \quad \lambda(\theta) = \lambda_{\text{eq}} + [\lambda(0) - \lambda_{\text{eq}}] \cos \theta. \quad (140)$$

Here θ is the exact normalized oscillation phase on $[0, \pi]$. The absolute sum is convex in $\cos \theta$, so its maximum lies at an endpoint. For $r = L_2/L_1 > 1$, the applicable growth boundaries are

$$\kappa_+(r) = \frac{\sqrt{2r-1}-1}{2\sqrt{r}}, \quad \kappa_-(r) = \frac{1}{\sqrt{r}}. \quad (141)$$

Growth occurs below the respective boundary and no growth above it in this stated connected model. To derive the positive-orientation result, set the endpoint candidate $-\lambda_1(\pi) + \lambda_2(\pi)$ equal to $I(L_1 + M)$; the condition reduces to $2M^2 + 2L_1M - L_1(L_2 - L_1) = 0$. For negative orientation the same comparison with $I(L_1 - M)$ reduces to $L_1 + M = 0$. Checking the endpoint signs supplies the stated $r > 1$ branches. At the threshold equality holds exactly. At $\kappa = 1$ the invertible-state derivation ceases to apply.

A first zero of a component satisfies a scalar cosine equation from (140); its exact existence condition is that the resulting cosine lies in $[-1, 1]$. Tangencies and zero modal amplitudes need their own event classification. This distinguishes an analytic enclosure of a root from a finite observation bracket or a dense-interpolation estimate. Loaded receivers and other initial states need new guards; a no-growth maximum can have an initial-time plateau, so differentiating that maximum is not a regular surface continuation.

An entry fraction also needs a measure. For a rectangle $r \in [r_a, r_b]$, $\kappa \in [0, K]$, the uniform-product fraction is exactly

$$\frac{1}{(r_b - r_a)K} \int_{r_a}^{r_b} \min\{K, \kappa_\sigma(r)\} dr. \quad (142)$$

while a uniform logarithmic-ratio measure replaces $dr/(r_b - r_a)$ by $dr/[r \log(r_b/r_a)]$. An illustrative prior can instead use a truncated lognormal ratio with logarithmic center $\log 4$ and width $3/4$, independently of a beta density $6u(1-u)$ for $u = \kappa/K$. A finite equal-weight set defines yet another, discrete measure. None is a measure-free probability of counterflow. Joint parameter enclosures, untraced folds, more general preparations and actual prior calibration remain open.

14.4 Incidence, scalar annihilators and observation bounds

For a connected oriented graph with n_v vertices and n_e edges, let $B_v \in \mathbb{R}^{n_v \times n_e}$ be its vertex-edge incidence matrix, $H = \text{diag}(1/L_j)$, and take unit edge capacitances in normalized units. Vertex linkages $\lambda_v \in \mathbb{R}^{n_v}$ and edge charges $q_e \in \mathbb{R}^{n_e}$ obey

$$\dot{\lambda}_v = -B_v q_e, \quad \dot{q}_e = B_v^T H \lambda_v, \quad \ddot{\lambda}_v = -B_v B_v^T H \lambda_v. \quad (143)$$

They imply $\mathbf{1}^T \lambda_v = \text{constant}$ and, for a vertex subset U , $\Delta(\mathbf{1}_U^T \lambda_v) + \int \mathbf{1}_U^T B_v q_e dt = 0$. More generally, $h_v^T \lambda_v$ is conserved for every edge state exactly when $B_v^T h_v = 0$, since its derivative is $-h_v^T B_v q_e$. For a connected graph this kernel is spanned by $\mathbf{1}$. The store is $(\lambda_v^T H \lambda_v + q_e^T q_e)/2$; its internal powers cancel by the same incidence identity. If p is the characteristic polynomial of $B_v B_v^T H$, Cayley–Hamilton gives $p(-D^2) \lambda_{v,j} = 0$. This is generally a dense, nonminimal higher-order annihilator. Repeated, zero and unobservable factors need reduction for the chosen output. An exact graph nullspace is a zero-frequency mode, not an extremely slow nonzero oscillator.

The mechanical counterpart has $\dot{p} = -B_v f$, $\dot{e} = B_v^T M^{-1} p$, and $f = Ke + CB_v^T M^{-1} p$ for elastic and viscous edges. Damper powers are separately $-c_e(v_{\text{tail}} - v_{\text{head}})^2$. A unilateral edge uses an active-contact law such as $k_e \max(e_e, 0)$. Its active graph changes at engagement and release; there is no single global constant-coefficient scalar elimination. Initial zero gaps, simultaneous activation, chatter, restitution, plasticity and rigid limits require event laws and limiting paths. Finite compliant states have no impulse merely because the active graph changes continuously at zero force.

A distinct circuit construction places linkage on branches. Let B_e be that circuit's node-branch incidence matrix, ϕ its node potentials, $v_e = B_e^T \phi$ its branch voltages and $B_e i_e = 0$ its Kirchhoff current law. Then $v_e^T i_e = 0$, even if the two vectors were evaluated at different times on the same graph. For branches whose voltage is the linkage derivative, $\dot{\lambda}_e = v_e$, an edge functional $h_e^T \lambda_e$ is conserved under all node

potentials precisely when $B_e h_e = 0$. This follows from $(h_e^T \lambda_e)' = (B_e h_e)^T \phi$. With winding resistance, $\dot{\lambda}_e = v_e - R_e i_e$ instead gives the additional term $-h_e^T R_e i_e$; a cycle vector alone no longer establishes conservation.

The two kernels concern different spaces. For a two-vertex, one-edge tree, $B_v = (1, -1)^T$ gives the vertex invariant $\lambda_{v,1} + \lambda_{v,2}$, whereas the corresponding node-branch matrix has trivial edge-cycle kernel. Across different circuit graphs the cross-power identity requires inclusion of the voltage subspace in the orthogonal complement of the current subspace. A switch can destroy it. It does not replace the event work audit.

Magnitude bounds are useful diagnostics once the physical state and metric are known. If $E = x^T H x / 2$, $H > 0$, and $y = Cx$ with full observation rank, constrained minimization gives

$$E_{\min}(y) = \frac{1}{2} y^T Q y, \quad Q = (CH^{-1}C^T)^{-1}, \quad \|y\|_1 \leq \sqrt{2E \max_{\sigma_j=\pm 1} \sigma^T CH^{-1}C^T \sigma}. \quad (144)$$

The proof completes the square in x under $Cx = y$, then uses $\|y\|_1 = \max_{\sigma} \sigma^T y$. This is a current-state envelope using independently evaluated $E(t)$, not an assumption that initial energy stays constant. Additional invariants restrict the ellipsoid further, as derived in Appendix D. A Euclidean $\sqrt{N}$ scaling is an isotropic reference only, not a universal network ceiling.

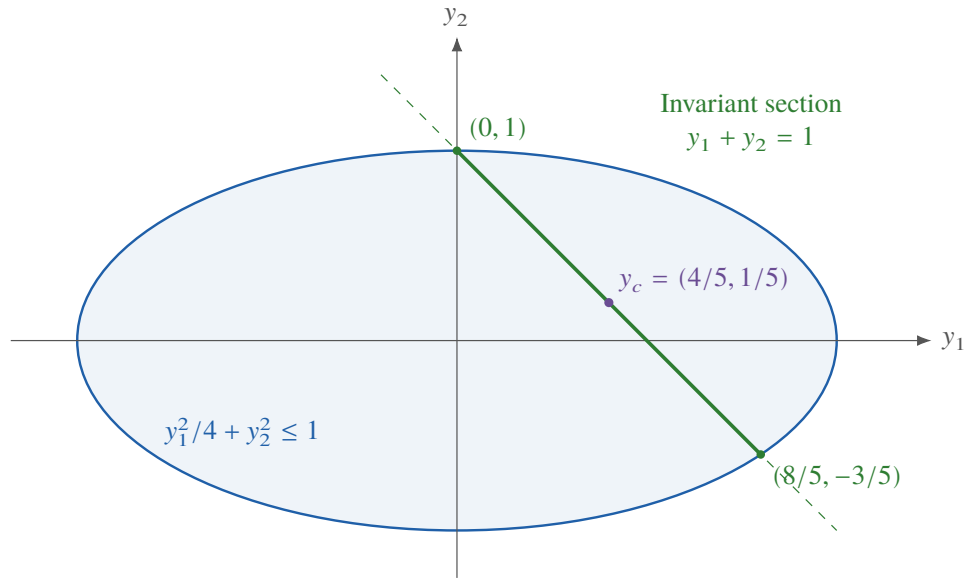


Figure 3. Schematic observation geometry. The illustrative ellipse $y_1^2/4 + y_2^2 \leq 1$ is restricted by $y_1 + y_2 = 1$. Its center and endpoints follow from Appendix D. Magnitude extrema depend on the metric, observation and invariant.

For a fixed linear system and componentwise source limits $|u_j(t)| \leq U_j$, a direction a has exact reachable support

$$\sup a^T x(T) = a^T e^{FT} x_0 + \sum_j U_j \int_0^T |a^T e^{F(T-t)} b_j| dt. \quad (145)$$

Choose the input signs to attain each integrand; constrained bandwidth or slew requires a smaller reachable set. Matrix logarithmic norms also give trajectory bounds in a declared metric. These bounds certify reachable observations and sensitivity, not an energy-based design rule.

The weighted one-norm has the exact logarithmic norm

$$\mu_{1,W}(F) = \max_j \left[F_{jj} + \sum_{i \neq j} \frac{W_i}{W_j} |F_{ij}| \right], \quad \|x(t)\|_{1,W} \leq e^{\mu_{1,W} t} \|x(0)\|_{1,W}, \quad (146)$$

for constant F and positive weights. A nonpositive value certifies contraction in that norm; a positive value still supplies a finite bound. The observation map must be included before comparing it with $\|y\|_1$. Total variation of each observed coordinate, the time fraction in counterflow, and $\|y\|_1 - |\sum y_j|$ describe different trajectory features. Absolute port throughput $\sum_\ell \int |P_\ell| dt$ is a separate audit diagnostic after the signed works are retained.

A physical ideal transformer can also change a magnitude diagnostic. With secondary-to-primary voltage ratio ν , $\lambda_s = \nu \lambda_p$, $i_s = i_p / \nu$, and $L_s = \nu^2 L_p$. In the canonical pair the conserved functional is $\lambda_1 + \lambda_{2,s} / \nu$, while the raw secondary-coordinate peak is $\max(10, 9 + 19\nu)$. The two transformer port products cancel and it stores no energy. An ideal gyrator has $v_a = R_g i_b$, $v_b = -R_g i_a$, so its two inward powers likewise sum to zero. The mapping $q = -\lambda / R_g$, $C = L / R_g^2$ preserves the corresponding component energy and gives canonical charge peak $28 / R_g$. It changes units and components; it does not supply a new invariant absolute sum. Integrate both two-port powers on the same interval before using their cancellation.

Orthogonal common/differential or polyphase transformations preserve the corresponding quadratic form but not a componentwise absolute sum. Physical branch states must be reconstructed before interpreting stress. Hidden zero-sequence motion requires neutral or internal sensors; adding parasitic RL or RLC branches adds physical states, while a capacitor alone does not define a resonance. A reduced passive terminal model need not preserve any internal magnitude sum. Phase imbalance, noncommensurate harmonics, nonlinear or asymmetric parasitics and sensor placement require separate identification.

An ordered sequence of component signs, with zeros retained as boundary events, is invariant under positive diagonal scaling. It is generally changed by modal mixing or affine observer shifts and reverses under time reversal. Winding counts of an open observation path are not general invariants. A closed-path topological statement requires a specified excluded point, orientation, physical frame and homotopy class. These qualifications matter when a proposed higher-order signature is based on sign crossings rather than on the operator itself.

14.5 Interruption, perfect coupling and a third-order clamp

Two mutually coupled windings with a secondary capacitor have inductance matrix $\mathbf{L} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix}$, $\Delta = L_1 L_2 - M^2 > 0$. During a constant opposing primary clamp voltage $V_c > 0$,

$$\mathbf{L} \begin{pmatrix} \dot{i}_1 \\ \dot{i}_2 \end{pmatrix} = \begin{pmatrix} -V_c \\ v_C \end{pmatrix}, \quad C \dot{v}_C = -i_2. \quad (147)$$

The system has three state coordinates. Eliminating the secondary variables gives

$$C \Delta \dot{i}_1^{(3)} + L_1 \dot{i}_1 + V_c = 0. \quad (148)$$

For $i_1(0) = I$, $i_2(0) = v_C(0) = 0$, put $\omega^2 = L_1 / (C \Delta)$. The exact solution is

$$v_C(t) = \frac{M V_c}{L_1} (\cos \omega t - 1),$$

$$\begin{aligned}
i_2(t) &= \frac{CMV_c\omega}{L_1} \sin \omega t, \\
i_1(t) &= I - \frac{V_c t}{L_1} - \frac{M^2 V_c}{L_1 \Delta \omega} \sin \omega t.
\end{aligned} \tag{149}$$

At $M = 0$ the primary has first-order dynamics, so the third-order annihilator is not minimal. Otherwise opening the primary at its first current zero switches to the two-state secondary LC interval. The capacitor voltage and secondary current are transported continuously under the declared ideal opening; this is an actual change of governing order.

The store is $E = (L_1 i_1^2 + 2M i_1 i_2 + L_2 i_2^2 + C v_C^2)/2$. The sole external power during the ideal clamp is $-V_c i_1$. Its integral to any t in that interval is

$$W_c = -V_c I t + \frac{V_c^2 t^2}{2L_1} + \frac{M^2 V_c^2}{L_1 \Delta \omega^2} (1 - \cos \omega t), \tag{150}$$

and independently evaluating (149) in E verifies $\Delta E = W_c$. This assigns work to the clamp; it does not decide whether real switch hardware returns it to a supply, heats, arcs or radiates.

An alternative rapid primary-current removal preserves secondary linkage only under a declared limiting topology. If capacitor voltage remains continuous and secondary winding voltage has no impulse, then $L_2 \Delta i_2 + M \Delta i_1 = 0$. From the canonical initial state this gives $i_2^+ = MI/L_2$ after $i_1^+ = 0$. Direct integration along the finite limiting path assigns the remaining outward work $(L_1 - M^2/L_2)I^2/2$ to the switch boundary. A different clamp, parasitic capacitor or edge sequence can change the partition. A prescribed current reset is not the same model as a finite constant-voltage clamp.

At perfect coupling $\Delta = 0$, write $L = aa^T$ and let $n^T a = 0$. For a connection vector b , the descriptor equation $L\dot{i} = b v_C$ imposes $n^T b v_C = 0$. If $n^T b \neq 0$, compatibility requires $v_C = 0$ and its derivative imposes the corresponding current constraint. An inconsistent initial state is not made admissible by using an inverse matrix at nearby coupling. A named parasitic path must select any impulsive projection and its work. For one constraint $i_1 = i_2$ and $a = (1, \sqrt{19})^T$, conserving $a^T i$ from $(10, 0)$ gives the compatible common current $10/(1 + \sqrt{19})$; this is conditional on that projection law, not a universal perfect-coupling limit.

Motion can itself introduce further states and coupling terms. For a position-dependent symmetric inductance matrix $L(q)$, a consistent winding law and magnetic force are

$$v = Ri + L(q)\dot{i} + L_q(q)i\dot{q}, \quad F_{\text{mag}} = \frac{1}{2}i^T L_q(q)i. \tag{151}$$

Consequently $\dot{E}_{\text{mag}} = i^T v - i^T Ri - F_{\text{mag}}\dot{q}$ for $E_{\text{mag}} = i^T L(q)i/2$. Coupling a mechanical mass adds displacement and momentum, and the same force–velocity product enters its store with the opposite sign. Keeping the motional voltage while changing or omitting this reaction force leaves an unmatched force–velocity contribution. A controlled nonreciprocal realization must establish that transfer and its supply independently. A permanent magnet held on a prescribed path needs a holding/mechanical port; a relaxing material coordinate adds memory and its conjugate loss, as in (116). A near-match of two current traces does not establish equality of their forces, stores or signed work.

Spatial field cancellation is a different observation from temporal state order. A local gradient and a distant field can change independently with winding geometry and orientation. A finite-core dipole model or a finite spatial integration domain does not identify the field store of surveyed windings and magnetic material. Local/exterior geometry, core radius, field boundaries and calibrated internal sensors remain

necessary before promoting a terminal or coordinate magnitude into a component rating.

Two coupled LC resonators have four continuous states. A memoryless threshold gap switches their graph without adding a continuous state; a gap-conductance evolution equation adds one. A distributed winding has further field modes, so even an exact finite clamp law does not validate a lumped description of an interruption front. Winding geometry, mutual orientation, arc ignition and restrike, winding capacitance, clamp release, core material and exterior-field ports remain unresolved physical inputs. Zero capacitance, zero switching time and simultaneous perfect coupling must be analyzed separately. Interruption work can depend on their relative limiting rates.

14.6 A loaded transformer is a three-state physical model

An explicit finite transformer circuit clarifies the difference between a voltage reference and an attached return. Take $L_2 = n^2 L_1$, $M = \sigma \kappa n L_1$, $\sigma = \pm 1$, $|\kappa| < 1$, winding resistances R_1, R_2 , source resistance R_s , core conductance G_c , output capacitance C_o and load/probe conductances G_ℓ, G_p . Let $\mu = 1$ for a tapped connection and $\mu = 0$ for an isolated secondary. The equations are

$$\begin{aligned} v_a &= \frac{u - R_s(i_1 - \mu i_2)}{1 + R_s G_c}, \quad i_s = i_1 - \mu i_2 + G_c v_a, \\ \mathbb{L} \begin{pmatrix} \dot{i}_1 \\ \dot{i}_2 \end{pmatrix} &= \begin{pmatrix} v_a \\ v_o - \mu v_a \end{pmatrix} - \text{diag}(R_1, R_2) \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}, \\ C_o \dot{v}_o &= -i_2 - (G_\ell + G_p) v_o. \end{aligned} \quad (152)$$

This has generic scalar order three. The complete store is $i^T \mathbb{L} i / 2 + C_o v_o^2 / 2$; its separately integrated inward powers are $u i_s$, $-R_s i_s^2$, $-R_1 i_1^2$, $-R_2 i_2^2$, $-G_c v_a^2$, $-G_\ell v_o^2$ and $-G_p v_o^2$. Substitution proves their sum equals the store derivative. A drive-to-zero event is not an opened primary conductor, and removal of the resistive load does not remove the output capacitor. Finite state continuity gives zero ideal impulse under those specific switch actions; omitted switch-control work remains a model limitation.

Changing a voltage origin is a coordinate operation. Connecting a finite probe or a different return changes G_p , μ or the circuit graph and therefore the trajectory and order conditions. A common voltage shift leaves the total terminal power unchanged only when total terminal current is zero; it need not preserve individual terminal contributions. Linkage reconstruction requires $\lambda_j(t) = \lambda_j(t_0) + \int_{t_0}^t (v_j - R_j i_j) dt$ and an independently known initial linkage. Perfect coupling alone does not produce an ideal transformer if finite magnetizing dynamics remain. For a pair of winding voltage integrals $\Psi_j = \int v_j dt$, the ideal ratio discrepancy satisfies exactly

$$\Psi_2 - \sigma n \Psi_1 = \Delta(\lambda_2 - \sigma n \lambda_1) + \int (R_2 i_2 - \sigma n R_1 i_1) dt. \quad (153)$$

Linkage endpoints and copper integrals are evaluated independently. The ideal comparison is $v_2 = \sigma n v_1$ in these winding orientations; reversing a winding changes σ , not the physical conclusion. This volt-second discrepancy is not an energy residual. Physical grounding, probe common-mode paths, winding-to-chassis capacitance, core saturation and remanence, frequency-dependent loss and thermal evolution require more states or ports. Zero source resistance, zero output capacitance, zero turns ratio, an exact output short, tap commutation and prepared nonzero states require separate constraints and finite switching laws. Appendix G derives specific constrained and prepared limits; arbitrary joint limits remain open.

A mechanical gear ratio and a winding ratio may share a kinematic relation without sharing dynamics. Under $V = \alpha \omega$, $i = \tau / \alpha$, rotational inertia maps to capacitance J / α^2 , not automatically to inductance.

In a rotating frame, kinetic energy transforms as $E' = E - \Omega H_z + \Omega^2 I_z/2$ for rotation about a fixed axis. Its derivative includes the frame-rate and angular-momentum terms; a moving frame can change shaft, support and field powers separately. Euler and centrifugal terms must be retained when their assumptions apply. Thus a readout reference or gear count cannot identify a higher-order transformer realization. The finite correspondence below establishes a specified elastic network and physical reference cell. It leaves broader joint-force identification, non-rigid extensions and measured supply correspondence open.

14.6.1 Loading and return paths as measurable work questions

How does the separately integrated source, winding and receiver work change when a physical return or probe is connected, and what residual survives the complete finite-state account [OP-TRF-01]? A low-voltage pair of coupled coils with a finite source resistor, load and output capacitor provides accessible winding voltages, shunt currents and capacitor endpoints. Hold the winding orientation, component values, initial currents and capacitor voltage fixed while changing a declared load or return. Integrate each of the seven powers accompanying (152) over the same finite interval; evaluate both magnetic and electric endpoint stores independently. Current preparation or initial linkage must be measured, not inferred from the final work balance. Simultaneous low-frequency acquisition makes polarity, phase and resistor heating accessible with ordinary instruments.

Zero commanded source voltage and an opened conductor are different conditions. An opened winding needs a finite commutation path; it is not a substitution into the closed-circuit equations. An observed load-work increase may accompany greater source work, release of initially stored energy, or a resolved residual. Their signs and endpoint changes discriminate these alternatives. A probe-induced trajectory change is a physical loading result, whereas a common voltage-origin shift with zero total terminal current leaves total power unchanged. Neither the volt-second comparison nor a load-only integral answers the energy question.

Can independently calibrated winding and probe parameters predict those local works and endpoint states across both finite-return configurations [OP-TRF-07]? Keep parameter identification separate from the final conditions and retain the full joint uncertainty in L_1, L_2, M when evaluating $i^T Li/2$. Section 16.2 supplies channel-level bounds. Agreement within them supports this specified realization; a signed excess outside them remains an open result. Saturation, chassis current, thermal drift and an unmeasured initial flux remain distinct hypotheses requiring their own observations.

14.7 A finite physical reference and its supply

Which independently measured supply transfers realize a moving reference in a finite compound electromechanical correspondence [OP-TRF-12]? A coordinate change by itself supplies none. Begin with a physical capacitor cell: a node V_A and driven reference V_R bound a capacitor $C = J/\alpha^2$, where J is an inertia and α converts angular velocity to voltage. Let $w = V_A - V_R$, $V_R = \alpha\Omega$, and install two actual current-source ports,

$$I_\tau = \tau/\alpha, \quad I_r = -C\dot{V}_R, \quad C\dot{w} = I_\tau + I_r, \quad E_C = \frac{1}{2}Cw^2. \quad (154)$$

The compatible map $V_A = \alpha\omega$ gives $J\dot{\omega} = \tau$. The separately integrated capacitor powers are $wI_\tau = \tau(\omega - \Omega)$ and $wI_r = -J\dot{\Omega}(\omega - \Omega)$. These are declared source ports and reproduce the accelerating-frame terms exactly. Omitting I_r defines a different physical cell. With $J = \alpha = 1$ in consistent units, $\tau = 0$, $\Omega = 2t$ and zero initial velocities on $[0, 1\text{ s}]$, the compensated cell has $w = -2t$, $W_\tau = 0$, $W_r = 2\text{ J}$,

and independently $E_0 = 0$, $E_1 = 2$ J. The uncompensated floating cell instead has $w = 0$ and zero store throughout. Each model has a zero full residual; they are different trajectories and different sources.

A separate finite reference driver obeys $C_d \dot{r} = j_d = (u - r)/R_d$, with $r = V_R$. Its capacitor and resistor are physical even when the ideal floating receiver draws zero net bias current. For a finite linear ramp on $[0, T]$, prescribe $r = r_- + \Delta r t/T$ and $u = r + R_d C_d \Delta r/T$. Enclosing only the driver capacitor gives the independently evaluated account

$$\begin{aligned} W_u &= \int_0^T u j_d dt = C_d r_- \Delta r + \frac{1}{2} C_d (\Delta r)^2 + \frac{R_d C_d^2 (\Delta r)^2}{T}, \\ W_{R_d} &= - \int_0^T R_d j_d^2 dt = - \frac{R_d C_d^2 (\Delta r)^2}{T}, \\ E_{d,0} &= \frac{1}{2} C_d r_-^2, \quad E_{d,1} = \frac{1}{2} C_d (r_- + \Delta r)^2, \quad r_d = E_{d,1} - E_{d,0} - W_u - W_{R_d} = 0. \end{aligned} \quad (155)$$

There is no ideal event within the interval; extending the ramp requires an explicit adjoining drive. The cell's compensation-source work is separate from this driver account, not supplied for free by its bias node.

For $C_d = 1$ mF, $R_d = 100 \Omega$, $r_- = 0$ and $\Delta r = 1$ V, a one-second ramp gives $j_d = 1$ mA, $W_u = 3/5000$ J, $W_{R_d} = -1/10000$ J, and $E_{d,1} = 1/2000$ J. A half-second ramp gives $j_d = 2$ mA, $W_u = 7/10000$ J and $W_{R_d} = -1/5000$ J, with unchanged capacitor endpoints. These finite voltage, current and timing choices are set before the work audit. Ordinary voltage and shunt channels directly distinguish duration-dependent works from duration-independent endpoint change. Independently observe the compensation-source outputs and their supplies when adding the receiver; their work is not the driver remainder.

The unit-scale reference-cell example also permits the distinct driver choice $C_d = 1/200$ F, $R_d = 3/4 \Omega$, $r = 2t$. It gives $j_d = 1/100$ A, $W_u = 403/40000$ J, $W_{R_d} = -3/40000$ J, and endpoint change $1/100$ J. None of these equals the receiver's 2 J compensation work. This separates three boundaries that a formal change of reference can conceal.

14.7.1 Compound states, controls and physical boundaries

A finite compound network has nodes A, B, C, P and four oriented winding branches $A \rightarrow C$, $P \rightarrow C$, $B \rightarrow C$, $P \rightarrow C$. Let B be its node-branch incidence matrix, positive at each branch origin, C its positive diagonal node-capacitance matrix and j its external node-current vector. For two coupled pairs $X = A, B$, take signed nonzero ratios $h_X, L_0, R_0 > 0$ and $|\kappa| < 1$:

$$\begin{aligned} C \dot{v} &= j - B i, & L \dot{i} &= B^T v - R i, \\ L_X &= L_0 \begin{pmatrix} h_X^2 & \kappa h_X \\ \kappa h_X & 1 \end{pmatrix}, & R_X &= R_0 \text{diag}(h_X^2, 1). \end{aligned} \quad (156)$$

The pair matrices form block diagonals. Four node voltages and four winding currents give eight independent continuous states when all nodes are dynamic. Holding one node at a compatible ideal reference removes one capacitive coordinate and gives seven; several capacitors sharing a node contribute stores without creating additional independent voltages. Generic scalar order is at most the state dimension and still depends on controllability, observation and initialized cancellations.

With $\omega = v/\alpha$, $z = Li/\alpha$, $J = \alpha^2 C$, $K = \alpha^2 L^{-1}$ and $\tau = \alpha j$, the mechanical equations are

$$J\dot{\omega} = \tau - BKz, \quad \dot{z} = B^T \omega - \frac{R}{\alpha^2} Kz. \quad (157)$$

This is an exact finite inertia–elastic-state correspondence, not a rigid ideal gear. Its store is both $v^T C v/2 + i^T L i/2$ and $\omega^T J \omega/2 + z^T K z/2$. Each node power $v_X j_X = \omega_X \tau_X$ and each winding heat $-R_b i_b^2$ is separately integrated. Direct differentiation proves the full conditional residual zero. Shaft and mesh diagnostics alone omit inertia, elastic and resistive accounts; a support constraint can carry force or torque while doing zero work at its held coordinate.

For example, a carrier source behind R_s and two receiver paths, one to ground and one to the carrier, contribute $u i_s$, $-R_s i_s^2$, $-G_g v_F^2$ and $-G_n (v_F - v_C)^2$. Four winding losses, a probe $-G_p v_F^2$ and a reset path $-G_r v_F^2$ complete ten separately integrated powers. Retain distinct preparation, operation, receiver-change, relaxation and reset intervals with independently evaluated endpoint stores. Changing finite conductances with all current paths retained leaves the states continuous and has no ideal impulsive work. Rewiring a charged capacitor or opening an energized winding is a different event.

For a physical common reference r , let $w = v - \mathbf{1}r$ and install four compensation currents $-\mathbf{C}\mathbf{1}\dot{r}$. Then

$$\mathbf{C}\dot{w} = j - \mathbf{B}i - \mathbf{C}\mathbf{1}\dot{r}, \quad L\dot{i} = \mathbf{B}^T w - \mathbf{R}i. \quad (158)$$

Since $\mathbf{B}^T \mathbf{1} = 0$, this reproduces (156) for $v = w + \mathbf{1}r$ from compatible initial data. Each receiver capacitor sees a node-source power $w_X j_X$ and a compensation power $-C_X w_X \dot{r}$. The reference conductor's net ideal bias current is zero by node balance, not by assuming these individual powers vanish. A prescribed $r(t)$ is distinct from a state-dependent command $r = \chi(t)v_S$: the latter needs the actual selected node state and $\dot{r} = \chi \dot{v}_S + \dot{\chi} v_S$. Its exact algebraic tracking relation restricts initialization and is not an additional free state.

Finite-bandwidth sensors, inductive current actuators, the command capacitor, reference-driver capacitor and supply rail replace that constraint by dynamical equations. Appendix F gives a 28-state realization with separately declared receiver, actuator, sensor, driver, converter and rail accounts. Its finite capacitances, current limits and conversion losses need not preserve ideal reference tracking or the ideal scalar order. At supply cutoff its remaining stores do not disappear. A resolved local residual survives even if receiver and supply remainders cancel in a combined account.

The capacitor-cell correspondence and finite ramp integrals are established exact model results. The physically unresolved questions concern actual compensation currents, reference tracking, independent supply work, initial-state preparation, nonideal return paths and signed local residuals. A compound measurement adds independently calibrated node and winding channels to the same volt/current/endpoints approach; it cannot borrow the small cell's uncertainty bound. Finite apparatus limits, nonlinear driver events and general electromechanical correspondence remain open. The exact constrained and prepared limits in Appendix G delimit, rather than erase, these questions.

14.8 Contact applications with additional states

A rigid teeterboard with known actuator forcing can be represented by angle and rate. Adding a bending amplitude adds a second-order mode; adding a finite activation time adds a first-order state. Leg compression, changing lever arms and unilateral contacts then produce a nonlinear hybrid system. A model that imposes the measured endpoint cannot independently validate the launch. Timing, recoil, support force and actuator/leg extension should be predicted on separate conditions. Pivot and stop impulses belong in

the momentum ledger even when their ideal work vanishes. Whether fixture coupling, bending or actuator delay explains a discrepancy remains an identification question.

For a club–ball encounter with impact normal n and offset r from the head center of mass, rigid-body mechanics gives

$$\frac{1}{m_{\text{eff}}} = \frac{1}{m_h} + (r \times n)^T I_h^{-1} (r \times n). \quad (159)$$

In a planar case with offset d and radius of gyration k_g , this is $m_h/(1 + d^2/k_g^2)$. The normal impulse with restitution e is $J = (1 + e)v_{\text{rel}}/(1/m_{\text{eff}} + 1/m_b)$ for initially closing contact and a declared fixed normal. A compliant realization resolves its duration and signed powers. Shaft bending in two planes, torsion and longitudinal motion add independent modal states or a distributed impedance. A handle-response exclusion requires comparing contact duration with ℓ/c_{mode} for each relevant wave class. A vibration node, center of percussion and a wave-arrival condition are different mechanisms. Grip wrench, acceleration, jerk and impulse must be referred to colocated channels. Modal convergence, grip compliance, acoustic loss and actual hardware discrimination remain unresolved.

A hammer–nail boundary can accumulate penetration. A depth-memory coordinate such as $s_{\text{max}}(t) = \max_{u \leq t} s(u)$ distinguishes advancing and retreating force laws and cannot be replaced by an elastic spring without changing the model. Support impedance can reduce relative deformation without absorbing work at a fixed support. For a rod, the ratio of transfer duration to round-trip wave time, $\kappa_t = t_{\text{transfer}}/(2\ell/c)$, distinguishes a slow regime from a first-transient regime requiring distributed or modal dynamics. Rate dependence, friction, plasticity, orientation under matched impact speed and finite support motion require explicit laws. A small striker mass ratio alone does not identify the penetration work or the derivative order.

Reversible, accumulating and terminal internal states are likewise distinct. A fuse, fracture or threshold device changes its connection graph when an internal condition is met. Transmitted, reflected and diverted outputs then depend on the post-event topology. A released store requires separately integrated finite event ports or an independently justified impulsive limit; a partition residual cannot be assigned by subtraction and called a measured destination. Failure-threshold distributions, nonmonotone damage histories and fragment or acoustic partition remain open. A single fixed-order smooth ODE cannot replace an unspecified threshold law.

Across these applications, simultaneous contacts, initially touching edges, chatter, support-port disappearance and terminal thresholds can make event ordering consequential. Four distinct discrepancies should be kept: the equation residual, the port-integration residual, the constitutive/model residual and the independently evaluated endpoint balance residual; impulse and partition identities add their own checks. Exact smooth-interval formulas do not settle convergence or physical meaning at an unresolved event. Accessible measurements differ by apparatus, and no hardware evidence is inferred from a proposed model.

15 Signed physical work and equation-term identities

The coefficient laws and physical trajectories are specified before the audits in this section. Signed work tests their declared boundaries, ports and retained stores. An equation-term identity has a physical interpretation only when the realization supplies it; no work outcome selects coefficient weights or derivative order.

15.1 Boundary, event sides and residuals

For a declared system boundary, positive power enters through a port. An electrical port has $P = vi$, a translational port $P = Fv$, and a rotational port $P = \tau\omega$, with conjugate signs fixed together. On each smooth interval $[t_0, t_1]$ define

$$W_\ell(t_0, t_1) = \int_{t_0}^{t_1} e_\ell(t) f_\ell(t) dt, \quad r_E = E(x(t_1)) - E(x(t_0)) - \sum_\ell W_\ell. \quad (160)$$

Each signed port integral is evaluated before summation. End stores come independently from the component states. Switching work is evaluated on a finite regularization or as a separately justified impulse; one must distinguish t_e^- and t_e^+ . A fixed support can deliver impulse with zero work because its velocity is zero. Nonzero reaction force does not imply energy transfer.

For the exact integrations below, the numerical integration residual is exactly zero. The endpoint balance residual r_E is still evaluated for the ports actually declared: an unassigned event or coupling contribution may make it nonzero. The model discrepancy r_{model} names what a physical implementation adds or changes, including constitutive error and unresolved event destinations; its value is unknown unless a stated assumption determines it. A surviving discrepancy remains open with its sign, magnitude and conditions of occurrence. Reversing a sensor orientation requires reversing the associated incidence map. Voltages and currents from different connection graphs or different event sides cannot be paired without proving a common physical port.

15.2 The collision ledger

During (15) with $u = 0$, let $K_h = J_h \dot{\theta}_h^2/2$, $K_a = J_a \dot{\theta}_a^2/2$, $U_j = k_j \theta_a^2/2$. Define positive transfer powers

$$P_{hc} = \tau_c \dot{\theta}_h, \quad P_{ca} = \tau_c \dot{\theta}_a, \quad P_{\text{rel}} = \tau_c \dot{\delta}, \quad P_j = \tau_j \dot{\theta}_a. \quad (161)$$

These orientations denote transfer out of the hammer, into the anvil, into the contact deformation, and into the joint, respectively. Direct integration of the separate products yields

$$\begin{aligned} W_{hc} &= -\Delta K_h, & W_{ca} &= \Delta K_a + W_j, \\ W_{\text{rel}} &= \Delta U_c + \int_{t_0}^{t_1} d_c \dot{\delta}^2 dt, & W_j &= \Delta U_j + \int_{t_0}^{t_1} d_j \dot{\theta}_a^2 dt. \end{aligned} \quad (162)$$

For the complete two-body, two-spring boundary, the external loss powers are $-d_c \dot{\delta}^2$ and $-d_j \dot{\theta}_a^2$; contact transfer cancels internally. If a drive acts, its separate inward power is $u \dot{\theta}_h$. Initial preparation belongs to an earlier interval and cannot be omitted when comparing a complete operating cycle.

Return is retained by splitting W_{hc} into the integrals over $P_{hc} > 0$ and $P_{hc} < 0$. They have opposite signs and need not be small separately. If the hammer reaches its first zero speed monotonically from the illustrative initial state of (17), its outward contact work to that instant is exactly 144/5 J. A later return changes the final hammer store; it does not invalidate the earlier signed integral.

15.3 The unresolved store at contact release

The local impact-driver question in Section 3.7 establishes the positive remaining store in (33) and the negative deletion residual in (34). Here the same declared event is extended to destination accounts, repeated contacts and a finite observation window. No release law or destination follows from the coefficient representation alone.

Assigning $W_{\text{release}} = -U_c(t_e^-)$ makes an augmented mathematical ledger close. It specifies the transfer that would be needed; it does not identify that transfer. Four exclusive counterfactual accounts can assign the whole amount to acoustic radiation, heat, fracture or fixture motion. They are alternatives, not four simultaneous works, and cannot be summed or averaged into an identified partition. A mixed physical destination requires separately evaluated channel works and a remaining unassigned part. A finite release path must supply each effort–flow product, including acoustic pressure times volume flow, support torque times angular speed, or a specified thermal or fracture transfer.

For several deleted contact stores, the event contribution is exactly

$$r_{E,\text{gates}} = - \sum_j U_c(t_j^-). \quad (163)$$

Motor work, damper work and fixture transfer on the intervening finite intervals are integrated separately. A fixed fixture has zero work and can still supply the impulse needed in the momentum account. At deformation-zero release the elastic store is zero, but the Kelvin–Voigt force may already have become tensile. A compressive-only law needs its own state and release definition; force clipping alone does not specify what happens to stored deformation. The nonlinear potential in (28) gives the corresponding event deficit $-U_c(t_e^-)$ under the same deletion assumptions. Restitution above unity requires evaluating any prepared spring change and active-port work on the stated interval rather than deciding its interpretation from the restitution value alone.

A slower compliant torsional contact with measured torque, angles and spring deflection can first distinguish retained deformation from an imposed state deletion. Ordinary encoders and a torque sensor can resolve that limited question without measuring a short acoustic pulse. Fixture transfer requires synchronized support torque and speed. Full acoustic, thermal or fracture attribution requires its own calibrated transfer measurement; a microphone amplitude or temperature change alone does not determine the corresponding work. Let A_{out} be the positive outward work magnitude obtained from independent signed channel integrals, and let $E_{\text{ret}} \geq 0$ be an independently evaluated retained store replacing the deleted coordinate. If the other event-side stores remain unchanged, the remaining residual is

$$r_{E,e} = E_{\text{ret}} - U_c(t_e^-) + A_{\text{out}}. \quad (164)$$

All additional measured store changes must be included if that continuity assumption fails. Apply the uncertainty enclosure in Section 16.1 to determine whether a deficit remains. The destination and criterion stay open when no such independent account is available; difficulty measuring them does not remove the problem.

For a finite observation window $[t_e - h_-, t_e + h_+]$, with positive h_-, h_+ , record both event sides and integrate the regular drive, support and loss powers separately on each side. Retain contact transfers only in the corresponding subsystem accounts. A bound $|P_\ell| \leq M_\ell$ gives a work bound $M_\ell \delta t$ for an uncertain interval endpoint displaced by at most δt ; a separately justified impulse retains its own uncertainty. Apply the state-error enclosure of Section 16.2 to the spring deflection and stiffness. Resolving $U_c(t_e^-) > 0$ at force zero is the first attainable outcome. A full destination account additionally requires retained-state changes and each outward channel; a measured temperature rise alone does not determine thermal work across the chosen boundary.

15.4 What multiplication by a derivative does, and does not, prove

Multiplying (2) by $\dot{y}$ gives $(a\dot{y}^2/2 + c\dot{y}^2/2)' = f\dot{y} - b\dot{y}^2$. This is a physical work identity only when the variables and coefficients belong to a declared realization. Multiplication of an arbitrary equation by a constant rescales every term integral without changing its solutions.

For a homogeneous fourth-order scalar equation, integration by parts gives the exact identity

$$\begin{aligned} Q &= \frac{B_0 y^2}{2} + \frac{B_2 \dot{y}^2}{2} + B_3 \ddot{y} \dot{y} + B_4 \left(y^{(3)} \dot{y} - \frac{\ddot{y}^2}{2} \right), \\ \dot{Q} &= -\Pi, \quad \Pi = B_1 \dot{y}^2 - B_3 \ddot{y}^2. \end{aligned} \quad (165)$$

For $B_1, B_3 > 0$, this identity predicts increasing Q precisely when $B_3 \ddot{y}^2 > B_1 \dot{y}^2$. In particular, a velocity zero with nonzero acceleration gives $\dot{Q} > 0$. Its effective-rate zeros satisfy $\ddot{y}/\dot{y} = \pm \sqrt{B_1/B_3}$ when those expressions are defined; they need not coincide with an internal component's zero power. Meanwhile the undriven collision store obeys $\dot{E} = -d_c \delta^2 - d_j \dot{\theta}_a^2$. The formal increase and the physical store change are distinct results to compare. Appendix B derives the arbitrary-order identity.

The physical comparison can be made directly in scalar-jet coordinates. For $q = \theta_h$, put $a_1 = J_h \ddot{q} + k_c q + d_c \dot{q}$ and $a_2 = J_h q^{(3)} + k_c \dot{q} + d_c(1 + J_h/J_a) \ddot{q}$. Inverting (29) when $\Delta \neq 0$ gives the reconstructed angle and rate

$$\begin{aligned} q_a^* &= \frac{(k_c - d_c d_j/J_a) a_1 - d_c a_2}{\Delta}, & w_a^* &= \frac{(d_c k_j/J_a) a_1 + k_c a_2}{\Delta}, \\ E_{\text{jet}} &= \frac{1}{2} J_h \dot{q}^2 + \frac{1}{2} J_a (w_a^*)^2 + \frac{1}{2} k_c (q - q_a^*)^2 + \frac{1}{2} k_j (q_a^*)^2. \end{aligned} \quad (166)$$

This is the positive physical store transported through the invertible state map, not the integration-by-parts form Q . Its derivative follows the component balance because the map reconstructs both original state equations. On $\Delta = 0$, the hidden state must instead be retained independently. The unresolved realization question is which declared normalization, state map and physical port, if any, makes a higher-derivative boundary form an accessible store or transfer. Its indefinite form, arbitrary equation scaling and lack of an identified component assignment constrain that question. They do not remove the formal sign reversal or supply its physical attribution.

The physical mobility in (16) is passive because the independently derived state balance has nonnegative dissipation. For the artificial mobility $s/P(s)$, writing $P(s) = \sum_{k=0}^4 p_k s^k$ instead gives

$$\text{Re} \frac{i\omega}{P(i\omega)} = \frac{\omega^2(p_1 - p_3 \omega^2)}{|P(i\omega)|^2}. \quad (167)$$

It is negative above $\sqrt{p_1/p_3}$. A stable denominator can thus be used in a nonpassive port map. Stability, positive-realness and a component realization are distinct assertions.

For a real rational impedance with no poles in $\text{Re } s > 0$, positive-realness requires $\text{Re } Z(s) \geq 0$ there, including the appropriate nonnegative residues at allowed imaginary-axis poles. Nonnegative real part on a finite frequency interval is not a global certificate. A cubic $ds^3 + as^2 + bs + c$ with positive coefficients is Hurwitz exactly when $ab > dc$. Hence $1 + s + s^2 + \alpha s^3$ is strictly stable for $0 < \alpha < 1$, marginal at $\alpha = 1$, and unstable for $\alpha > 1$; for $\alpha < 0$ continuity on the positive real axis already gives a positive real root. At $\alpha = 0$ the order changes, so the extra initial derivative is no longer free.

16 Signed residual outcomes and measurement uncertainty

A comparison of synthesized coefficient laws requires independently bounded observation errors. Physical-accounting comparisons additionally require all signed works and endpoint states. The following enclosures identify the conclusions that those distinct measurements can support.

16.1 Which residual survives the independent account?

A positive or negative energy residual is reported for the declared boundary and interval before any additional transfer is assigned. For each case retain the initial and final constitutive stores, every separately integrated signed work, event-side states, and the checks on polarity, timing, dimensions and constitutive assumptions. The exact contact deficit (34) and coupling residual (127) are nonzero examples with zero numerical integration error. An independently demonstrated transfer or store correction changes the account by its own signed contribution; report the remaining residual as well.

For measurement estimates $\widehat{E}_j, \widehat{W}_\ell$, suppose calibration and the declared observation model establish $|\widehat{E}_j - E_j| \leq \epsilon_{E_j}$ and $|\widehat{W}_\ell - W_\ell| \leq \epsilon_{W_\ell}$. Then

$$\begin{aligned}\hat{r}_E &= \widehat{E}_1 - \widehat{E}_0 - \sum_{\ell} \widehat{W}_\ell, \\ |\hat{r}_E - r_E| &\leq \epsilon_{E_0} + \epsilon_{E_1} + \sum_{\ell} \epsilon_{W_\ell} \equiv \epsilon_r.\end{aligned}\tag{168}$$

The proof is the triangle inequality; it requires no statistical independence. Correlated error sets can give a tighter enclosure when justified. Include event-time and event-side uncertainty in the corresponding store and work bounds. A covariance matrix alone supplies no deterministic bound. Unquantified omitted physics remains an unquantified model limitation, rather than an adjustable allowance chosen after seeing the residual.

Table 15. Signed residual outcomes. Each result retains its interval, preparation, ports and model scope.

Exact condition on the estimated residual interval	Outcome for the declared boundary and assumptions
$\hat{r}_E - \epsilon_r > 0$	Resolved positive residual
$\hat{r}_E + \epsilon_r < 0$	Resolved negative residual
$\hat{r}_E - \epsilon_r \leq 0 \leq \hat{r}_E + \epsilon_r$	Sign unresolved at the stated resolution

If no independently established transfer or store change accounts for a resolved excess, retain it as an open problem with magnitude, sign, conditions and completed checks. A residual interval containing zero gives a bound at that resolution; it is not a missing-destination assignment. Finite source, switching and thermal accounts can fail locally even when a combined interval appears to close. For contiguous intervals with consistent shared endpoint states, the residuals add exactly, so opposite local residuals can cancel. For nested boundaries, cancellation of a declared internal transfer likewise does not certify either subsystem's endpoint reconstruction.

Residual sign patterns are worth investigating, but their interpretation needs an error model. A gain error with a fixed sign can bias every integrated work in a family in the same direction. More generally, a deterministic integration discrepancy need not be symmetric. For an exact algebraic illustration, the

equal-subinterval trapezoidal expression for $f(t) = at^2$, $a > 0$, on $[0, T]$ satisfies

$$\frac{T}{n} \left[\frac{f(0) + f(T)}{2} + \sum_{j=1}^{n-1} f(jT/n) \right] - \int_0^T f(t) dt = \frac{aT^3}{6n^2} > 0. \quad (169)$$

This identity follows from $\sum_{j=1}^{n-1} j^2 = (n-1)n(2n-1)/6$; it is an exact finite-sum control, with no numerical integration performed. A proposed sign test therefore needs a justified null distribution and dependence model. Repeated or shared trajectories are not automatically independent trials. Sign bias alone does not identify a physical transfer, and sign symmetry alone does not resolve a discrepancy. Retain both the sign investigation and each signed residual until its uncertainty or physical account has been established.

16.2 From instrument channels to work and endpoint bounds

Suppose synchronized effort and flow estimates satisfy $|e - \hat{e}| \leq \epsilon_e(t)$ and $|f - \hat{f}| \leq \epsilon_f(t)$ on a fixed interval. Expand their product before applying the triangle inequality:

$$|W - \widehat{W}| \leq \int_{t_0}^{t_1} (|\hat{e}|\epsilon_f + |\hat{f}|\epsilon_e + \epsilon_e\epsilon_f) dt, \quad \widehat{W} = \int_{t_0}^{t_1} \hat{e}\hat{f} dt. \quad (170)$$

This retains the cross term and requires no independence assumption. Gain, offset, bandwidth and polarity belong in the channel model. A relative timing error bounded by δt contributes $M_f \delta t$ to ϵ_f when $|\dot{f}| \leq M_f$ on the required smooth extension. Such a bound cannot cross an unresolved discontinuity. Split at events; if regular power is bounded by M_P , moving an integration endpoint by at most δt adds $M_P \delta t$ to its work bound. Impulsive work and event-side stores require separate bounds.

For an independently calibrated capacitor, $|C - \hat{C}| \leq \epsilon_C$, $|v - \hat{v}| \leq \epsilon_v$ and $C > 0$ imply the exact endpoint bound

$$\left| \frac{1}{2} C v^2 - \frac{1}{2} \hat{C} \hat{v}^2 \right| \leq \frac{1}{2} \epsilon_C \hat{v}^2 + (\hat{C} + \epsilon_C) |\hat{v}| \epsilon_v + \frac{1}{2} (\hat{C} + \epsilon_C) \epsilon_v^2. \quad (171)$$

The same form applies to an inductor, translational mass or spring using the corresponding coefficient and state. Apply it independently at both endpoints. For coupled magnetic storage retain the full joint parameter set, including M and its sign. More generally, for $E = x^T H x / 2$, $\|H - \hat{H}\|_2 \leq \epsilon_H$ and $\|x - \hat{x}\|_2 \leq \epsilon_x$, after declaring a dimensionally consistent coordinate scaling,

$$|E - \hat{E}| \leq \frac{1}{2} \epsilon_H \|\hat{x}\|_2^2 + (\|\hat{H}\|_2 + \epsilon_H) \|\hat{x}\|_2 \epsilon_x + \frac{1}{2} (\|\hat{H}\|_2 + \epsilon_H) \epsilon_x^2. \quad (172)$$

Optimizing over a calibrated correlated set can tighten this enclosure; omitting cross-storage terms cannot. For the assumptions in Section 8.4.2, (171) bounds each capacitor endpoint by 1432301/200000000000 J. Equation (170) bounds each one-second resistor work by 2201/1000000000 J. Four endpoints and two works give (80) exactly. The example demonstrates a falsifiable resolution requirement, not observed accuracy.

Recorded-waveform integration adds its own independently bounded acquisition and quadrature error to each work. Exact analytic integrations in this article have zero numerical integration error; that does not eliminate physical calibration or omitted-model uncertainty. A residual bound is complete only for its declared boundary, all named ports, both endpoint states, event sides and observation model. Report an unknown contribution as unknown rather than choosing its size to contain the residual.

16.3 Projection residuals and remaining physical limits

A residual against a second-order projection is a question about that projection, not a new energy law. For a declared physical power P_{exc} and an independently chosen template, define

$$r_T(t) = P_{\text{exc}}(t) - [a\dot{v}(t)v(t) + bv(t)^2 + cy(t)v(t)], \quad y' = v, \quad W_T = \int_{t_0}^{t_1} r_T dt. \quad (173)$$

A dimensionless magnitude can use $\Phi_T = \int |r_T| dt / \int (|P_{\text{exc}}| + |a\dot{v}v| + |bv^2| + |cyv|) dt$ when its denominator is positive. State the normalization explicitly. Signed cancellation can make W_T small while Φ_T remains large. Neither quantity replaces (160).

A disciplined interpretation first establishes the observation and arithmetic uncertainty on the same interval and port, then considers omitted higher-order states, time-varying coefficients, declared event partitions and other actual ports. A higher-order equation that explains a residual still needs a realization. Local coefficient variation can absorb discrepancy without identifying physical parameter variation. A multi-port projection must show its dependence on the chosen port; selecting a port by integrated throughput would make work part of selection and is excluded here. A matched exact-model control has zero numerical floor. Actual instrumentation supplies its own nonzero uncertainty and cannot borrow a floor from another scale, interval or apparatus.

There are three possible failures of such a residual program: the residual may be indistinguishable from its measurement floor; every excess may already be explained by the declared finite model; or the conclusion may depend mainly on arbitrary port choice. In those cases the quantity has, respectively, no discriminating power, only a bounded audit role, or no robust cross-port interpretation. A small residual, a large residual and an unperformed comparison establish different facts. None establishes an unspecified energy destination.

Physical sensor uncertainty includes channel gain, offset, polarity, bandwidth, phase, trigger timing, loading and cross-channel covariance. For an explicitly affine observation-error model $g = g_0 + J\epsilon$ with covariance Σ of ϵ , the covariance is exactly $J\Sigma J^T$; a nonlinear observation requires propagation of the full declared error set or distribution. In particular, polarity faults are discrete alternatives, not a small Gaussian perturbation. A prior probability of reversed polarity or an alarm threshold is an illustrative assumption until calibrated. No synthetic error model establishes a real sensor distribution.

A useful exact gain example avoids linearization: if measured effort and flow are $(1 + \epsilon_e)e$ and $(1 + \epsilon_f)f$, their measured power is $(1 + \epsilon_e)(1 + \epsilon_f)ef$. With offsets and timing shifts the cross terms and phase errors must also be retained. For sinusoidal peak phasors a relative phase error $\Delta\phi$ replaces $\text{Re}(VI^*)$ by $|V||I|\cos(\arg V - \arg I + \Delta\phi)$. Near quadrature, a small phase error can reverse the sign of inferred real work. Internal branch voltage integration additionally requires an initial-flux reference and drift control. Extra sign crossings induced by noise are not physical contact events.

The unresolved boundaries are specific. Coefficient identification needs independently variable physical coordinates, realistic covariance, full interactions among partition, estimator, phase and fixture changes, and certified roots at stability or passivity boundaries. Finite frequency atlases do not cover intermediate coordinates, arbitrary signed combinations, large condition numbers or the right-half-plane interior. Duality comparisons need transported candidate sets and fully mapped ports. A general six-term combination has no certificate merely because a single cubic continuation is understood.

Preparation and history need attainable source limits, finite clamps, correlated initial states, longer history bases and well-defined simultaneous-event limits. Relaxation over widely separated time constants requires interval bounds that resolve each scale. Exact identities here do not retroactively validate

unresolved finite-precision calculations of short preparation, coincident-event work, long-base compatibility, near-neutral modes or unstable prefixes. Those are still uncertainty questions, with no numerical magnitude asserted as an exact physical result.

Driven systems additionally need material identification, thermal and sensor realizations, finite-supply return paths, pole-coalescence analysis, exact singular coupling constraints, noncommensurate forcing and closed-loop robustness. Repeated operation with heating changes resistance and can create self-limiting or runaway behavior only under a declared thermal feedback law. A cycle map $T_{n+1} = \Psi(T_n)$ has a locally stable fixed point when $|\Psi'(T_*)| < 1$; $\Psi'(T_*) = 1$ is a neutral boundary requiring higher-order analysis. A finite collection of positive-temperature states does not certify that boundary or continued sign-crossing surfaces. Preparation and reset of the full plant cannot be validated by a reduced two-store template shared by several applications.

Passive network questions still include construction throughout correlated signed-residue regions, minimal physical storage on their boundaries, available and required storage bounds, resistor-loss allocation and finite stress under nonideal transformer constraints. Distributed questions still include continuous spectral error bounds, nonmonotone finite-order improvements, delay history and fields, exterior ports and infinite-band impossibility. Mechanical questions still include order-transition surfaces, nonlinear contacts, independent inertia and stiffness ratios, more complicated re-engagement, absorber coscalings, grip and fixture loading, and thermal/acoustic release destinations. A declared finite model can resolve a local mathematical question without resolving any of these broader physical ones.

For actual validation, ordinary synchronized effort and flow channels should be accompanied by the smallest internal sensor set that separates the proposed alternatives, plus a full internal control where feasible. Keep device calibration, coefficient estimation, model choice and final conditions separate. A full-rank normalized representative map does not guarantee raw-unit identifiability, predictability under a different model family or successful physical probe placement. Coupled gain/phase drift, non-Gaussian noise, sensor saturation, component-temperature-aging dependence and geometry/material covariance require their own uncertainty specification. A finite set of parameter vertices does not enclose a continuous nonlinear region without an interval or monotonicity proof. Hardware apparatus, calibration, authority to operate it and actual observations remain external requirements; no proposed measurement is reported as completed.

17 Conclusions

Higher-order coefficient synthesis begins with a declared observation, forcing and reference chart. The two dimensional constraints give every monomial in the positive triple (a, b, c) as $A_{r,k} = S\tau^k \rho^{-r}$. They leave the dimensionless coefficient functions undetermined. A finite support, exponent class and constant weights are additional claims whose domain, normalization and source of information must be stated. Reference and duality transformations transport those coordinates; they do not select a physical operator.

The rotary impact driver makes the construction concrete. Eliminating the anvil from the two-body component laws gives coefficients depending on four independent dimensionless component ratios. Restricting the parameter domain gives the nine-atom two-coordinate and seven-atom one-coordinate identities, with their forcing operators transported under the same normalization. The distinct powers prove support requirements on an open stiffness interval. These results establish representation and coefficient prediction across their specified domains without refitting. Recovery from noisy motion measurements remains a separate identifiability question. Changing the joint supplies a prediction to check with the same law; synchronized hammer, anvil and torque observations distinguish contact from joint dynamics.

Additional internal contact or joint states require their own construction and preparation.

Adding a declared polarization branch constructs the next complete operator through $Q_{N+1} = (1 + \tau_{N+1}s)Q_N$ and $\mathcal{P}_{N+1} = (1 + \tau_{N+1}s)\mathcal{P}_N + R_{N+1}Q_N$. The resulting coefficient recurrence changes lower orders as well as the leading term and has an explicit finite representation in the dimensional family with additional independent component ratios. By contrast, the rational-series recurrence generates higher expansion coefficients of a fixed state model. Neither it nor a dimensional lattice shift establishes a new physical state merely by increasing an index.

Independent coordinates and an observation model determine what a measurement can recover. Distinct finite Laurent monomials are independent on an open domain of independent coordinates, but restricted parameter paths can make them coincide. Exact rank, practical sensitivity and prediction on independent conditions have separate requirements. The impact example shows why an internal anvil observation can distinguish preparations invisible in hammer angle and contact torque. The complementary circuit and RF proposals retain deliberate null controls and uncertainty-based separation of competing fixed laws. A good terminal waveform fit alone does not establish a weak coefficient, an exponent or a particular internal realization.

The synthesized equation's meaning also depends on its forcing and initialization. Four physical collision states can yield a third-order initialized output and a second-order zero-state transfer. A compound receiver has seven or eight independent states under its declared constraints, while a finite sensor–actuator–rail implementation has 28. A vanishing common current can retain a finite prepared store in the specified rigid family. Passive realizations can have identical terminal functions and different internal currents, stresses and preparations. Stability, port passivity and component realization are additional conditions; the direct polynomial driving-point obstruction does not exclude a passive rational higher-order realization. Distributed, fractional and delayed laws retain field or history information and require explicit domains for every finite representation.

Signed work audits the physical systems after their coefficient laws have been specified. The impact driver supplies the leading question: what remains stored, returns to the mechanism or leaves through the support when contact releases? Encoders and torque sensing on a slow torsional analogue can resolve the remaining deformation and mechanical transfer. Deleting the contact store at force-zero release gives the negative model residual $-d_c^2 \dot{\delta}(t_c^-)^2 / (2k_c)$ under the stated continuity and event assumptions. Its actual thermal, acoustic, fracture or fixture destination requires independent evidence. The unequal-coupling cubic increases its declared store by 9/2 J and has two signed loss works totaling -3 J, leaving $+15/2$ J absent from its listed ports on the stated interval. An explicitly powered reciprocal-actuator control supplies that work in its own model; the different finite-bias realization retains separate coupling, coil, DC-link, thermal and interval questions. The fourth-order integration-by-parts form has its own exact increase condition, distinct from the positive collision store reconstructed through compatible scalar jets.

Hidden RC relaxation has a complete resistor-work account despite a quiet terminal. Its unresolved observation question concerns the energy ceiling allowed by uncertain coupling and preparation. The practical controls retain separately measured internal states, actuator output and supply channels, reference-driver works at identical capacitor endpoints, and transformer winding orientations and magnetic endpoints. Resolving one internal change does not close every transfer account. State, response, work and endpoint convergence are distinct claims. A positive or negative measured residual that survives its independently established uncertainty remains open with its magnitude, conditions and completed checks. None of these audit outcomes is used to choose coefficients or derivative order.

A Complete coefficient array and elementary stability checks

For the electrical template $(a, b, c) = (L, R, 1/C)$,

$$A_{r,k} = L^{k-1+r} R^{2-k-2r} C^{-r}. \quad (174)$$

The following two tables reproduce the full array for $-3 \leq r \leq 3$ and $-4 \leq k \leq 6$. Negative derivative indices retain the initialized-history interpretation of Section 7. Every entry follows from (174); the tables introduce no selected dynamics.

Table 16. Electrical coefficients for derivative indices -4 through 1.

r	$k = -4$	$k = -3$	$k = -2$	$k = -1$	$k = 0$	$k = 1$
3	$\frac{1}{C^3 L^2}$	$\frac{1}{C^3 L R}$	$\frac{1}{C^3 R^2}$	$\frac{L}{C^3 R^3}$	$\frac{L^2}{C^3 R^4}$	$\frac{L^3}{C^3 R^5}$
2	$\frac{R^2}{C^2 L^3}$	$\frac{R}{C^2 L^2}$	$\frac{1}{C^2 L}$	$\frac{1}{C^2 R}$	$\frac{L}{C^2 R^2}$	$\frac{L^2}{C^2 R^3}$
1	$\frac{R^4}{C L^4}$	$\frac{R^3}{C L^3}$	$\frac{R^2}{C L^2}$	$\frac{R}{C L}$	$\frac{1}{C}$	$\frac{L}{C R}$
0	$\frac{R^6}{L^5}$	$\frac{R^5}{L^4}$	$\frac{R^4}{L^3}$	$\frac{R^3}{L^2}$	$\frac{R^2}{L}$	R
-1	$\frac{C R^8}{L^6}$	$\frac{C R^7}{L^5}$	$\frac{C R^6}{L^4}$	$\frac{C R^5}{L^3}$	$\frac{C R^4}{L^2}$	$\frac{C R^3}{L}$
-2	$\frac{C^2 R^{10}}{L^7}$	$\frac{C^2 R^9}{L^6}$	$\frac{C^2 R^8}{L^5}$	$\frac{C^2 R^7}{L^4}$	$\frac{C^2 R^6}{L^3}$	$\frac{C^2 R^5}{L^2}$
-3	$\frac{C^3 R^{12}}{L^8}$	$\frac{C^3 R^{11}}{L^7}$	$\frac{C^3 R^{10}}{L^6}$	$\frac{C^3 R^9}{L^5}$	$\frac{C^3 R^8}{L^4}$	$\frac{C^3 R^7}{L^3}$

Table 17. Electrical coefficients for derivative indices 2 through 6.

r	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
3	$\frac{L^4}{C^3 R^6}$	$\frac{L^5}{C^3 R^7}$	$\frac{L^6}{C^3 R^8}$	$\frac{L^7}{C^3 R^9}$	$\frac{L^8}{C^3 R^{10}}$
2	$\frac{L^3}{C^2 R^4}$	$\frac{L^4}{C^2 R^5}$	$\frac{L^5}{C^2 R^6}$	$\frac{L^6}{C^2 R^7}$	$\frac{L^7}{C^2 R^8}$
1	$\frac{L^2}{C R^2}$	$\frac{L^3}{C R^3}$	$\frac{L^4}{C R^4}$	$\frac{L^5}{C R^5}$	$\frac{L^6}{C R^6}$
0	L	$\frac{L^2}{R}$	$\frac{L^3}{R^2}$	$\frac{L^4}{R^3}$	$\frac{L^5}{R^4}$
-1	$C R^2$	$C L R$	$C L^2$	$\frac{C L^3}{R}$	$\frac{C L^4}{R^2}$
-2	$\frac{C^2 R^4}{L}$	$C^2 R^3$	$C^2 L R^2$	$C^2 L^2 R$	$C^2 L^3$
-3	$\frac{C^3 R^6}{L^2}$	$\frac{C^3 R^5}{L}$	$C^3 R^4$	$C^3 L R^3$	$C^3 L^2 R^2$

A finite signed-atom illustration can include the following distinct derivative and exponent choices.

Table 18. Illustrative signed support spanning initialized integrals and high derivatives. These atoms define a candidate family; no physical realization or conditioning result is implied.

r	k	Weight w
-6	-4	-5/4
-3	-1	3/8
0	0	-9/40
1	2	4/5

r	k	Weight w
4	8	$-3/5$
6	20	$11/10$

Subsets of this list distinguish activation count from derivative order. A finite frequency domain, however wide, cannot certify intermediate cancellation surfaces or the positive-real condition on the entire right half-plane. Nor does a large range of individual term magnitudes establish a bound on (14).

For completeness, the cubic Routh array for $ds^3 + as^2 + bs + c$ has first column $d, a, (ab - dc)/a, c$. With $a, b, c, d > 0$, all entries are positive exactly when $ab > dc$. Equality gives the imaginary pair in (10). This condition addresses the homogeneous poles; numerator zeros and the chosen effort–flow port still decide positive-realness. Higher-order polynomials require their full Hurwitz conditions or another rigorous stability argument, such as the independently constructed passive state model with no undamped invariant subspace.

The economical coefficients also have a closed finite structure. Put $u = (a/c)s^2$. Then

$$\begin{aligned} P_{2m}(s) &= c \sum_{j=0}^m u^j + bs \sum_{j=0}^{m-1} u^j, \\ P_{2m+1}(s) &= (c + bs) \sum_{j=0}^m u^j. \end{aligned} \quad (175)$$

The finite sums, rather than a quotient with a removable singularity, define the expression at $u = 1$. For odd order, every nontrivial root of $1 + u + \dots + u^m$ supplies additional roots in s . This makes explicit why the coefficient convention is no universal stability construction.

B Integration by parts at arbitrary derivative order

Let y have the required continuous derivatives on a smooth interval. For $m \geq 1$ define

$$\begin{aligned} H_{2m} &= \sum_{j=0}^{m-2} (-1)^j y^{(2m-1-j)} y^{(1+j)} + \frac{(-1)^{m-1}}{2} (y^{(m)})^2, \\ H_{2m+1} &= \sum_{j=0}^{m-1} (-1)^j y^{(2m-j)} y^{(1+j)}, \quad H_0 = y^2/2, \quad H_1 = 0. \end{aligned} \quad (176)$$

An empty sum is zero. The product rule gives

$$y^{(2m)} \dot{y} = \dot{H}_{2m}, \quad y^{(2m+1)} \dot{y} = \dot{H}_{2m+1} + (-1)^m (y^{(m+1)})^2. \quad (177)$$

Adjacent terms telescope; the last product in the even case is the derivative of a square. This proves the identities without a physical interpretation. For $P(D)y = f$ they yield

$$\frac{d}{dt} \left(\sum_k B_k H_k \right) = f \dot{y} - \sum_{m \geq 0} (-1)^m B_{2m+1} (y^{(m+1)})^2, \quad (178)$$

with only indices present in P included. The fourth-order expression (165) follows immediately. Multiplying P and f by a nonzero constant multiplies every term here without changing y . Positivity, physical

dimensions of a selected port and a component realization cannot be inferred from this algebra.

C A finite connection event with nonunique port partitions

Let two capacitors $C_1, C_2 > 0$ have voltages v_1, v_2 and be joined during $0 \leq t \leq \varepsilon$ by two parallel, time-dependent conductances $g_A, g_B \geq 0$. Define $C_e = C_1 C_2 / (C_1 + C_2)$, $d = v_1 - v_2$, and total charge $Q = C_1 v_1 + C_2 v_2$. Then

$$\dot{Q} = 0, \quad C_e \dot{d} = -(g_A + g_B)d, \quad E = \frac{Q^2}{2(C_1 + C_2)} + \frac{1}{2}C_e d^2. \quad (179)$$

The two independently signed resistor powers are $P_A = -g_A d^2$ and $P_B = -g_B d^2$. Put $g_j(t) = K C_e \phi_j(t/\varepsilon)/\varepsilon$ with $\int_0^1 \phi_A du = \int_0^1 \phi_B du = 1$. Exact integration gives

$$d^+ = e^{-2K} d^-, \quad W_A + W_B = -\frac{C_e (d^-)^2}{2} (1 - e^{-4K}). \quad (180)$$

For sequential conductances, A then B , the separate integrals are

$$W_A = -\frac{C_e (d^-)^2}{2} (1 - e^{-2K}), \quad W_B = -\frac{C_e (d^-)^2}{2} e^{-2K} (1 - e^{-2K}). \quad (181)$$

Reversing the order exchanges the physical port assignments while preserving the endpoint. For identical coincident profiles each integral is half of (180). The scalar transition maps commute, but their work partitions need not coincide. Multidimensional reset generators need not commute at all.

For the illustrative $C_1 = C_2 = 1$ F, $d^- = 2$ V, the total loss tends to 1 J as $K \rightarrow \infty$. The initial and final common-voltage store is unchanged; the independently evaluated difference store falls by that amount. Taking $\varepsilon \rightarrow 0$ at fixed K yields a finite jump with path-dependent port assignment. A bounded voltage contributes a vanishing $O(\varepsilon)$ amount to an initialized integral history during that shrinking interval; the bound follows directly from $|\int y dt| \leq \varepsilon \sup |y|$, without a series approximation.

After the event, a fixed conductance $G_0 = C_e/\tau$ leaves $d' = -d/\tau$ and constant Q . Each capacitor voltage therefore satisfies $v_j'' + v_j'/\tau = 0$. Before connection the isolated voltage satisfies $v_j' = 0$. The derivative order changes because the accessible difference mode changes. Jets must be reconstructed from the post-event state, and a discarded difference store must not be concealed in an arbitrary derivative reset.

D Invariant sections and exact bounds on observed magnitude

For (144), impose independent linear invariants $A^T y = b$, as illustrated in Figure 3. Set

$$\begin{aligned} y_c &= Q^{-1} A (A^T Q^{-1} A)^{-1} b, \\ E_c &= \frac{1}{2} b^T (A^T Q^{-1} A)^{-1} b, \\ K_c &= Q^{-1} - Q^{-1} A (A^T Q^{-1} A)^{-1} A^T Q^{-1}. \end{aligned} \quad (182)$$

Completing the square gives, for $E \geq E_c$,

$$\max a^T y = a^T y_c + \sqrt{2(E - E_c) a^T K_c a}. \quad (183)$$

If $a^T K_c a > 0$, an attaining observation is $y_c + \sqrt{2(E - E_c)/(a^T K_c a) K_c a}$. If that scalar is zero, the objective is constant on the section. If $E < E_c$, no compatible state exists. Taking the maximum over all sign vectors a gives the exact one-norm envelope for the ellipsoid section. Restricting to a specific sign cell additionally requires its inequalities and boundary faces. These are geometric audits of the current independently determined store and constraints, not a prescription for choosing a physical trajectory.

A reachable observation can lie strictly inside this geometric envelope because topology, input limits and finite time constrain the state. Conversely, appending an independently prepared hidden store can enlarge the full physical energy without changing the observed section. This is why a terminal rational function and a geometric upper bound do not identify an actual internal stress trajectory.

E A finite bias supply and its separate store accounts

The constant-coupling cubic in (125) and a transducer with a finite bias supply are different dynamical systems. This appendix constructs one conditional seven-state model so that source-off coil growth, DC-link transfer and thermal change can be checked separately. The construction states the required port laws; it does not identify a physical implementation of the unequal coupling or settle an unexplained apparatus residual.

E.1 State equations and finite control laws

Take the plant states (i, x, v) from (123), the bias current i_b from (134), the DC-link voltage V_b , sensor voltage z and thermal energy $E_\theta \geq 0$. All capacitances and inductances are positive. Internal plant damping is $c \geq 0$, load damping is $c_\ell \geq 0$, and additional finite reset resistances and dampings are $R_r(t), c_r(t) \geq 0$. On a declared smooth interval the plant equations are

$$\begin{aligned} L\dot{i} &= v_s - (R + R_r)i - \bar{g}_e(i_b/I_*)v, \\ \dot{x} &= v, \quad m\dot{v} = \bar{g}_m(i_b/I_*)i - (c + c_r + c_\ell)v - kx + F_s. \end{aligned} \quad (184)$$

The bias reaction effort A , loss voltage v_ℓ and current equation are (134). They define $P_a = i_b A$ positive into the plant and $-Ai_b$ into the bias winding. A finite converter command can, for example, be

$$u_b = \text{clip}(R_b I_* \alpha + K_b(I_* \alpha - i_b), -V_b, V_b), \quad V_b > 0, \quad K_b \geq 0, \quad 0 \leq \alpha \leq 1. \quad (185)$$

Here K_b has resistance units and $\text{clip}(q, a, b) = \min\{b, \max\{a, q\}\}$. The dimensionless command $\alpha(t)$ is prescribed by the operation or preparation path. A smooth finite connection can use $\alpha = [1 - \cos(\pi(t - t_0)/T)]/2$ on $[t_0, t_0 + T]$; disconnection reverses this path. On the specific source-off interval set $u_b = 0$ as a separate converter mode. Setting $\alpha = 0$ in the feedback expression alone would generally apply a braking voltage instead.

Let an ideal charger effort U feed the DC link through resistance $R_s > 0$ and a symmetric current limit $I_{\max} > 0$. Its constitutive path is

$$j = \text{clip}\left(\frac{U - V_b}{R_s}, -I_{\max}, I_{\max}\right), \quad v_{\text{lim}} = U - V_b - R_s j, \quad P_{\text{line}} = (R_s j)j + v_{\text{lim}} j. \quad (186)$$

Inside the current limit $v_{\text{lim}} = 0$; outside it the limiter voltage and current have the same sign. Thus each loss product is nonnegative and $Uj = V_b j + P_{\text{line}}$ by the declared voltage law. At $U = 0$ the source work is zero but the attached resistance/limiter can still discharge the bus: source-off and open charger are different topologies.

For a dimensionally explicit sensor signal define $h = a_i i + a_v v$, with a_i in volts per ampere and a_v in volt seconds per metre. Let $C_s > 0$, $G_s \geq 0$ and $\tau_s > 0$. In the powered sensing mode set $\dot{z} = (h - z)/\tau_s$; in the source-off mode set $\dot{z} = -G_s z/C_s$. The injected sensor current, signed sensor-supply power and nonnegative logic loss are

$$j_s = C_s \dot{z} + G_s z, \quad P_s = z j_s, \quad P_{\text{logic}} = u_{\text{logic}} j_{\text{logic}}, \quad j_{\text{logic}} = G_{\text{logic}} u_{\text{logic}}, \quad G_{\text{logic}} \geq 0. \quad (187)$$

The sensor power may be negative during return. In the off mode $j_s = 0$; disabling the logic sets $u_{\text{logic}} = 0$. The sensor and logic effort laws describe a candidate implementation and require physical identification. In particular, motion and current are never added without their voltage conversion scales.

For DC-link leakage $G_b \geq 0$, its state equation is

$$C_b \dot{V}_b = j - G_b V_b - \frac{u_b i_b + P_s + P_{\text{logic}}}{V_b}, \quad V_b > 0. \quad (188)$$

The demand contains coil-terminal, sensor and logic powers. Actuator reaction and converter loss already enter the coil equation; adding them again to this DC-link demand would count the same account twice. At $V_b = 0$, nonzero demanded power is outside this realization. A zero-demand continuation requires a separately defined converter law, not division by a vanishing voltage.

Define plant heat rates $H_e = (R + R_r) i^2$, $H_m = (c + c_r) v^2$, bias copper heat $H_b = R_b i_b^2$, link leakage heat $H_g = G_b V_b^2$ and sensor heat $H_s = G_s z^2$. All are effort–flow products in their corresponding electrical or mechanical components. A conditional thermal reservoir obeys

$$\dot{E}_\theta = H_e + H_m + H_b + H_g + H_s + \ell(P_a) + P_{\text{logic}} + P_{\text{line}} - Q_{\text{amb}}, \quad Q_{\text{amb}} = \gamma_\theta E_\theta, \quad \gamma_\theta \geq 0. \quad (189)$$

One constitutive temperature assignment is $\Theta = \Theta_a + E_\theta/C_\theta$ with $\Theta_a, C_\theta > 0$. The outward thermal port has power $\Theta \dot{S}_{\text{out}} = Q_{\text{amb}}$, where $\dot{S}_{\text{out}} = Q_{\text{amb}}/\Theta$ is the specified entropy flow at that boundary. The temperature law, heat destinations and thermal conductance remain material assumptions to identify. Load power $-c_\ell v^2$ crosses to the external load and is not also included in this thermal store.

There are seven continuous coordinates, with local uniqueness on the declared regular domain. The products involving i_b make the combined model nonlinear; a constant third-order characteristic polynomial no longer describes it globally. Scalar observable order requires its own local elimination and state-image analysis. Finite preparation, connection, operation, disconnection, source-off relaxation and resistive reset have distinct vector fields. Continuous state transport gives zero ideal impulse for these finite changes. A finite reset generally retains an endpoint store, and that remainder is carried into the next interval rather than deleted.

E.2 Separate works, endpoints and unresolved local checks

For an explicit finite interval $[t_0, t_1]$ lying within one mode, write $\mathcal{W}[p] = \int_{t_0}^{t_1} p(t) dt$. Every occurrence below denotes a separately evaluated signed port integral. Evaluate the stores independently at both ends:

$$E_p = \frac{Li^2 + mv^2 + kx^2}{2}, \quad E_b = \frac{L_b i_b^2}{2}, \quad E_{\text{dc}} = \frac{C_b V_b^2}{2}, \quad E_s = \frac{C_s z^2}{2}, \quad E_{\text{all}} = E_p + E_b + E_{\text{dc}} + E_s + E_\theta. \quad (190)$$

The separate inward power products for the four reactive sub-boundaries are

Table 19. Nested boundaries for the conditional finite supply. Thermal receiving powers are given separately in (189).

Store boundary	Inward port powers, each integrated separately
Plant E_p	$v_s i, F_s v, -H_e, -H_m, -(c_\ell v)v, P_a$
Bias coil E_b	$u_b i_b, -H_b, -A i_b, -v_\ell i_b$
DC link E_{dc}	$V_b j, -H_g, -u_b i_b, -z j_s, -u_{logic} j_{logic}$
Sensor E_s	$z j_s, -H_s$

Define each residual as its independent store change minus the sum of those individual works. In particular,

$$\begin{aligned} r_{dc} &= \Delta E_{dc} - (\mathcal{W}[V_b j] + \mathcal{W}[-H_g] + \mathcal{W}[-u_b i_b] + \mathcal{W}[-P_s] + \mathcal{W}[-P_{logic}]), \\ r_s &= \Delta E_s - \mathcal{W}[P_s] - \mathcal{W}[-H_s]. \end{aligned} \quad (191)$$

The bias residual is (135), with $\mathcal{W}[u_b i_b]$ additionally retained when the converter is on. The thermal residual compares ΔE_θ with the separate integrals of each receiving heat and the outward ambient port. The plant residual includes $\mathcal{W}[P_a]$ only for this declared candidate realization; the listed-port residual (127) remains a separate comparison.

Multiplying each state equation by its conjugate effort proves these conditional identities. Adding only after integration cancels the common internal works in pairs. The boundary containing the plant, supply, sensor, line losses and thermal reservoir then has

$$r_{all} = \Delta E_{all} - (\mathcal{W}[U j] + \mathcal{W}[v_s i] + \mathcal{W}[F_s v] + \mathcal{W}[-c_\ell v^2] + \mathcal{W}[-Q_{amb}]). \quad (192)$$

This is zero for an exact solution of the full declared model. The controller boundary containing only the bias coil and DC link instead retains charger-terminal work, actuator reaction, converter/copper/leakage losses, sensor supply and logic loss separately. Thus $V_b j$, rather than $U j$, is its charger port; the source-line loss is outside that smaller boundary. This distinction remains relevant when the source command is zero.

The exact derivation establishes the model's internal consistency without a numerical integration error. It leaves specific physical and uncertainty questions open: whether the coil reaction is realized; whether each DC-link interval and its integrated demand agree; whether thermal endpoints can be independently reconstructed; and whether finite work evaluations resolve each transfer on stable or unstable prefixes. A coil identity or paired transfer cancellation does not resolve DC-link or thermal uncertainty. A combined complete-path result cannot replace those local checks. Retain positive and negative residuals and absolute as well as relative uncertainty on growing prefixes; finite supply ratings, depletion, saturation and unmodeled converter dynamics delimit how far any such prefix represents a physical system. No convergence or hardware result follows from writing an exact conditional identity.

F Finite reference controllers and nested port accounts

The receiver (156) and reference cell (158) specify different levels of implementation. An ideal state-dependent controller imposes $r = \chi v_s$ with compatible initial data, and four ideal node-current sources plus four compensation sources drive the receiver. A ninth ideal bidirectional converter supplies the

reference driver's command voltage $u = r + R_d C_d \dot{r}$. If P_k is the separately declared output power of converter k , a finite rail with capacitance C_s and controller conductance G_q obeys

$$C_s V_s \dot{V}_s = - \sum_{k=1}^9 P_k - G_q V_s^2, \quad E_{\text{rail}} = \frac{1}{2} C_s V_s^2, \quad V_s > 0. \quad (193)$$

For the receiver outputs, $P_k = w_X j_X$ or $-C_X w_X \dot{r}$; the driver output is $u j_d$. Input and output work agree only for these explicitly lossless converter laws. Tracking is an imposed constitutive relation, not a consequence of the rail balance. Controller heat and each converter transfer are integrated separately; the rail endpoint is evaluated from V_s . Source and controller ratings constrain the admissible trajectory before its energy audit.

F.1 Finite sensing, actuation and conversion

A finite implementation instead has four receiver voltages w_X , four winding currents, nine RC sensor voltages z_a , eight actuator currents j_b , command voltage u , reference voltage r and rail voltage V_s : 28 independent regular coordinates before any additional ideal constraints. The nine sensed quantities are four absolute node voltages $w_X + r$, four winding currents and driver current. Current channels require a calibrated conversion factor β with units V A^{-1} ; for example $\beta = 1/10 \text{ V A}^{-1}$ is an illustrative assumption, not a dimensionless current-to-voltage identity. With rail-limited sensor targets ξ_a ,

$$I_a = \frac{\xi_a - z_a}{R_a}, \quad C_a \dot{z}_a = I_a, \quad E_a = \frac{1}{2} C_a z_a^2. \quad (194)$$

The separately integrated inward powers are $\xi_a I_a$ and $-R_a I_a^2$. The transduction input is nonloading under this model; a physical input impedance requires additional receiver ports. Equivalently, on a smooth target interval one may use $\ell_a = (\xi_a - z_a)/(R_a C_a)$ with $\dot{\ell}_a = (\dot{\xi}_a - \dot{\ell}_a)/(R_a C_a)$ and $z_a = \xi_a - R_a C_a \ell_a$. Compatibility of this changed coordinate is essential; it does not change sensor storage or eliminate its state.

From a selected voltage sensor, set $r_* = \chi z_S$ and $\dot{r}_* = \chi \dot{z}_S + \dot{\chi} z_S$. External-current targets use the sensed receiver state; a finite holding servo uses the sensed primary current minus a specified conductance times the sensed held-node voltage. Four other targets are $-C_X \dot{r}_*$. Thus finite holding error and finite compensation error are part of the dynamics, not suppressed constraints.

Each of the eight actual source currents passes through its own inductor L_b and resistor R_b . A declared local feedback law with time constant τ_b gives

$$e_{b,\text{raw}} = w_X + R_b j_b + \frac{L_b}{\tau_b} (j_{b,*} - j_b), \quad L_b \dot{j}_b = e_b - w_X - R_b j_b. \quad (195)$$

$$e_b = \text{clip}(e_{b,\text{raw}}, -\gamma_b V_s, \gamma_b V_s),$$

Here $0 < \gamma_b \leq 1$ is a declared voltage limit. The store is $L_b j_b^2/2$, and its three inward powers are $e_b j_b$, $-w_X j_b$ and $-R_b j_b^2$. Adding four external and four compensation currents into the receiver yields $C \dot{w} = j_{\text{ext}} + j_{\text{comp}} - B i$. Source/load coefficients now command regenerative actuators; they are not silently counted as passive source or load resistors. The separate passive circuit remains the one preceding (158). The command capacitor and physical driver satisfy

$$C_u \dot{u} = J_u - j_d, \quad J_u = \hat{j}_d + \frac{C_u}{\tau_u} (u_* - u),$$

$$u_* = \text{clip}(r_* + R_d C_d \dot{r}_*, -\gamma_u V_s, \gamma_u V_s), \quad j_d = (u - r)/R_d, \quad C_d \dot{r} = j_d. \quad (196)$$

The sensed driver current $\hat{j}_d$ is reconstructed using its calibrated β . The command boundary has store $C_u u^2/2$ and powers $uJ_u, -uj_d$. The driver boundary has store $C_d r^2/2$ and powers $uj_d, -R_d j_d^2$; ideal floating-network bias transfer is zero. The driver capacitor is additional to the receiver's carrier capacitance.

Eighteen converters supply eight actuators, nine sensors and the command capacitor. For each separately declared output power $P_{\text{out},k}$, let $0 < \eta_k \leq 1$ and

$$P_{\text{in},k} = \begin{cases} P_{\text{out},k}/\eta_k, & P_{\text{out},k} \geq 0, \\ \eta_k P_{\text{out},k}, & P_{\text{out},k} < 0. \end{cases} \quad H_k = P_{\text{in},k} - P_{\text{out},k} \geq 0. \quad (197)$$

Each converter has separately integrated input, negative output and negative heat powers. It has no store in this constitutive idealization; switching filters or semiconductor thermal dynamics would add states. This efficiency law specifies the transfer before any residual is evaluated.

Writing $q = V_s^2$ avoids division by rail voltage inside the stated positive-voltage range:

$$\dot{q} = -\frac{2}{C_s} \left(\sum_{k=1}^{18} P_{\text{in},k} + G_q q \right), \quad E_{\text{rail}} = \frac{1}{2} C_s q, \quad V_s \geq V_c > 0. \quad (198)$$

The inward rail powers are the eighteen $-P_{\text{in},k}$ and $-G_q q$. The combined circuit exports 41 distinct heats: four winding, eight actuator, one driver, nine sensor, eighteen converter and one controller heat. This port count is not its state dimension. Independent receiver, actuator, sensor, command, driver, converter and rail accounts are retained before canceling internal transfers in the combined boundary. The ideal mechanical comparator has its own shaft, accelerating-reference and mobility-loss integrals and independent endpoints. A zero residual in each model does not imply equal port works between the models.

F.2 Preparation, cutoff and fast transfers

Prepare the isolated rail from zero voltage using constant current $j_p = C_s V_0/T_p$ through a resistor R_p for a finite chosen time T_p . The source voltage is $u_p = V_0 t/T_p + R_p j_p$. Direct integration gives

$$W_p = \frac{1}{2} C_s V_0^2 + \frac{R_p C_s^2 V_0^2}{T_p}, \quad W_{R_p} = -\frac{R_p C_s^2 V_0^2}{T_p}, \quad E_0 = 0, \quad E_1 = \frac{1}{2} C_s V_0^2. \quad (199)$$

The exact preparation residual is zero; preparation is not inferred from a desired energy budget. Receiver, sensor and actuator initial states must be specified separately.

An exact zero-drive cutoff control initializes every other state at zero. Then all converter powers vanish and

$$q(t) = V_0^2 e^{-2G_q t/C_s}, \quad t_c = \frac{C_s}{G_q} \log \frac{V_0}{V_c}, \quad W_q(0, t) = -\frac{1}{2} C_s V_0^2 (1 - e^{-2G_q t/C_s}). \quad (200)$$

Independent endpoints are $C_s V_0^2/2$ and $C_s V_c^2/2$ at cutoff, so the residual is zero and the remaining store is positive. For $C_s = 1$ F and $V_c = 1/100$ V it is 1/20000 J. An active-drive cutoff generally retains receiver and actuator stores too. Shutdown, reset and the event's timing uncertainty require their own intervals; termination of the operating equations does not set their endpoints to zero.

Fast same-sign transfers need not have an interior power zero. For example the exact positive pulse $P(t) = P_0 e^{-t/\tau}$ has work $P_0 \tau (1 - e^{-T/\tau})$ on $[0, T]$. A procedure that finds every sign change but does not resolve this scale can miss its work. Opposite actuator and rail errors may cancel in the combined account. Likewise an inaccurate rail endpoint can dominate a residual even when the port-work sum changes little. These are distinct uncertainty mechanisms, not evidence for a new transfer; a resolved physical residual still requires an independent account.

A specified approach to the ideal controller may take sensor and actuator time constants proportional to ε^2 , actuator inductances and command capacitance proportional to ε , sensor capacitances proportional to ε^2 , actuator resistances proportional to ε , and $\eta = 1 - c\varepsilon$ with $0 \leq c\varepsilon < 1$. Receiver coefficients, the RC driver and finite rail remain fixed. These simultaneous choices do not prove uniform tracking or work convergence: clipping, prepared fast states, rail depletion and persistent tracking error require bounds on the complete interval. The linear event enclosure in Appendix H is not a certificate for this nonlinear limited driver. Hardware error and global nonlinear event completeness remain open.

G Constrained and prepared limits of the compound network

Singular component values change the equation class. A descriptor system $E\dot{x} = Ax + b$ at singular E requires its algebraic constraints and compatible initial charge/flux before an ODE order is assigned. Setting a capacitance to zero imposes node-current balance; setting source resistance to zero imposes a voltage clamp. Neither operation allows arbitrary initial voltage. A zero signed winding ratio changes topology rather than giving the regular inverse in (156).

G.1 Perfect coupling retains a magnetizing state

For one pair at $\kappa = 1$, put $a = (h, 1)^T$ so $L = L_0 a a^T$. Compatible flux is $\lambda = a\phi$. With positive winding resistance matrix R and winding effort vector u , KVL and $a^T i = \phi/L_0$ imply

$$\dot{\phi} = \frac{a^T R^{-1} u - \phi/L_0}{a^T R^{-1} a}, \quad i = R^{-1} (u - a\dot{\phi}), \quad E_\phi = \frac{\phi^2}{2L_0}. \quad (201)$$

This still has a dynamic magnetic state. Its source works are the individual $\int u_j i_j dt$ and its heat works the individual $-\int R_j i_j^2 dt$; endpoint flux determines the store independently. Reversing the coupling sign changes a consistently. Removing resistance, magnetizing inductance and capacitance in a joint limit is a different constrained problem.

An energized voltage clamp illustrates why continuity and event work cannot be guessed. Connect a capacitor initially at v_- to fixed voltage U through $\rho > 0$. For $t \geq 0$,

$$v(t) = U + (v_- - U)e^{-t/(\rho C)}, \quad i(t) = (U - v_-)\rho^{-1}e^{-t/(\rho C)}, \\ W_U(0, T) = CU(U - v_-)(1 - e^{-T/(\rho C)}), \quad W_\rho(0, T) = -\frac{C}{2}(U - v_-)^2(1 - e^{-2T/(\rho C)}). \quad (202)$$

Independently use $E_0 = Cv_-^2/2$ and $E_T = Cv(T)^2/2$; their difference minus both works is zero. For a completed layer, the source share is $CU(U - v_-)$ and the heat share $-C(U - v_-)^2/2$, while the endpoint change is $C(U^2 - v_-^2)/2$. Taking $\rho \rightarrow 0$ after retaining these integrals does not erase them. A bounded-voltage inductor retains continuous flux; a discontinuous flux needs its own impulse law. Mechanical rate clamps have the corresponding momentum and kinetic-store constraints. Finite locking friction, impact and magnetic saturation are additional physical laws.

Similarly, (155) retains a resistor work proportional to $-1/T$ at fixed $R_d, C_d, \Delta r$. A zero-duration reference shift is not a finite-work limit of that fixed driver. The measured finite-duration predictions remain valid under their stated component and supply limits.

G.2 A specified rigid family and its hidden preparations

For $0 < \varepsilon \leq 1$ set $L_0 = \ell_0/\varepsilon$, $\kappa = 1 - \varepsilon^2$, $R_0 = r_0\varepsilon^2$, and node capacitances $C = \varepsilon C_0$, with fixed positive ℓ_0, r_0, C_0 and fixed nonzero ratios. For one pair define $p = hi_1$, $q = i_2$, common current $s_c = p + q$ and differential current $d_c = p - q$. The exact independent winding equations and store are

$$\begin{aligned} L_+ \dot{s}_c &= S - R_0 s_c, & S &= u_1/h + u_2, & L_+ &= \ell_0(2 - \varepsilon^2)/\varepsilon, \\ L_- \dot{d}_c &= D - R_0 d_c, & D &= u_1/h - u_2, & L_- &= \ell_0\varepsilon, \\ E_{\text{pair}} &= \frac{1}{4}(L_+ s_c^2 + L_- d_c^2). \end{aligned} \quad (203)$$

Here d_c denotes this appendix's differential current, not a contact damping coefficient. Both modes retain their independently prepared initial values:

$$y(t) = y(0)e^{-R_0 t/L_{\pm}} + \frac{1}{L_{\pm}} \int_0^t e^{-R_0(t-\xi)/L_{\pm}} f(\xi) d\xi, \quad (y, f) = (s_c, S) \text{ or } (d_c, D). \quad (204)$$

This exact convolution, not a truncated expansion, determines what survives.

For matched constant winding efforts $u_1/h = u_2 = U$ and $s_c(0) = 0$, $s_c = 2U(1 - e^{-R_0 t/L_+})/R_0$ and $d_c = d_c(0)e^{-R_0 t/L_-}$. The identity $1 - e^{-z} = \int_0^z e^{-u} du \leq z$ gives, uniformly on $[0, T]$,

$$|s_c(t)| \leq \frac{2|U|T}{L_+}, \quad |d_c(t) - d_c(0)| \leq |d_c(0)| \frac{R_0 T}{L_-}. \quad (205)$$

For bounded initial differential current both magnetic stores vanish along this matched family. Bounded voltages also make the capacitive stores vanish. This is a conditional reduction, not a claim about arbitrary preparation or drive.

For an explicit finite preparation, ramp isolated winding currents from zero to i_0 linearly over $[0, T_p]$, with source efforts $u = Li_0/T_p + Ri_0 t/T_p$. Each source and heat work is

$$W_{u,j} = \frac{1}{2}i_{0,j}(Li_0)_j + \frac{T_p}{3}R_j i_{0,j}^2, \quad W_{R,j} = -\frac{T_p}{3}R_j i_{0,j}^2. \quad (206)$$

Independent magnetic endpoints are zero and $i_0^T Li_0/2$. A separate linear voltage ramp from zero to $v_{0,X}$ at each capacitor has signed source work $C_X v_{0,X}^2/2$ and the same independently evaluated final store. The complete preparation residual is zero under these ideal finite drives. Reconnection with continuous capacitor voltages and winding fluxes and compatible secondary currents has no imposed impulse; real switching needs a measured finite path. On the matched family the extra planet clamp and copper losses tend to zero under (205), but their finite works remain part of every finite account.

An independently imposed common preparation $s_c(0) = s_* \varepsilon^p$ instead gives exactly

$$E_+(0) = \frac{\ell_0 s_*^2}{4}(2 - \varepsilon^2)\varepsilon^{2p-1}. \quad (207)$$

It vanishes for $p > 1/2$, tends to $\ell_0 s_*^2/2$ for $p = 1/2$, and grows without bound for $p < 1/2$ when $s_* \neq 0$. At $p = 1/2$ the current tends to zero while a finite store remains; at $p = 0$ a fixed common current carries a diverging store in this ideal parameter family. Opposed common preparations in a compound assembly may cancel in selected terminal observations without canceling either store. Their finite preparation paths, component ratings and signed source-return works must be evaluated, not inferred from apparent rigid motion.

A small drive mismatch can survive as well. With $d_c(0) = 0$ and constant $D = D_* \varepsilon^q$,

$$d_c(t) = \frac{D_* \varepsilon^q}{R_0} (1 - e^{-R_0 t/L_-}), \quad \frac{L_- d_c(t)}{D_* \varepsilon^q t} = \frac{1 - e^{-z}}{z}, \quad z = R_0 t/L_- . \quad (208)$$

For fixed $t > 0$ and $D_* \neq 0$, the last ratio tends to one. Consequently the current tends to zero for $q > 1$, to $D_* t/\ell_0$ for $q = 1$, and grows for $q < 1$. Its differential store tends to $D_*^2 t^2/(4\ell_0)$ when $q = 1/2$, even though the effort mismatch tends to zero. For nonconstant drive, the signed convolution, including cancellation and initialization, replaces this constant-drive conclusion.

G.3 Passive loading and fast-layer conditions

For a held member of ratio $h_H \neq 0$ and free member of ratio h_F , the ideal rigid voltage relation is $v_F = a v_C$, $a = 1 - h_F/h_H$. Retain a passive source resistor R_s , a receiver conductance G_g to ground and G_n to the carrier. Their matched static solution is

$$v_C = \frac{u}{1 + R_s[G_g a^2 + G_n(a-1)^2]}, \quad v_F = a v_C, \quad i_F = -[G_g a + G_n(a-1)]v_C. \quad (209)$$

This follows directly from constrained node balance and each receiver's current law. Different receiver connections change both the loading and finite dynamics. Matching a prescribed winding drive does not establish this passive-source limit.

For the differential fast subsystem, let B_- be the differential-mode incidence after the held coordinate is eliminated and let G describe the retained passive conductances. Its scaled matrix and positive metric are

$$F_f = \begin{pmatrix} -C_0^{-1}G & -C_0^{-1}B_- \\ (2/\ell_0)B_-^T & 0 \end{pmatrix}, \quad H_f = \text{diag}(C_0, (\ell_0/2)I). \quad (210)$$

Direct multiplication gives $F_f^T H_f + H_f F_f = \text{diag}(-2G, 0)$. Strict decay requires the absence of a nonzero invariant undamped subspace; positive semidefinite dissipation alone is insufficient. If this condition makes F_f Hurwitz, constants $M, \gamma > 0$ give $\|\exp(F_f t/\varepsilon)\| \leq M e^{-\gamma t/\varepsilon}$. Variation of constants then bounds forced deviations when the forcing and compatible slow states are bounded. Initial layers remain at $t = 0$ unless prepared away. Degenerate geometry, unmatched forcing, precharged fast stores or a changing limiting constraint need separate analysis.

These limits distinguish convergence of node rates, of winding currents, of separately integrated port works and of endpoint stores. None implies the others without its own bounds. A favorable signed-work comparison can coexist with large hidden stores or a different current history. Exact finite initial-state formulas retain those alternatives as physical preparation and measurement questions.

H Exact events and their model-specific scope

On any fixed linear switching interval let $\dot{x} = F(t)x + b(t)$ with affine analytic coefficients. Augmenting by a constant coordinate reduces it to a homogeneous analytic system. An instantaneous power is then a

linear or quadratic analytic observable. Lifting the quadratic monomials and adjoining work coordinates also gives a linear analytic system when discussing equalities between integrated works. A zero of an individual power, equality of two absolute powers, and a physical change of circuit law are different events.

To enclose events exactly, expand an observable f about a center c in its convergent Taylor series. Suppose it is analytic on the complex disk $|t - c| \leq \rho$ with $|f| \leq M_f$. For an affine matrix $F(t) = F(c) + (t - c)F_1$, a state norm bound on that disk is $\|x(c)\| \exp(\|F(c)\|\rho + \|F_1\|\rho^2/2)$; the observable's declared linear or quadratic form gives M_f . Cauchy's coefficient bound yields, for $q = |t - c|/\rho < 1$,

$$\begin{aligned} \left| f(t) - \sum_{k=0}^N a_k (t - c)^k \right| &\leq \frac{M_f q^{N+1}}{1 - q}, \\ \left| f'(t) - \sum_{k=1}^N k a_k (t - c)^{k-1} \right| &\leq \frac{M_f q^N [(N + 1) - Nq]}{\rho (1 - q)^2}. \end{aligned} \quad (211)$$

These are exact convergent-series remainder bounds, not an asserted unbounded-domain polynomial approximation. A value enclosure excluding zero proves no root in its interval. Opposite endpoint signs together with a derivative enclosure excluding zero certify one crossing. Tangencies require a separate multiplicity or factor argument; a sign-change search alone misses them.

For a power product, retain zeros of both effort and flow. For absolute-power comparisons, factor the equality into $P_j - P_k = 0$ or $P_j + P_k = 0$. Identical factors give persistent ties rather than isolated roots. Overlapping enclosures of different factors do not prove coincident roots; an exact relation $f(t) = c(t)g(t)$ with c finite and nonzero throughout the interval does preserve their zeros and multiplicities. Between a complete set of enclosed events, signs and maximizing branches remain fixed; each signed port work is integrated before any maximum or ratio is formed. Integrals over the root enclosures themselves remain in the partition and add with the root-free pieces. A ratio with zero denominator is undefined, including zero divided by zero. Extra apparent crossings from finite observation error do not supply physical switching times.

For example define terminal and internal diagnostics $T(t) = \max_X |v_X j_X|$ and $C(t) = \max_b |e_b f_b|$ over explicitly named mechanical core ports. Their ratio compares those lists, not whole-system energy. A finite network's shaft, mesh and pin powers need not exhaust its store/heat account. For pair X , denote its primary and secondary mechanical branch torques by f_X and $\tilde{f}_X$. The extra mesh couple is $m_X = \tilde{f}_X + h_X f_X$, while the carrier pin torque includes $J_{\text{orb}} \dot{\omega}_C + (h_A - 1)f_A + (h_B - 1)f_B$, with orbital inertia J_{orb} . Orbital acceleration, these extra couples and the chosen frame must be retained before comparing the core diagnostics. Summed charging current requires the corresponding reference return when node-power reference invariance is invoked.

Exact reversal histories supply controls independent of root searching. For a single inertia $J = 1$ with zero initial velocity, $\tau = 1 - 2t$ on $[0, 1]$ gives $\omega = t - t^2$, zero endpoint stores and signed shaft works $+1/32$ and $-1/32$ on its two half intervals. Thus zero total work does not mean zero intermediate transfers.

For the lossless compound receiver, choose $v = u a(t)$, where u is a fixed node-voltage vector and the scalar polynomial a vanishes at both endpoints with $\int_0^T a(t) dt = 0$. Put $A(t) = \int_0^t a(\xi) d\xi$ and initialize zero winding flux. Then

$$\lambda = B^T u A, \quad i = L^{-1} B^T u A, \quad j = C u \dot{a} + B L^{-1} B^T u A. \quad (212)$$

Each node's source power is a fixed linear combination of $a\dot{a}$ and aA ; its integral is zero separately since a and A vanish at both endpoints. Every capacitor and winding endpoint store is independently zero. Subinterval works can be nonzero. On normalized $[0, 1]$, $a = t(1-t)(1-2t)$ gives $A = t^2(1-t)^2/2$. The distinct choice $a = t(1-t)(3t-1)^2(7t-5)/63$ has an interior double zero at $t = 1/3$ and primitive $A = -t^2(t-1)^2(21t^2 - 18t + 5)/126$; the identically zero profile is a third case. A planet node can be left without injection if both $e_P^T C u = 0$ and $e_P^T B L^{-1} B^T u = 0$. These are explicit preparation and forcing conditions, not a universal zero-work statement. A physical moving reference additionally requires the compensation, driver and supply ports already derived; zero receiver work does not assign their work zero.

Linear drive-sign reversal with reversed compatible initial data changes states' signs but leaves quadratic powers unchanged. It supplies an exact symmetry, not an independent physical trial. Analytic event enclosures certify only the specified linear laws, coefficients and intervals. The finite-rail clipped controller is nonlinear and piecewise smooth; its saturation branches, fast same-sign transfers, cutoff and all possible tangencies require separate enclosures. Finite pointwise agreement and empirical accuracy estimates cannot establish global root completeness or hardware accuracy. The exact constructions resolve the stated bounded mathematical questions while retaining those broader open ones.

I Exact verification identities and comparison obligations

The principal algebraic checks are summarized below to make the derivations reproducible without computational output. They support the coefficient constructions in Sections 2–5 and distinguish those exact results from further claims about initialization, realization and observation.

Table 20. Independent checks that determine the scope of the exact results.

Construction	Exact check	Boundary that must remain separate
Coefficient monomial	$\alpha + \beta + \gamma = 1, 2\alpha + \beta = k$	Zero references and nonconvergent tails
Collision elimination	Determinant of the two-body dynamic stiffness matrix	$\Delta = 0$, fixed or massless bodies
Branch coefficient recurrence	Split the added branch in (36) to obtain (39)	Unreduced degree, repeated branch times and prepared states
Support uniqueness	Polynomial independence after clearing Laurent denominators, Lemma 1	Restricted parameter domains and finite-observation uncertainty
Cubic stability	First Routh column $d, a, (ab - dc)/a, c$	Equality and leading-coefficient zero
State image	$\mathcal{O}_m x_0 + a$ and its exact nullspace	Affine forcing and unobservable factors
RC positive-realness	Polynomial $q_N(x) \geq 0$ for $x \geq 0$	Signed regions without a supplied synthesis
Finite line	Telescoping node/branch power products	Open/short, field preparation and radiation

Construction	Exact check	Boundary that must remain separate
Clamp	Substitution of (149) into (147)	First-zero opening and perfect coupling
Absorber	Two-by-two determinant and Schur elimination	Zero mass/damping coscalings
Transducer	Cubic determinant and Hermitian port matrix	Bias/controller supply and unstable prefixes
Connection event	Integrate $-g_A d^2$ and $-g_B d^2$ separately	Edge ordering and incompatible initial states

There is also an exact discrete algebraic control for a lossless linear state equation. If $F^T H + H F = 0$ and an implicit midpoint update is defined by

$$x_+ - x = hF \frac{x_+ + x}{2}, \quad (213)$$

then $E(x_+) - E(x) = h[(x_+ + x)/2]^T H F [(x_+ + x)/2] = 0$ whenever the update exists. This identity is independent of arithmetic implementation. It does not supply a finite-precision drift bound, an event rule, or a physical model certificate; those remain separate uncertainty questions. Likewise, a null singular value and a divergent unstable continuation are exact features before any arithmetic error is assessed.

The proved complete-model balances follow from separate port integrals and endpoint states. Exact integration has zero numerical residual; it does not make every declared-boundary balance zero. The contact deletion and listed-port coupling examples retain their signed nonzero residuals, while the expanded supply accounts remain conditional on their state and port laws. Unspecified material, sensor, switching, thermal, acoustic and exterior-field effects remain r_{model} , whose sign and magnitude are not invented. Unresolved interval, DC-link, thermal-endpoint and stochastic-work checks retain their own scope. Every surviving discrepancy is reported with its interval, orientation, preparation, event sides and uncertainty rather than absorbed into a presumed destination.

References

1. Wettstein, A., Graubeger, P., and Matthiesen, S. (2021). [Modeling dynamic mechanical system behavior using sequence modeling of embodiment function relations: case study on a hammer mechanism](#). *SN Applied Sciences* **3**, article 128. DOI: 10.1007/s42452-021-04149-8.
2. Bible, S. (2002). [Crystal Oscillator Basics and Crystal Selection for rfPIC and PICmicro Devices](#). Microchip Technology, Application Note AN826, DS00826A, pp. 1–14.
3. Coilcraft (n.d.). [Measuring Self Resonant Frequency](#). Technical application note.
4. Keysight Technologies (n.d.). [Impedance Measurement Handbook](#). Application note 5950-3000.
5. Kalman, R. E. (1963). [Mathematical Description of Linear Dynamical Systems](#). *Journal of the Society for Industrial and Applied Mathematics, Series A: Control* **1**(2), 152–192. DOI: 10.1137/0301010.
6. Willems, J. C. (1972). [Dissipative dynamical systems Part II: Linear systems with quadratic supply rates](#). *Archive for Rational Mechanics and Analysis* **45**, 352–393. DOI: 10.1007/BF00276494.
7. Foster, R. M. (1924). [A Reactance Theorem](#). *Bell System Technical Journal* **3**(2), 259–267. DOI: 10.1002/j.1538-7305.1924.tb01358.x.
8. Rice, S. O. (1945). [Mathematical Analysis of Random Noise](#). *Bell System Technical Journal* **24**(1), 46–156. DOI: 10.1002/j.1538-7305.1945.tb00453.x.